7 Transcendental Functions



OVERVIEW Exponential functions model a wide variety of phenomena of interest in science, engineering, mathematics, and economics. These functions are useful in studying the growth of a biological population, the spread of a disease or of information throughout a human community, drug dosages, radioactive elements and their role in dating fossils, the earth's atmospheric pressure, temperature changes, waves, electrical circuits, the vibrations in bridges, interest rates, and probabilities.

In this chapter, we use the methods of calculus to obtain rigorous and precise definitions and properties of the exponential functions. We first define the natural logarithm function $y = \ln x$ as a certain integral, and then the natural exponential function $y = e^x$ as its inverse function. These two foundational functions form the basis for the full array of logarithmic and exponential functions. We also introduce inverse trigonometric functions, as well as hyperbolic functions and their inverses, and investigate their applications. Along with the trigonometric functions, all of these functions belong to the category of *transcendental functions*.

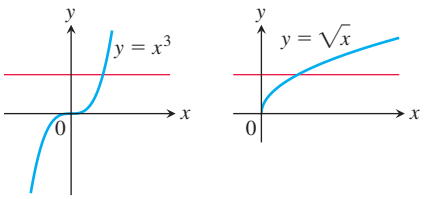
7.1 Inverse Functions and Their Derivatives

A function that undoes, or inverts, the effect of a function f is called the *inverse* of f. Many common functions, though not all, are paired with an inverse. Important inverse functions often show up in applications. Inverse functions also play a key role in the development and properties of the exponential functions. To have an inverse, a function must possess a special property over its domain.

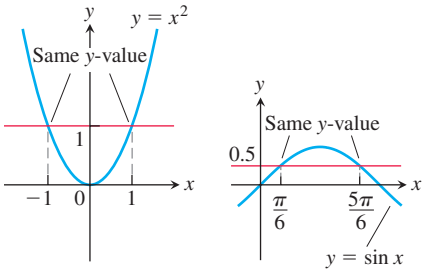
One-to-One Functions

A function is a rule that assigns a value from its range to each element in its domain. Some functions assign the same range value to more than one element in the domain. The function $f(x) = x^2$ assigns the same value, 1, to both of the numbers -1 and $+1$. Similarly the sines of $\pi/3$ and $2\pi/3$ are both $\sqrt{3}/2$. Other functions assume each value in their range no more than once. The square roots and cubes of different numbers are always different. A function that has distinct values at distinct elements in its domain is called *one-to-one*.

DEFINITION A function $f(x)$ is **one-to-one** on a domain D if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ in D.



(a) One-to-one: Graph meets each horizontal line at most once.



(b) Not one-to-one: Graph meets one or more horizontal lines more than once.

FIGURE 7.1 (a) $y = x^3$ and $y = \sqrt{x}$ are one-to-one on their domains $(-\infty, \infty)$ and $[0, \infty)$. (b) $y = x^2$ and $y = \sin x$ are not one-to-one on their domains $(-\infty, \infty)$.

EXAMPLE 1 Some functions are one-to-one on their entire natural domain. Other functions are not one-to-one on their entire domain, but by restricting the function to a smaller domain we can create a function that is one-to-one. The original and restricted functions are not the same functions, because they have different domains. However, the two functions have the same values on the smaller domain.

(a) $f(x) = \sqrt{x}$ is one-to-one on any domain of nonnegative numbers because $\sqrt{x_1} \neq \sqrt{x_2}$ whenever $x_1 \neq x_2$.

(b) $g(x) = \sin x$ is *not* one-to-one on the interval $[0, \pi]$ because $\sin(\pi/6) = \sin(5\pi/6)$. In fact, for each element x_1 in the subinterval $[0, \pi/2)$ there is a corresponding element x_2 in the subinterval $(\pi/2, \pi]$ satisfying $\sin x_1 = \sin x_2$. The sine function *is* one-to-one on $[0, \pi/2]$, however, because it is an increasing function on $[0, \pi/2]$ and therefore gives distinct outputs for distinct inputs in that interval. ■

The graph of a one-to-one function $y = f(x)$ can intersect a given horizontal line at most once. If the function intersects the line more than once, then it assumes the same y-value for at least two different x-values and is therefore not one-to-one (Figure 7.1).

The Horizontal Line Test for One-to-One Functions
A function $y = f(x)$ is **one-to-one** if and only if its graph intersects each horizontal line at most once.

Inverse Functions

Since each output of a one-to-one function comes from just one input, the effect of the function can be inverted to send each output back to the input from which it came.

DEFINITION Suppose that f is a one-to-one function on a domain D with range R. The **inverse function** f^{-1} is defined by

$$f^{-1}(b) = a \quad \text{if} \quad f(a) = b.$$

The domain of f^{-1} is R and the range of f^{-1} is D.

Caution
Do not confuse the inverse function f^{-1} with the reciprocal function $1/f$.

The symbol f^{-1} for the inverse of f is read "f inverse." The "-1" in f^{-1} is *not* an exponent; $f^{-1}(x)$ does not mean $1/f(x)$. Notice that the domains and ranges of f and f^{-1} are interchanged.

EXAMPLE 2 Suppose a one-to-one function $y = f(x)$ is given by a table of values

x	1	2	3	4	5	6	7	8
$f(x)$	3	4.5	7	10.5	15	20.5	27	34.5

A table for the values of $x = f^{-1}(y)$ can then be obtained by simply interchanging the values in each column of the table for f:

y	3	4.5	7	10.5	15	20.5	27	34.5
$f^{-1}(y)$	1	2	3	4	5	6	7	8

■

If we apply f to send an input x to the output $f(x)$ and follow by applying f^{-1} to $f(x)$, we get right back to x, just where we started. Similarly, if we take some number y in the range of f, apply f^{-1} to it, and then apply f to the resulting value $f^{-1}(y)$, we get back the value y from which we began. Composing a function and its inverse has the same effect as doing nothing.

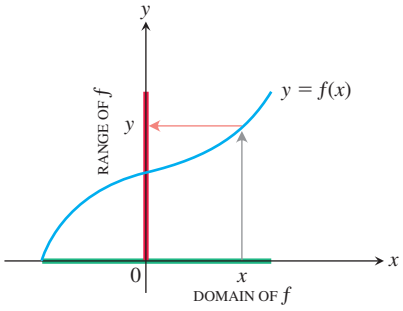
$$(f^{-1} \circ f)(x) = x, \qquad \text{for all } x \text{ in the domain of } f$$
$$(f \circ f^{-1})(y) = y, \qquad \text{for all } y \text{ in the domain of } f^{-1} \text{ (or range of } f)$$

Only a one-to-one function can have an inverse. The reason is that if $f(x_1) = y$ and $f(x_2) = y$ for two distinct inputs x_1 and x_2, then there is no way to assign a value to $f^{-1}(y)$ that satisfies both $f^{-1}(f(x_1)) = x_1$ and $f^{-1}(f(x_2)) = x_2$.

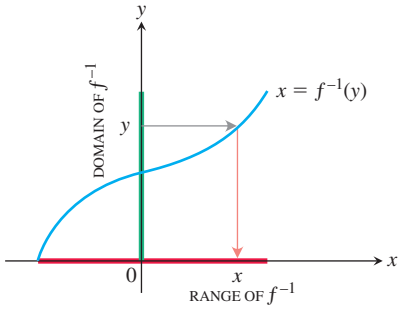
A function that is increasing on an interval satisfies the inequality $f(x_2) > f(x_1)$ when $x_2 > x_1$, so it is one-to-one and has an inverse. A function that is decreasing on an interval also has an inverse. Functions that are neither increasing nor decreasing may still be one-to-one and have an inverse. An example is the function $f(x) = 1/x$ for $x \neq 0$ and $f(0) = 0$, defined on $(-\infty, \infty)$ and passing the horizontal line test.

Finding Inverses

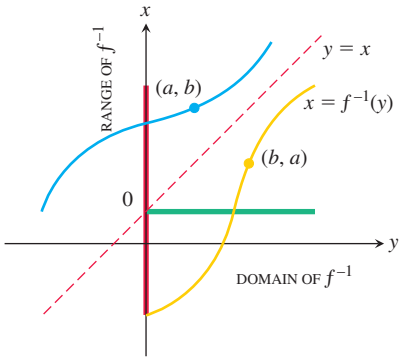
The graphs of a function and its inverse are closely related. To read the value of a function from its graph, we start at a point x on the x-axis, go vertically to the graph, and then move horizontally to the y-axis to read the value of y. The inverse function can be read from the graph by reversing this process. Start with a point y on the y-axis, go horizontally to the graph of $y = f(x)$, and then move vertically to the x-axis to read the value of $x = f^{-1}(y)$ (Figure 7.2).



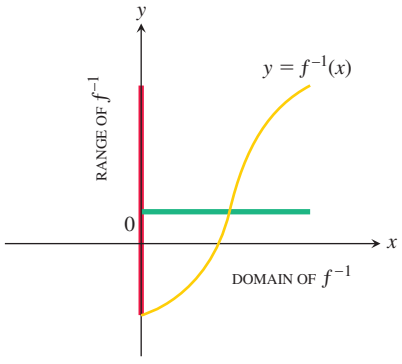
(a) To find the value of f at x, we start at x, go up to the curve, and then over to the y-axis.



(b) The graph of f^{-1} is the graph of f, but with x and y interchanged. To find the x that gave y, we start at y and go over to the curve and down to the x-axis. The domain of f^{-1} is the range of f. The range of f^{-1} is the domain of f.



(c) To draw the graph of f^{-1} in the more usual way, we reflect the system across the line $y = x$.



(d) Then we interchange the letters x and y. We now have a normal-looking graph of f^{-1} as a function of x.

FIGURE 7.2 The graph of $y = f^{-1}(x)$ is obtained by reflecting the graph of $y = f(x)$ about the line $y = x$.

We want to set up the graph of f^{-1} so that its input values lie along the x-axis, as is usually done for functions, rather than on the y-axis. To achieve this we interchange the x- and y-axes by reflecting across the 45° line $y = x$. After this reflection we have a new graph that represents f^{-1}. The value of $f^{-1}(x)$ can now be read from the graph in the usual way, by starting with a point x on the x-axis, going vertically to the graph, and then horizontally to the y-axis to get the value of $f^{-1}(x)$. Figure 7.2 indicates the relationship between the graphs of f and f^{-1}. The graphs are interchanged by reflection through the line $y = x$.

The process of passing from f to f^{-1} can be summarized as a two-step procedure.

1. Solve the equation $y = f(x)$ for x. This gives a formula $x = f^{-1}(y)$ where x is expressed as a function of y.
2. Interchange x and y, obtaining a formula $y = f^{-1}(x)$ where f^{-1} is expressed in the conventional format with x as the independent variable and y as the dependent variable.

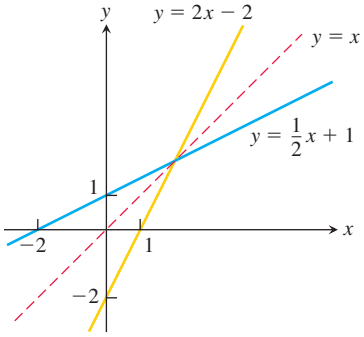


FIGURE 7.3 Graphing the functions $f(x) = (1/2)x + 1$ and $f^{-1}(x) = 2x - 2$ together shows the graphs' symmetry with respect to the line $y = x$ (Example 3).

EXAMPLE 3 Find the inverse of $y = \frac{1}{2}x + 1$, expressed as a function of x.

Solution

1. *Solve for x in terms of y*: $y = \frac{1}{2}x + 1$ — The graph is a straight line satisfying the horizontal line test (Figure 7.3).

$$2y = x + 2$$
$$x = 2y - 2.$$

2. *Interchange x and y*: $y = 2x - 2$. — Expresses the function in the usual form where y is the dependent variable.

The inverse of the function $f(x) = (1/2)x + 1$ is the function $f^{-1}(x) = 2x - 2$. (See Figure 7.3.) To check, we verify that both compositions give the identity function:

$$f^{-1}(f(x)) = 2\left(\frac{1}{2}x + 1\right) - 2 = x + 2 - 2 = x$$

$$f(f^{-1}(x)) = \frac{1}{2}(2x - 2) + 1 = x - 1 + 1 = x.$$ ■

EXAMPLE 4 Find the inverse of the function $y = x^2, x \geq 0$, expressed as a function of x.

Solution For $x \geq 0$, the graph satisfies the horizontal line test, so the function is one-to-one and has an inverse. To find the inverse, we first solve for x in terms of y:

$$y = x^2$$
$$\sqrt{y} = \sqrt{x^2} = |x| = x \qquad |x| = x \text{ because } x \geq 0$$

We then interchange x and y, obtaining

$$y = \sqrt{x}.$$

The inverse of the function $y = x^2, x \geq 0$, is the function $y = \sqrt{x}$ (Figure 7.4). ■

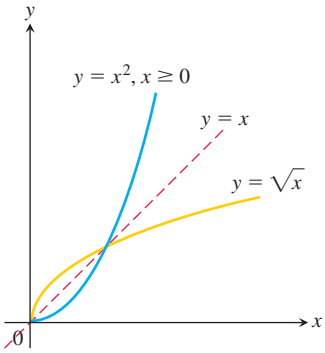


FIGURE 7.4 The functions $y = \sqrt{x}$ and $y = x^2, x \geq 0$, are inverses of one another (Example 4).

Notice that the function $y = x^2, x \geq 0$, with domain *restricted* to the nonnegative real numbers, *is* one-to-one (Figure 7.4) and has an inverse. On the other hand, the function $y = x^2$, with no domain restrictions, *is not* one-to-one (Figure 7.1b) and therefore has no inverse.

Derivatives of Inverses of Differentiable Functions

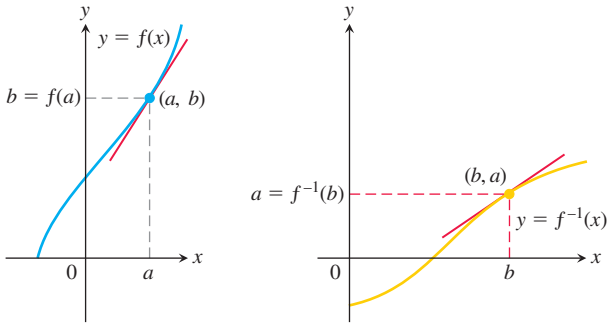
We calculated the inverse of the function $f(x) = (1/2)x + 1$ to be $f^{-1}(x) = 2x - 2$ in Example 3. If we calculate their derivatives, we see that

$$\frac{d}{dx}f(x) = \frac{d}{dx}\left(\frac{1}{2}x + 1\right) = \frac{1}{2}$$

$$\frac{d}{dx}f^{-1}(x) = \frac{d}{dx}(2x - 2) = 2.$$

The derivatives are reciprocals of one another, so the slope of one line is the reciprocal of the slope of its inverse line. (See Figure 7.3.)

This is not a special case. Reflecting any nonhorizontal or nonvertical line across the line $y = x$ always inverts the line's slope. If the original line has slope $m \neq 0$, the reflected line has slope $1/m$.



The slopes are reciprocal: $(f^{-1})'(b) = \dfrac{1}{f'(a)}$ or $(f^{-1})'(b) = \dfrac{1}{f'(f^{-1}(b))}$

FIGURE 7.5 The graphs of inverse functions have reciprocal slopes at corresponding points.

The reciprocal relationship between the slopes of f and f^{-1} holds for other functions as well, but we must be careful to compare slopes at corresponding points. If the slope of $y = f(x)$ at the point $(a, f(a))$ is $f'(a)$ and $f'(a) \neq 0$, then the slope of $y = f^{-1}(x)$ at the point $(f(a), a)$ is the reciprocal $1/f'(a)$ (Figure 7.5). If we set $b = f(a)$, then

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}.$$

If $y = f(x)$ has a horizontal tangent line at $(a, f(a))$, then the inverse function f^{-1} has a vertical tangent line at $(f(a), a)$, and this infinite slope implies that f^{-1} is not differentiable at $f(a)$. Theorem 1 gives the conditions under which f^{-1} is differentiable in its domain (which is the same as the range of f).

THEOREM 1—The Derivative Rule for Inverses

If f has an interval I as domain and $f'(x)$ exists and is never zero on I, then f^{-1} is differentiable at every point in its domain (the range of f). The value of $(f^{-1})'$ at a point b in the domain of f^{-1} is the reciprocal of the value of f' at the point $a = f^{-1}(b)$:

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))} \tag{1}$$

or

$$\left.\frac{df^{-1}}{dx}\right|_{x=b} = \frac{1}{\left.\dfrac{df}{dx}\right|_{x=f^{-1}(b)}}.$$

Theorem 1 makes two assertions. The first of these has to do with the conditions under which f^{-1} is differentiable; the second assertion is a formula for the derivative of f^{-1} when it exists. While we omit the proof of the first assertion, the second one is proved in the following way:

$$f(f^{-1}(x)) = x \qquad \text{Inverse function relationship}$$

$$\frac{d}{dx} f(f^{-1}(x)) = 1 \qquad \text{Differentiating both sides}$$

$$f'(f^{-1}(x)) \cdot \frac{d}{dx} f^{-1}(x) = 1 \qquad \text{Chain Rule}$$

$$\frac{d}{dx} f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}. \qquad \text{Solving for the derivative}$$

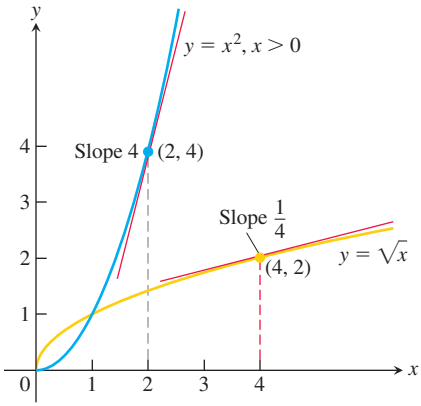


FIGURE 7.6 The derivative of $f^{-1}(x) = \sqrt{x}$ at the point (4, 2) is the reciprocal of the derivative of $f(x) = x^2$ at (2, 4) (Example 5).

EXAMPLE 5 The function $f(x) = x^2, x > 0$ and its inverse $f^{-1}(x) = \sqrt{x}$ have derivatives $f'(x) = 2x$ and $(f^{-1})'(x) = 1/(2\sqrt{x})$.

Let's verify that Theorem 1 gives the same formula for the derivative of $f^{-1}(x)$:

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

$$= \frac{1}{2(f^{-1}(x))} \qquad f'(x) = 2x \text{ with } x \text{ replaced by } f^{-1}(x)$$

$$= \frac{1}{2(\sqrt{x})}. \qquad f^{-1}(x) = \sqrt{x}$$

Theorem 1 gives a derivative that agrees with the known derivative of the square root function.

Let's examine Theorem 1 at a specific point. We pick $x = 2$ (the number a) and $f(2) = 4$ (the value b). Theorem 1 says that the derivative of f at 2, which is $f'(2) = 4$, and the derivative of f^{-1} at $f(2)$, which is $(f^{-1})'(4)$, are reciprocals. Thus

$$(f^{-1})'(4) = \frac{1}{f'(f^{-1}(4))} = \frac{1}{f'(2)} = \frac{1}{2x}\bigg|_{x=2} = \frac{1}{4}.$$

See Figure 7.6. ■

We will use the procedure illustrated in Example 5 to calculate formulas for the derivatives of many inverse functions throughout this chapter. Equation (1) sometimes enables us to find specific values of df^{-1}/dx without knowing a formula for f^{-1}.

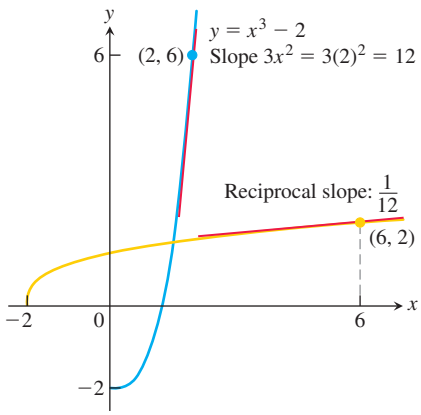


FIGURE 7.7 The derivative of $f(x) = x^3 - 2$ at $x = 2$ tells us the derivative of f^{-1} at $x = 6$ (Example 6).

EXAMPLE 6 Let $f(x) = x^3 - 2, x > 0$. Find the value of df^{-1}/dx at $x = 6 = f(2)$ without finding a formula for $f^{-1}(x)$. See Figure 7.7.

Solution We apply Theorem 1 to obtain the value of the derivative of f^{-1} at $x = 6$:

$$\frac{df}{dx}\bigg|_{x=2} = 3x^2\bigg|_{x=2} = 12$$

$$\frac{df^{-1}}{dx}\bigg|_{x=f(2)} = \frac{1}{\left.\dfrac{df}{dx}\right|_{x=2}} = \frac{1}{12}. \qquad \text{Eq. (1)}$$

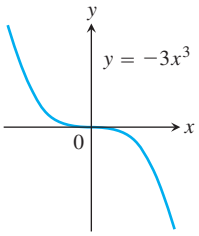
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EXERCISES 7.1

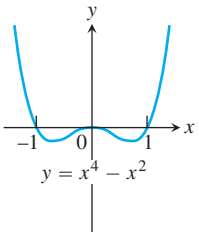
Identifying One-to-One Functions Graphically

Which of the functions graphed in Exercises 1–6 are one-to-one, and which are not?

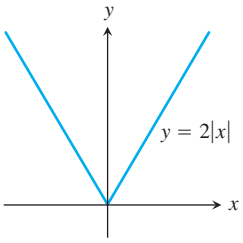
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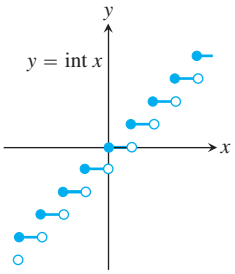
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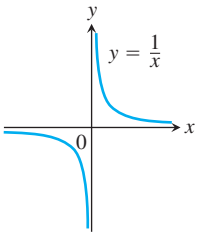
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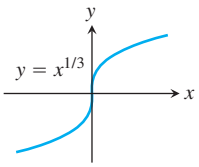
4.



5.



6.



In Exercises 7–10, determine from its graph if the function is one-to-one.

7. $f(x) = \begin{cases} 3 - x, & x < 0 \\ 3, & x \geq 0 \end{cases}$

8. $f(x) = \begin{cases} 2x + 6, & x \leq -3 \\ x + 4, & x > -3 \end{cases}$

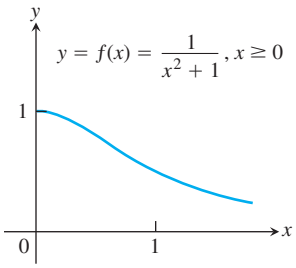
9. $f(x) = \begin{cases} 1 - \dfrac{x}{2}, & x \leq 0 \\ \dfrac{x}{x + 2}, & x > 0 \end{cases}$

10. $f(x) = \begin{cases} 2 - x^2, & x \leq 1 \\ x^2, & x > 1 \end{cases}$

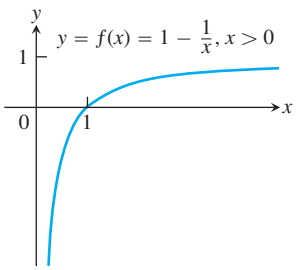
Graphing Inverse Functions

Each of Exercises 11–16 shows the graph of a function $y = f(x)$. Copy the graph and draw in the line $y = x$. Then use symmetry with respect to the line $y = x$ to add the graph of f^{-1} to your sketch. (It is not necessary to find a formula for f^{-1}.) Identify the domain and range of f^{-1}.

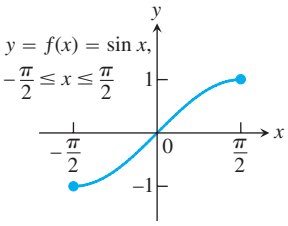
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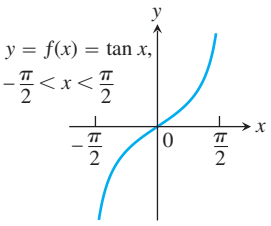
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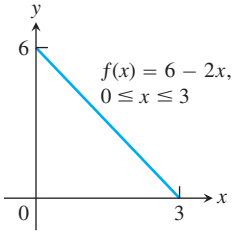
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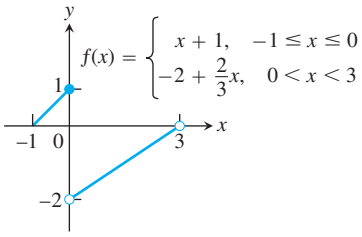
14.



15.



16.



17. a. Graph the function $f(x) = \sqrt{1 - x^2}, 0 \leq x \leq 1$. What symmetry does the graph have?

b. Show that f is its own inverse. (Remember that $\sqrt{x^2} = x$ if $x \geq 0$.)

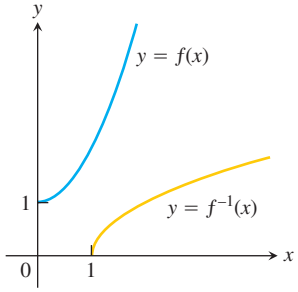
18. a. Graph the function $f(x) = 1/x$. What symmetry does the graph have?

b. Show that f is its own inverse.

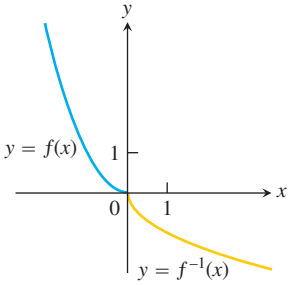
Formulas for Inverse Functions

Each of Exercises 19–24 gives a formula for a function $y = f(x)$ and shows the graphs of f and f^{-1}. Find a formula for f^{-1} in each case.

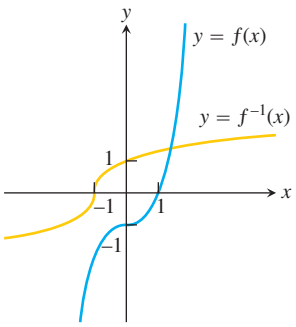
19. $f(x) = x^2 + 1, \quad x \geq 0$



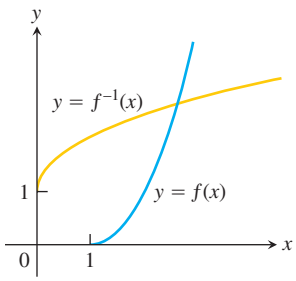
20. $f(x) = x^2, \quad x \leq 0$



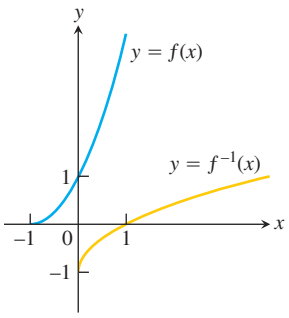
21. $f(x) = x^3 - 1$



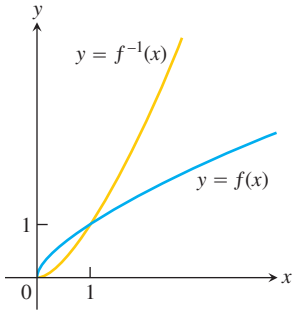
22. $f(x) = x^2 - 2x + 1, \quad x \geq 1$



23. $f(x) = (x + 1)^2, \quad x \geq -1$



24. $f(x) = x^{2/3}, \quad x \geq 0$



Derivatives of Inverse Functions

Each of Exercises 25–34 gives a formula for a function $y = f(x)$. In each case, find $f^{-1}(x)$ and identify the domain and range of f^{-1}. As a check, show that $f(f^{-1}(x)) = f^{-1}(f(x)) = x$.

25. $f(x) = x^5$

26. $f(x) = x^4, \quad x \geq 0$

27. $f(x) = x^3 + 1$

28. $f(x) = (1/2)x - 7/2$

29. $f(x) = 1/x^2, \quad x > 0$

30. $f(x) = 1/x^3, \quad x \neq 0$

31. $f(x) = \dfrac{x + 3}{x - 2}$

32. $f(x) = \dfrac{\sqrt{x}}{\sqrt{x} - 3}$

33. $f(x) = x^2 - 2x, \quad x \leq 1$ (*Hint:* Complete the square.)

34. $f(x) = (2x^3 + 1)^{1/5}$

In Exercises 35–38:

a. Find $f^{-1}(x)$.

b. Graph f and f^{-1} together.

c. Evaluate df/dx at $x = a$ and df^{-1}/dx at $x = f(a)$ to show that at these points $df^{-1}/dx = 1/(df/dx)$.

35. $f(x) = 2x + 3, \quad a = -1$

36. $f(x) = \dfrac{x + 2}{1 - x}, \quad a = 1/2$

37. $f(x) = 5 - 4x, \quad a = 1/2$

38. $f(x) = 2x^2, \quad x \geq 0, \quad a = 5$

39. a. Show that $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ are inverses of one another.

b. Graph f and g over an x-interval large enough to show the graphs intersecting at (1, 1) and $(-1, -1)$. Be sure the picture shows the required symmetry about the line $y = x$.

c. Find the slopes of the tangent lines to the graphs of f and g at (1, 1) and $(-1, -1)$ (four tangent lines in all).

d. What lines are tangent to the curves at the origin?

40. a. Show that $h(x) = x^3/4$ and $k(x) = (4x)^{1/3}$ are inverses of one another.

b. Graph h and k over an x-interval large enough to show the graphs intersecting at (2, 2) and $(-2, -2)$. Be sure the picture shows the required symmetry about the line $y = x$.

c. Find the slopes of the tangent lines to the graphs of h and k at (2, 2) and $(-2, -2)$.

d. What lines are tangent to the curves at the origin?

41. Let $f(x) = x^3 - 3x^2 - 1, x \geq 2$. Find the value of df^{-1}/dx at the point $x = -1 = f(3)$.

42. Let $f(x) = x^2 - 4x - 5, x > 2$. Find the value of df^{-1}/dx at the point $x = 0 = f(5)$.

43. Suppose that the differentiable function $y = f(x)$ has an inverse and that the graph of f passes through the point (2, 4) and has a slope of $1/3$ there. Find the value of df^{-1}/dx at $x = 4$.

44. Suppose that the differentiable function $y = g(x)$ has an inverse and that the graph of g passes through the origin with slope 2. Find the slope of the graph of g^{-1} at the origin.

Inverses of Lines

45. a. Find the inverse of the function $f(x) = mx$, where m is a constant different from zero.

b. What can you conclude about the inverse of a function $y = f(x)$ whose graph is a line through the origin with a nonzero slope m?

46. Show that the graph of the inverse of $f(x) = mx + b$, where m and b are constants and $m \neq 0$, is a line with slope $1/m$ and y-intercept $-b/m$.

47. a. Find the inverse of $f(x) = x + 1$. Graph f and its inverse together. Add the line $y = x$ to your sketch, drawing it with dashes or dots for contrast.

b. Find the inverse of $f(x) = x + b$ (b constant). How is the graph of f^{-1} related to the graph of f?

c. What can you conclude about the inverses of functions whose graphs are lines parallel to the line $y = x$?

48. a. Find the inverse of $f(x) = -x + 1$. Graph the line $y = -x + 1$ together with the line $y = x$. At what angle do the lines intersect?

b. Find the inverse of $f(x) = -x + b$ (b constant). What angle does the line $y = -x + b$ make with the line $y = x$?

c. What can you conclude about the inverses of functions whose graphs are lines perpendicular to the line $y = x$?

Increasing and Decreasing Functions

49. Show that increasing functions and decreasing functions are one-to-one. That is, show that for any x_1 and x_2 in I, $x_2 \neq x_1$ implies $f(x_2) \neq f(x_1)$.

Use the results of Exercise 49 to show that the functions in Exercises 50–54 have inverses over their domains. Find a formula for df^{-1}/dx using Theorem 1.

50. $f(x) = (1/3)x + (5/6)$

51. $f(x) = 27x^3$

52. $f(x) = 1 - 8x^3$

53. $f(x) = (1 - x)^3$

54. $f(x) = x^{5/3}$

Theory and Applications

55. If $f(x)$ is one-to-one, can anything be said about $g(x) = -f(x)$? Is it also one-to-one? Give reasons for your answer.

56. If $f(x)$ is one-to-one and $f(x)$ is never zero, can anything be said about $h(x) = 1/f(x)$? Is it also one-to-one? Give reasons for your answer.

57. Suppose that the range of g lies in the domain of f so that the composition $f \circ g$ is defined. If f and g are one-to-one, can anything be said about $f \circ g$? Give reasons for your answer.

58. If a composition $f \circ g$ is one-to-one, must g be one-to-one? Give reasons for your answer.

59. Assume that f and g are differentiable functions that are inverses of one another so that $(g \circ f)(x) = x$. Differentiate both sides of this equation with respect to x using the Chain Rule to express $(g \circ f)'(x)$ as a product of derivatives of g and f. What do you find? (This is not a proof of Theorem 1 because we assume here the theorem's conclusion that $g = f^{-1}$ is differentiable.)

60. Equivalence of the washer and shell methods for finding volume Let f be differentiable and increasing on the interval $a \le x \le b$, with $a > 0$, and suppose that f has a differentiable inverse, f^{-1}. Revolve about the y-axis the region bounded by the graph of f and the lines $x = a$ and $y = f(b)$ to generate a solid. Then the values of the integrals given by the washer and shell methods for the volume have identical values:

$$\int_{f(a)}^{f(b)} \pi((f^{-1}(y))^2 - a^2)\, dy = \int_a^b 2\pi x(f(b) - f(x))\, dx.$$

To prove this equality, define

$$W(t) = \int_{f(a)}^{f(t)} \pi((f^{-1}(y))^2 - a^2)\, dy$$

$$S(t) = \int_a^t 2\pi x(f(t) - f(x))\, dx.$$

Then show that the functions W and S agree at a point of $[a, b]$ and have identical derivatives on $[a, b]$. As you saw in Section 4.2, Corollary 2 of the Mean Value Theorem guarantees that $W(t) = S(t)$ for all t in $[a, b]$. In particular, $W(b) = S(b)$. (*Source:* "Disks and Shells Revisited," by Walter Carlip, *American Mathematical Monthly*, Vol. 98, No. 2, Feb. 1991, pp. 172–174.)

COMPUTER EXPLORATIONS

In Exercises 61–66, you will explore some functions and their inverses together with their derivatives and linear approximating functions at specified points. Perform the following steps using your CAS:

a. Plot the function $y = f(x)$ together with its derivative over the given interval. Explain why you know that f is one-to-one over the interval.

b. Solve the equation $y = f(x)$ for x as a function of y, and name the resulting inverse function g.

c. Find the equation for the tangent line to f at the specified point $(x_0, f(x_0))$.

d. Find the equation for the tangent line to g at the point $(f(x_0), x_0)$ located symmetrically across the 45° line $y = x$ (which is the graph of the identity function). Use Theorem 1 to find the slope of this tangent line.

e. Plot the functions f and g, the identity, the two tangent lines, and the line segment joining the points $(x_0, f(x_0))$ and $(f(x_0), x_0)$. Discuss the symmetries you see across the main diagonal.

61. $y = \sqrt{3x - 2}, \quad \frac{2}{3} \le x \le 4, \quad x_0 = 3$

62. $y = \dfrac{3x + 2}{2x - 11}, \quad -2 \le x \le 2, \quad x_0 = 1/2$

63. $y = \dfrac{4x}{x^2 + 1}, \quad -1 \le x \le 1, \quad x_0 = 1/2$

64. $y = \dfrac{x^3}{x^2 + 1}, \quad -1 \le x \le 1, \quad x_0 = 1/2$

65. $y = x^3 - 3x^2 - 1, \quad 2 \le x \le 5, \quad x_0 = \dfrac{27}{10}$

66. $y = 2 - x - x^3, \quad -2 \le x \le 2, \quad x_0 = \dfrac{3}{2}$

In Exercises 67 and 68, repeat the steps above to solve for the functions $y = f(x)$ and $x = f^{-1}(y)$ defined implicitly by the given equations over the interval.

67. $y^{1/3} - 1 = (x + 2)^3, \quad -5 \le x \le 5, \quad x_0 = -3/2$

68. $\cos y = x^{1/5}, \quad 0 \le x \le 1, \quad x_0 = 1/2$

7.2 Natural Logarithms

Historically, logarithms played important roles in arithmetic computations, making possible the great seventeenth-century advances in offshore navigation and celestial mechanics. In this section we define the natural logarithm as an integral using the Fundamental Theorem of Calculus. While this indirect approach may at first seem strange, it provides an elegant and rigorous way to obtain the key characteristics of logarithmic and exponential functions.

Definition of the Natural Logarithm Function

The natural logarithm of a positive number x, written as $\ln x$, is defined as an integral.

DEFINITION The **natural logarithm** is the function given by

$$\ln x = \int_1^x \frac{1}{t}\,dt, \qquad x > 0.$$

From the Fundamental Theorem of Calculus, $\ln x$ is a continuous function. Geometrically, if $x > 1$, then $\ln x$ is the area under the curve $y = 1/t$ from $t = 1$ to $t = x$ (Figure 7.8). For $0 < x < 1$, $\ln x$ gives the negative of the area under the curve from x to 1, and the function is not defined for $x \le 0$. From the Zero Width Interval Rule for definite integrals, we also have

$$\ln 1 = \int_1^1 \frac{1}{t}\,dt = 0.$$

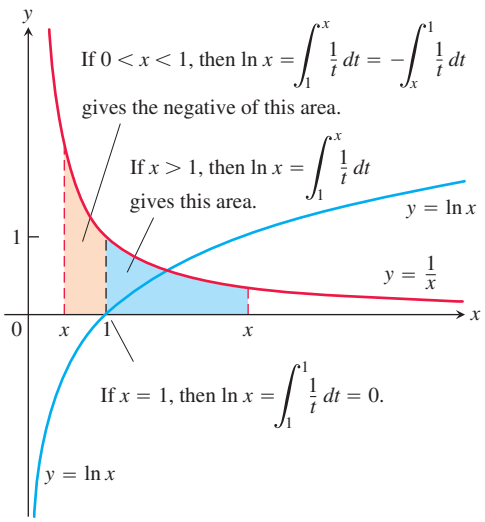


FIGURE 7.8 The graph of $y = \ln x$ and its relation to the function $y = 1/x, x > 0$. The graph of the logarithm rises above the x-axis as x moves from 1 to the right, and it falls below the axis as x moves from 1 to the left.

TABLE 7.1 Typical 2-place values of $\ln x$

x	$\ln x$
0	undefined
0.05	−3.00
0.5	−0.69
1	0
2	0.69
3	1.10
4	1.39
10	2.30

Notice that we show the graph of $y = 1/x$ in Figure 7.8 but use $y = 1/t$ in the integral. Using x for everything would have us writing

$$\ln x = \int_1^x \frac{1}{x}\,dx,$$

with x meaning two different things. So we change the variable of integration to t.

By using rectangles to obtain finite approximations of the area under the graph of $y = 1/t$ and over the interval between $t = 1$ and $t = x$, as in Section 5.1, we can approximate the values of the function $\ln x$. Several values are given in Table 7.1. There is an important number between $x = 2$ and $x = 3$ whose natural logarithm equals 1. This number, which we now define, exists because $\ln x$ is a continuous function and therefore satisfies the Intermediate Value Theorem on $[2, 3]$.

DEFINITION The **number e** is the number in the domain of the natural logarithm that satisfies

$$\ln(e) = \int_1^e \frac{1}{t}\,dt = 1.$$

Interpreted geometrically, the number e corresponds to the point on the x-axis for which the area under the graph of $y = 1/t$ and above the interval $[1, e]$ equals the area of the unit square. That is, the area of the region shaded blue in Figure 7.1 is 1 square unit when $x = e$. In the next section, we will see that the number e can be calculated as a limit and has the numerical value $e \approx 2.718281828459045$ to 15 decimal places.

The Derivative of $y = \ln x$

By the first part of the Fundamental Theorem of Calculus (Section 5.4),

$$\frac{d}{dx}\ln x = \frac{d}{dx}\int_1^x \frac{1}{t}\,dt = \frac{1}{x}.$$

For every positive value of x, we have

$$\frac{d}{dx}\ln x = \frac{1}{x}. \tag{1}$$

Therefore, the function $y = \ln x$ is a solution to the initial value problem $dy/dx = 1/x$, $x > 0$, with $y(1) = 0$. Notice that the derivative is always positive.

If u is a differentiable function of x whose values are positive, so that $\ln u$ is defined, then applying the Chain Rule we obtain

$$\frac{d}{dx}\ln u = \frac{1}{u}\frac{du}{dx}, \qquad u > 0. \tag{2}$$

EXAMPLE 1 We use Equation (2) to find derivatives.

(a) $\dfrac{d}{dx}\ln 2x = \dfrac{1}{2x}\dfrac{d}{dx}(2x) = \dfrac{1}{2x}(2) = \dfrac{1}{x}, \quad x > 0$

(b) Equation (2) with $u = x^2 + 3$ gives

$$\frac{d}{dx}\ln(x^2+3) = \frac{1}{x^2+3}\cdot\frac{d}{dx}(x^2+3) = \frac{1}{x^2+3}\cdot 2x = \frac{2x}{x^2+3}.$$

(c) Equation (2) with $u = |x|$ gives an important derivative:

$$\begin{aligned}
\frac{d}{dx}\ln|x| &= \frac{d}{du}\ln u \cdot \frac{du}{dx} && u = |x|,\ x \neq 0 \\
&= \frac{1}{u}\cdot\frac{x}{|x|} && \frac{d}{dx}(|x|) = \frac{x}{|x|} \\
&= \frac{1}{|x|}\cdot\frac{x}{|x|} && \text{Substitute for } u. \\
&= \frac{x}{x^2} \\
&= \frac{1}{x}.
\end{aligned}$$

Derivative of $\ln|x|$

$\dfrac{d}{dx}\ln|x| = \dfrac{1}{x}, \ x \neq 0$

So $1/x$ is the derivative of $\ln x$ on the domain $x > 0$, and the derivative of $\ln(-x)$ on the domain $x < 0$. ■

Notice from Example 1a that the function $y = \ln 2x$ has the same derivative as the function $y = \ln x$. This is true of $y = \ln bx$ for any constant b, provided that $bx > 0$:

$$\frac{d}{dx}\ln bx = \frac{1}{bx}\cdot\frac{d}{dx}(bx) = \frac{1}{bx}(b) = \frac{1}{x}.$$

$\dfrac{d}{dx}\ln bx = \dfrac{1}{x}, \quad bx > 0$

Properties of Logarithms

HISTORICAL BIOGRAPHY
John Napier
(1550–1617)
www.goo.gl/BvG0ua

Logarithms, invented by John Napier, were the single most important improvement in arithmetic calculation before the modern electronic computer. The properties of logarithms reduce multiplication of positive numbers to addition of their logarithms, division of positive numbers to subtraction of their logarithms, and exponentiation of a number to multiplication of its logarithm by the exponent.

THEOREM 2—Algebraic Properties of the Natural Logarithm
For any numbers $b > 0$ and $x > 0$, the natural logarithm satisfies the following rules:

1. *Product Rule:*	$\ln bx = \ln b + \ln x$	
2. *Quotient Rule:*	$\ln \frac{b}{x} = \ln b - \ln x$	
3. *Reciprocal Rule:*	$\ln \frac{1}{x} = -\ln x$	Rule 2 with $b = 1$
4. *Power Rule:*	$\ln x^r = r \ln x$	For r rational

For now we consider only rational exponents in Rule 4. In Section 7.3 we will see that the rule holds for all real exponents as well.

EXAMPLE 2 We apply the rules in Theorem 2.

(a) $\ln 4 + \ln \sin x = \ln (4 \sin x)$ Product Rule

(b) $\ln \frac{x+1}{2x-3} = \ln (x+1) - \ln (2x-3)$ Quotient Rule

(c) $\ln \frac{1}{8} = -\ln 8$ Reciprocal Rule

$= -\ln 2^3 = -3 \ln 2$ Power Rule ■

We now give the proof of Theorem 2. The properties are proved by applying Corollary 2 of the Mean Value Theorem to each rule.

Proof that $\ln bx = \ln b + \ln x$ The argument starts by observing that $\ln bx$ and $\ln x$ have the same derivative:

$$\frac{d}{dx}\ln (bx) = \frac{b}{bx} = \frac{1}{x} = \frac{d}{dx}\ln x.$$

According to Corollary 2 of the Mean Value Theorem, the functions must differ by a constant, which means that

$$\ln bx = \ln x + C$$

for some constant C.

Since this last equation holds for all positive values of x, it must hold for $x = 1$. Hence,

$$\ln (b \cdot 1) = \ln 1 + C$$
$$\ln b = 0 + C \qquad \ln 1 = 0$$
$$C = \ln b.$$

By substituting we conclude that

$$\ln bx = \ln b + \ln x.$$ ■

Proof that $\ln x^r = r \ln x$ (assuming r rational) We use the same-derivative argument again. For all positive values of x,

$$\frac{d}{dx}\ln x^r = \frac{1}{x^r}\frac{d}{dx}(x^r) \qquad \text{Eq. (2) with } u = x^r$$

$$= \frac{1}{x^r} r x^{r-1} \qquad \text{General Power Rule for derivatives, } r \text{ rational}$$

$$= r \cdot \frac{1}{x} = \frac{d}{dx}(r \ln x).$$

Since $\ln x^r$ and $r \ln x$ have the same derivative,

$$\ln x^r = r \ln x + C$$

for some constant C. Taking x to be 1 identifies C as zero, and we're done. (Exercise 48 in Section 3.7 indicates a proof of the General Power Rule for derivatives when r is rational.)

You are asked to prove Rule 2 in Exercise 90. Rule 3 is a special case of Rule 2, obtained by setting $b = 1$ and noting that $\ln 1 = 0$. This covers all cases of Theorem 2. ■

We have not yet proved Rule 4 for r irrational; however, the rule does hold for all r, rational or irrational. We will show this in the next section after we define exponential functions and irrational exponents.

The Graph and Range of $\ln x$

The derivative $d(\ln x)/dx = 1/x$ is positive for $x > 0$, so $\ln x$ is an increasing function of x. The second derivative, $-1/x^2$, is negative, so the graph of $\ln x$ is concave down. (See Figure 7.9a.)

We can estimate the value of $\ln 2$ by considering the area under the graph of $y = 1/x$ and above the interval $[1, 2]$. In Figure 7.9(b) a rectangle of height $1/2$ over the interval $[1, 2]$ fits under the graph. Therefore the area under the graph, which is $\ln 2$, is greater than the area, $1/2$, of the rectangle. So $\ln 2 > 1/2$. Knowing this we have

$$\ln 2^n = n \ln 2 > n\left(\frac{1}{2}\right) = \frac{n}{2}.$$

This result shows that $\ln(2^n) \to \infty$ as $n \to \infty$. Since $\ln x$ is an increasing function, we get that

$$\lim_{x \to \infty} \ln x = \infty. \qquad \ln x \text{ increasing, not bounded above}$$

We also have

$$\lim_{x \to 0^+} \ln x = \lim_{t \to \infty} \ln t^{-1} = \lim_{t \to \infty} (-\ln t) = -\infty. \qquad x = 1/t = t^{-1}$$

We defined $\ln x$ for $x > 0$, so the domain of $\ln x$ is the set of positive real numbers. The above discussion and the Intermediate Value Theorem show that its range is the entire real line, giving the graph of $y = \ln x$ shown in Figure 7.9(a).

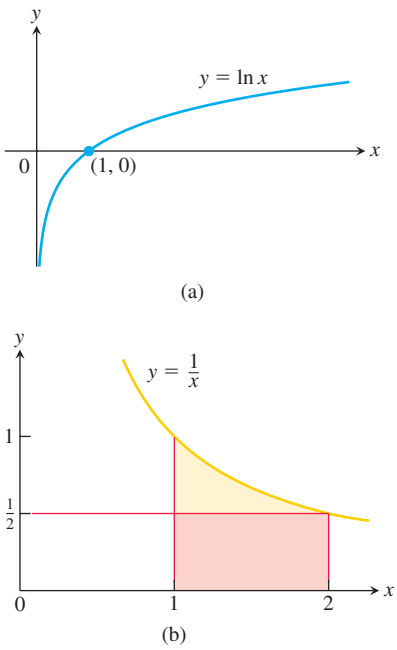


FIGURE 7.9 (a) The graph of the natural logarithm. (b) The rectangle of height $y = 1/2$ fits beneath the graph of $y = 1/x$ for the interval $1 \le x \le 2$.

The Integral $\int (1/u)\, du$

Example 2c leads to the following integral formula.

If u is a differentiable function that is never zero, then

$$\int \frac{1}{u}\, du = \ln|u| + C. \tag{3}$$

Equation (3) applies anywhere on the domain of $1/u$, which is the set of points where $u \neq 0$. It says that integrals that have the form $\int du/u$ lead to logarithms. Whenever $u = f(x)$ is a differentiable function that is never zero, we have that $du = f'(x)\,dx$ and

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C.$$

EXAMPLE 3 We rewrite an integral in the form $\int \frac{du}{u}$.

$$\int_{-\pi/2}^{\pi/2} \frac{4\cos\theta}{3 + 2\sin\theta}\,d\theta = \int_1^5 \frac{2}{u}\,du \qquad \begin{array}{l} u = 3 + 2\sin\theta, \quad du = 2\cos\theta\,d\theta, \\ u(-\pi/2) = 1, \quad u(\pi/2) = 5 \end{array}$$

$$= 2\ln|u|\Big]_1^5$$

$$= 2\ln|5| - 2\ln|1| = 2\ln 5$$

Note that $u = 3 + 2\sin\theta$ is always positive on $[-\pi/2, \pi/2]$, so Equation (3) applies. ■

The Integrals of tan x, cot x, sec x, and csc x

Equation (3) tells us how to integrate these trigonometric functions.

$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = \int \frac{-du}{u} \qquad \begin{array}{l} u = \cos x > 0 \text{ on } (-\pi/2, \pi/2), \\ du = -\sin x\,dx \end{array}$$

$$= -\ln|u| + C = -\ln|\cos x| + C$$

$$= \ln\frac{1}{|\cos x|} + C = \ln|\sec x| + C. \qquad \text{Reciprocal Rule}$$

For the cotangent,

$$\int \cot x\,dx = \int \frac{\cos x\,dx}{\sin x} = \int \frac{du}{u} \qquad \begin{array}{l} u = \sin x, \\ du = \cos x\,dx \end{array}$$

$$= \ln|u| + C = \ln|\sin x| + C = -\ln|\csc x| + C.$$

To integrate sec x, we multiply and divide by $(\sec x + \tan x)$ as an algebraic form of 1.

$$\int \sec x\,dx = \int \sec x\,\frac{(\sec x + \tan x)}{(\sec x + \tan x)}\,dx = \int \frac{\sec^2 x + \sec x\tan x}{\sec x + \tan x}\,dx$$

$$= \int \frac{du}{u} = \ln|u| + C = \ln|\sec x + \tan x| + C \qquad \begin{array}{l} u = \sec x + \tan x, \\ du = (\sec x\tan x + \sec^2 x)dx \end{array}$$

For csc x, we multiply and divide by $(\csc x + \cot x)$.

$$\int \csc x\,dx = \int \csc x\,\frac{(\csc x + \cot x)}{(\csc x + \cot x)}\,dx = \int \frac{\csc^2 x + \csc x\cot x}{\csc x + \cot x}\,dx$$

$$= \int \frac{-du}{u} = -\ln|u| + C = -\ln|\csc x + \cot x| + C \qquad \begin{array}{l} u = \csc x + \cot x, \\ du = (-\csc x\cot x - \csc^2 x)\,dx \end{array}$$

In summary, we have the following results.

Integrals of the tangent, cotangent, secant, and cosecant functions

$$\int \tan u\,du = \ln|\sec u| + C \qquad \int \sec u\,du = \ln|\sec u + \tan u| + C$$

$$\int \cot u\,du = \ln|\sin u| + C \qquad \int \csc u\,du = -\ln|\csc u + \cot u| + C$$

EXAMPLE 4

$$\int_0^{\pi/6} \tan 2x\,dx = \int_0^{\pi/3} \tan u\,\frac{du}{2} = \frac{1}{2}\int_0^{\pi/3} \tan u\,du$$

Substitute $u = 2x$, $dx = du/2$, $u(0) = 0$, $u(\pi/6) = \pi/3$

$$= \frac{1}{2}\ln|\sec u|\Big]_0^{\pi/3} = \frac{1}{2}(\ln 2 - \ln 1) = \frac{1}{2}\ln 2$$

Logarithmic Differentiation

The derivatives of positive functions given by formulas that involve products, quotients, and powers can often be found more quickly if we take the natural logarithm of both sides before differentiating. This enables us to use the laws of logarithms to simplify the formulas before differentiating. The process, called **logarithmic differentiation**, is illustrated in the next example.

EXAMPLE 5 Find dy/dx if

$$y = \frac{(x^2 + 1)(x + 3)^{1/2}}{x - 1}, \qquad x > 1.$$

Solution We take the natural logarithm of both sides and simplify the result with the properties of logarithms:

$$\ln y = \ln\frac{(x^2 + 1)(x + 3)^{1/2}}{x - 1}$$

$$= \ln((x^2 + 1)(x + 3)^{1/2}) - \ln(x - 1) \qquad \text{Quotient Rule}$$

$$= \ln(x^2 + 1) + \ln(x + 3)^{1/2} - \ln(x - 1) \qquad \text{Product Rule}$$

$$= \ln(x^2 + 1) + \frac{1}{2}\ln(x + 3) - \ln(x - 1). \qquad \text{Power Rule}$$

We then take derivatives of both sides with respect to x, using Equation (2) on the left:

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x^2 + 1}\cdot 2x + \frac{1}{2}\cdot\frac{1}{x + 3} - \frac{1}{x - 1}.$$

Next we solve for dy/dx:

$$\frac{dy}{dx} = y\left(\frac{2x}{x^2 + 1} + \frac{1}{2x + 6} - \frac{1}{x - 1}\right).$$

Finally, we substitute for y from the original equation:

$$\frac{dy}{dx} = \frac{(x^2 + 1)(x + 3)^{1/2}}{x - 1}\left(\frac{2x}{x^2 + 1} + \frac{1}{2x + 6} - \frac{1}{x - 1}\right).$$

The computation in Example 5 would be much longer if we used the product, quotient, and power rules.

EXERCISES 7.2

Using the Algebraic Properties—Theorem 2

1. Express the following logarithms in terms of $\ln 2$ and $\ln 3$.

a. $\ln 0.75$ **b.** $\ln (4/9)$ **c.** $\ln (1/2)$

d. $\ln \sqrt[3]{9}$ **e.** $\ln 3\sqrt{2}$ **f.** $\ln \sqrt{13.5}$

2. Express the following logarithms in terms of $\ln 5$ and $\ln 7$.

a. $\ln (1/125)$ **b.** $\ln 9.8$ **c.** $\ln 7\sqrt{7}$

d. $\ln 1225$ **e.** $\ln 0.056$

f. $(\ln 35 + \ln (1/7))/(\ln 25)$

Use the properties of logarithms to simplify the expressions in Exercises 3 and 4.

3. a. $\ln \sin\theta - \ln\left(\dfrac{\sin\theta}{5}\right)$ **b.** $\ln (3x^2 - 9x) + \ln\left(\dfrac{1}{3x}\right)$

c. $\dfrac{1}{2}\ln (4t^4) - \ln 2$

4. a. $\ln \sec\theta + \ln \cos\theta$ **b.** $\ln (8x + 4) - 2\ln 2$

c. $3\ln\sqrt[3]{t^2 - 1} - \ln (t + 1)$

In Exercises 5 and 6, solve for t.

5. $\ln\left(\dfrac{t}{t-1}\right) = 2$ **6.** $\ln(t - 2) = \ln 8 - \ln t$

Finding Derivatives

In Exercises 7–38, find the derivative of y with respect to x, t, or θ, as appropriate.

7. $y = \ln 3x$ **8.** $y = \ln kx$, k constant

9. $y = \ln (t^2)$ **10.** $y = \ln (t^{3/2})$

11. $y = \ln \dfrac{3}{x}$ **12.** $y = \ln \dfrac{10}{x}$

13. $y = \ln (\theta + 1)$ **14.** $y = \ln (2\theta + 2)$

15. $y = \ln x^3$ **16.** $y = (\ln x)^3$

17. $y = t(\ln t)^2$ **18.** $y = t\sqrt{\ln t}$

19. $y = \dfrac{x^4}{4}\ln x - \dfrac{x^4}{16}$ **20.** $y = (x^2 \ln x)^4$

21. $y = \dfrac{\ln t}{t}$ **22.** $y = \dfrac{1 + \ln t}{t}$

23. $y = \dfrac{\ln x}{1 + \ln x}$ **24.** $y = \dfrac{x \ln x}{1 + \ln x}$

25. $y = \ln (\ln x)$ **26.** $y = \ln (\ln (\ln x))$

27. $y = \theta(\sin (\ln \theta) + \cos (\ln \theta))$

28. $y = \ln (\sec\theta + \tan\theta)$

29. $y = \ln \dfrac{1}{x\sqrt{x+1}}$ **30.** $y = \dfrac{1}{2}\ln \dfrac{1+x}{1-x}$

31. $y = \dfrac{1 + \ln t}{1 - \ln t}$ **32.** $y = \sqrt{\ln \sqrt{t}}$

33. $y = \ln (\sec (\ln \theta))$ **34.** $y = \ln\left(\dfrac{\sqrt{\sin\theta \cos\theta}}{1 + 2\ln\theta}\right)$

35. $y = \ln\left(\dfrac{(x^2+1)^5}{\sqrt{1-x}}\right)$ **36.** $y = \ln \sqrt{\dfrac{(x+1)^5}{(x+2)^{20}}}$

37. $y = \displaystyle\int_{x^2/2}^{x^2} \ln \sqrt{t}\, dt$ **38.** $y = \displaystyle\int_{\sqrt{x}}^{\sqrt[3]{x}} \ln t\, dt$

Evaluating Integrals

Evaluate the integrals in Exercises 39–56.

39. $\displaystyle\int_{-3}^{-2} \frac{dx}{x}$ **40.** $\displaystyle\int_{-1}^{0} \frac{3\,dx}{3x - 2}$

41. $\displaystyle\int \frac{2y\,dy}{y^2 - 25}$ **42.** $\displaystyle\int \frac{8r\,dr}{4r^2 - 5}$

43. $\displaystyle\int_0^{\pi} \frac{\sin t}{2 - \cos t}\, dt$ **44.** $\displaystyle\int_0^{\pi/3} \frac{4\sin\theta}{1 - 4\cos\theta}\, d\theta$

45. $\displaystyle\int_1^2 \frac{2\ln x}{x}\, dx$ **46.** $\displaystyle\int_2^4 \frac{dx}{x\ln x}$

47. $\displaystyle\int_2^4 \frac{dx}{x(\ln x)^2}$ **48.** $\displaystyle\int_2^{16} \frac{dx}{2x\sqrt{\ln x}}$

49. $\displaystyle\int \frac{3\sec^2 t}{6 + 3\tan t}\, dt$ **50.** $\displaystyle\int \frac{\sec y \tan y}{2 + \sec y}\, dy$

51. $\displaystyle\int_0^{\pi/2} \tan\frac{x}{2}\, dx$ **52.** $\displaystyle\int_{\pi/4}^{\pi/2} \cot t\, dt$

53. $\displaystyle\int_{\pi/2}^{\pi} 2\cot\frac{\theta}{3}\, d\theta$ **54.** $\displaystyle\int_0^{\pi/12} 6\tan 3x\, dx$

55. $\displaystyle\int \frac{dx}{2\sqrt{x} + 2x}$ **56.** $\displaystyle\int \frac{\sec x\, dx}{\sqrt{\ln (\sec x + \tan x)}}$

Logarithmic Differentiation

In Exercises 57–70, use logarithmic differentiation to find the derivative of y with respect to the given independent variable.

57. $y = \sqrt{x(x+1)}$ **58.** $y = \sqrt{(x^2+1)(x-1)^2}$

59. $y = \sqrt{\dfrac{t}{t+1}}$ **60.** $y = \sqrt{\dfrac{1}{t(t+1)}}$

61. $y = \sqrt{\theta + 3}\sin\theta$ **62.** $y = (\tan\theta)\sqrt{2\theta + 1}$

63. $y = t(t+1)(t+2)$ **64.** $y = \dfrac{1}{t(t+1)(t+2)}$

65. $y = \dfrac{\theta + 5}{\theta\cos\theta}$ **66.** $y = \dfrac{\theta\sin\theta}{\sqrt{\sec\theta}}$

67. $y = \dfrac{x\sqrt{x^2+1}}{(x+1)^{2/3}}$ **68.** $y = \sqrt{\dfrac{(x+1)^{10}}{(2x+1)^5}}$

69. $y = \sqrt[3]{\dfrac{x(x-2)}{x^2+1}}$ **70.** $y = \sqrt[3]{\dfrac{x(x+1)(x-2)}{(x^2+1)(2x+3)}}$

Theory and Applications

71. Locate and identify the absolute extreme values of

a. $\ln (\cos x)$ on $[-\pi/4, \pi/3]$,

b. $\cos (\ln x)$ on $[1/2, 2]$.

72. a. Prove that $f(x) = x - \ln x$ is increasing for $x > 1$.

b. Using part (a), show that $\ln x < x$ if $x > 1$.

73. Find the area between the curves $y = \ln x$ and $y = \ln 2x$ from $x = 1$ to $x = 5$.

74. Find the area between the curve $y = \tan x$ and the x-axis from $x = -\pi/4$ to $x = \pi/3$.

Identifying Extrema

In Exercises 75 and 76:

a. Find the open intervals on which the function is increasing and decreasing.

b. Identify the function's local and absolute extreme values, if any, saying where they occur.

75. $g(x) = x(\ln x)^2$ **76.** $g(x) = x^2 - 2x - 4\ln x$

77. The region in the first quadrant bounded by the coordinate axes, the line $y = 3$, and the curve $x = 2/\sqrt{y+1}$ is revolved about the y-axis to generate a solid. Find the volume of the solid.

78. The region between the curve $y = \sqrt{\cot x}$ and the x-axis from $x = \pi/6$ to $x = \pi/2$ is revolved about the x-axis to generate a solid. Find the volume of the solid.

79. The region between the curve $y = 1/x^2$ and the x-axis from $x = 1/2$ to $x = 2$ is revolved about the y-axis to generate a solid. Find the volume of the solid.

80. In Section 6.2, Exercise 6, we revolved about the y-axis the region between the curve $y = 9x/\sqrt{x^3+9}$ and the x-axis from $x = 0$ to $x = 3$ to generate a solid of volume 36π. What volume do you get if you revolve the region about the x-axis instead? (See Section 6.2, Exercise 6, for a graph.)

81. Find the lengths of the following curves.

a. $y = (x^2/8) - \ln x, \quad 4 \le x \le 8$

b. $x = (y/4)^2 - 2\ln(y/4), \quad 4 \le y \le 12$

82. Find a curve through the point $(1, 0)$ whose length from $x = 1$ to $x = 2$ is

$$L = \int_1^2 \sqrt{1 + \frac{1}{x^2}}\, dx.$$

T **83. a.** Find the centroid of the region between the curve $y = 1/x$ and the x-axis from $x = 1$ to $x = 2$. Give the coordinates to two decimal places.

b. Sketch the region and show the centroid in your sketch.

84. a. Find the center of mass of a thin plate of constant density covering the region between the curve $y = 1/\sqrt{x}$ and the x-axis from $x = 1$ to $x = 16$.

b. Find the center of mass if, instead of being constant, the density function is $\delta(x) = 4/\sqrt{x}$.

85. Use a derivative to show that $f(x) = \ln(x^3 - 1)$ is one-to-one.

86. Use a derivative to show that $g(x) = \sqrt{x^2 + \ln x}$ is one-to-one.

Solve the initial value problems in Exercises 87 and 88.

87. $\dfrac{dy}{dx} = 1 + \dfrac{1}{x}, \quad y(1) = 3$

88. $\dfrac{d^2y}{dx^2} = \sec^2 x, \quad y(0) = 0 \quad \text{and} \quad y'(0) = 1$

T **89. The linearization of $\ln(1 + x)$ at $x = 0$** Instead of approximating $\ln x$ near $x = 1$, we approximate $\ln(1 + x)$ near $x = 0$. We get a simpler formula this way.

a. Derive the linearization $\ln(1 + x) \approx x$ at $x = 0$.

b. Estimate to five decimal places the error involved in replacing $\ln(1 + x)$ by x on the interval $[0, 0.1]$.

c. Graph $\ln(1 + x)$ and x together for $0 \le x \le 0.5$. Use different colors, if available. At what points does the approximation of $\ln(1 + x)$ seem best? Least good? By reading coordinates from the graphs, find as good an upper bound for the error as your grapher will allow.

90. Use the same-derivative argument, as was done to prove Rules 1 and 4 of Theorem 2, to prove the Quotient Rule property of logarithms.

T **91. a.** Graph $y = \sin x$ and the curves $y = \ln(a + \sin x)$ for $a = 2$, 4, 8, 20, and 50 together for $0 \le x \le 23$.

b. Why do the curves flatten as a increases? (*Hint:* Find an a-dependent upper bound for $|y'|$.)

T **92.** Does the graph of $y = \sqrt{x} - \ln x, x > 0$, have an inflection point? Try to answer this question **(a)** by graphing, **(b)** by using calculus.

7.3 Exponential Functions

Having developed the theory of the function $\ln x$, we introduce its inverse, the exponential function $\exp x = e^x$. We study its properties and compute its derivative and integral. We prove the power rule for derivatives involving general real exponents. Finally, we introduce general exponential functions, a^x, and general logarithmic functions, $\log_a x$.

The Inverse of $\ln x$ and the Number e

The function $\ln x$, being an increasing function of x with domain $(0, \infty)$ and range $(-\infty, \infty)$, has an inverse $\ln^{-1} x$ with domain $(-\infty, \infty)$ and range $(0, \infty)$. The graph of $\ln^{-1} x$ is the graph of $\ln x$ reflected across the line $y = x$. As you can see in Figure 7.10,

$$\lim_{x\to\infty} \ln^{-1} x = \infty \qquad \text{and} \qquad \lim_{x\to-\infty} \ln^{-1} x = 0.$$

The function $\ln^{-1} x$ is usually denoted as $\exp x$. We now show that $\exp x$ is an exponential function with base e.

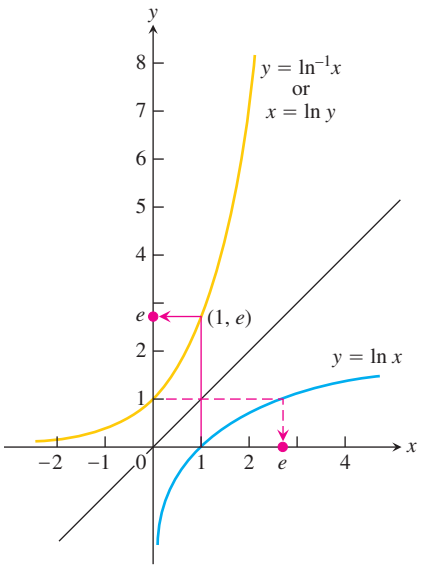


FIGURE 7.10 The graphs of $y = \ln x$ and $y = \ln^{-1} x = \exp x$. The number e is $\ln^{-1} 1 = \exp(1)$.

Typical values of e^x

x	e^x (rounded)
-1	0.37
0	1
1	2.72
2	7.39
10	22026
100	2.6881×10^{43}

The notations $\ln^{-1} x$, $\exp x$, and e^x all refer to the natural exponential function.

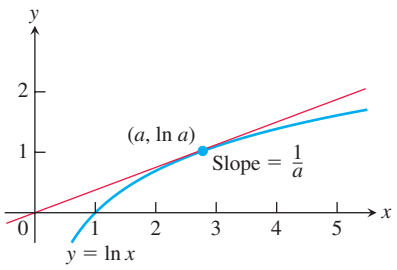


FIGURE 7.11 The tangent line intersects the curve at some point $(a, \ln a)$, where the slope of the curve is $1/a$ (Example 2).

The number e was defined to satisfy the equation $\ln(e) = 1$, so $e = \exp(1)$. We can raise the number e to a rational power r in the usual algebraic way:

$$e^2 = e \cdot e, \qquad e^{-2} = \frac{1}{e^2}, \qquad e^{1/2} = \sqrt{e},$$

and so on. Since e is positive, e^r is positive too, so we can take the logarithm of e^r. When we do, we find that for r rational

$$\ln e^r = r \ln e = r \cdot 1 = r. \qquad \text{Theorem 2, Rule 4}$$

Then applying the function $\ln^{-1}$ to both sides of the equation $\ln e^r = r$, we find that

$$e^r = \exp r \qquad \text{for } r \text{ rational.} \qquad \exp \text{ is } \ln^{-1}. \qquad (1)$$

We have not yet found a way to give an obvious meaning to e^x for x irrational. But $\ln^{-1} x$ has meaning for any x, rational or irrational. So Equation (1) provides a way to extend the definition of e^x to irrational values of x. The function $\exp x$ is defined for all x, so we use it to assign a value to e^x at every point.

DEFINITION For every real number x, we define the **natural exponential function** to be $e^x = \exp x$.

For the first time we have a precise meaning for an irrational exponent—we are raising a specific number e to any real power x, rational or irrational. Since the functions $\ln x$ and e^x are inverses of one another, we have the following relationships.

Inverse Equations for e^x and $\ln x$

$$e^{\ln x} = x \qquad (\text{all } x > 0)$$
$$\ln(e^x) = x \qquad (\text{all } x)$$

EXAMPLE 1 Solve the equation $e^{2x-6} = 4$ for x.

Solution We take the natural logarithm of both sides of the equation and use the second inverse equation:

$$\ln(e^{2x-6}) = \ln 4$$
$$2x - 6 = \ln 4 \qquad \text{Inverse relationship}$$
$$2x = 6 + \ln 4$$
$$x = 3 + \frac{1}{2}\ln 4 = 3 + \ln 4^{1/2}$$
$$x = 3 + \ln 2$$

EXAMPLE 2 A line with slope m passes through the origin and is tangent to the graph of $y = \ln x$. What is the value of m?

Solution Suppose the point of tangency occurs at the unknown point $x = a > 0$. Then we know that the point $(a, \ln a)$ lies on the graph and that the tangent line at that point has slope $m = 1/a$ (Figure 7.11). Since the tangent line passes through the origin, its slope is

$$m = \frac{\ln a - 0}{a - 0} = \frac{\ln a}{a}.$$

Setting these two formulas for m equal to each other, we have

$$\frac{\ln a}{a} = \frac{1}{a}$$
$$\ln a = 1$$
$$e^{\ln a} = e^1$$
$$a = e$$
$$m = \frac{1}{e}.$$

The Derivative and Integral of e^x

According to Theorem 1, the natural exponential function is differentiable because it is the inverse of a differentiable function whose derivative is never zero. We calculate its derivative using the inverse relationship and the Chain Rule:

$$\ln (e^x) = x \qquad \text{Inverse relationship}$$
$$\frac{d}{dx}\ln (e^x) = 1 \qquad \text{Differentiate both sides.}$$
$$\frac{1}{e^x}\cdot\frac{d}{dx}(e^x) = 1 \qquad \text{Eq. (2), Section 7.2, with } u = e^x$$
$$\frac{d}{dx}e^x = e^x. \qquad \text{Solve for the derivative.}$$

That is, for $y = e^x$, we find that $dy/dx = e^x$ so the natural exponential function e^x is its own derivative. Moreover, if $f(x) = e^x$, then $f'(0) = e^0 = 1$. This means that the natural exponential function e^x has slope 1 as it crosses the y-axis at $x = 0$.

The Chain Rule extends the derivative result for the natural exponential function to a more general form involving a function $u(x)$:

If u is any differentiable function of x, then

$$\frac{d}{dx}e^u = e^u\frac{du}{dx}. \tag{2}$$

EXAMPLE 3 We find derivatives of the exponential using Equation (2).

(a) $\dfrac{d}{dx}(5e^x) = 5\dfrac{d}{dx}e^x = 5e^x$

(b) $\dfrac{d}{dx}e^{-x} = e^{-x}\dfrac{d}{dx}(-x) = e^{-x}(-1) = -e^{-x}$ Eq. (2) with $u = -x$

(c) $\dfrac{d}{dx}e^{\sin x} = e^{\sin x}\dfrac{d}{dx}(\sin x) = e^{\sin x}\cdot\cos x$ Eq. (2) with $u = \sin x$

(d) $\dfrac{d}{dx}\left(e^{\sqrt{3x+1}}\right) = e^{\sqrt{3x+1}}\cdot\dfrac{d}{dx}\left(\sqrt{3x+1}\right)$ Eq. (2) with $u = \sqrt{3x+1}$

$$= e^{\sqrt{3x+1}}\cdot\frac{1}{2}(3x+1)^{-1/2}\cdot 3 = \frac{3}{2\sqrt{3x+1}}e^{\sqrt{3x+1}}$$

Since e^x is its own derivative, it is also its own antiderivative. So the integral equivalent of Equation (2) is the following.

The general antiderivative of the exponential function

$$\int e^u\,du = e^u + C$$

EXAMPLE 4

(a) $\displaystyle\int_0^{\ln 2} e^{3x}\,dx = \int_0^{\ln 8} e^u \cdot \frac{1}{3}\,du$ $\quad u = 3x,\ \frac{1}{3}du = dx,\ u(0) = 0,\ u(\ln 2) = 3\ln 2 = \ln 2^3 = \ln 8$

$$= \frac{1}{3}\int_0^{\ln 8} e^u\,du$$

$$= \frac{1}{3}e^u\Big]_0^{\ln 8}$$

$$= \frac{1}{3}(8 - 1) = \frac{7}{3}$$

(b) $\displaystyle\int_0^{\pi/2} e^{\sin x}\cos x\,dx = e^{\sin x}\Big]_0^{\pi/2}$ $\quad$ Antiderivative from Example 2c

$$= e^1 - e^0 = e - 1$$

The derivative of e^x exists and is everywhere positive, confirming that it is a continuous and increasing function as shown in Figure 7.10. Since the second derivative of e^x is also e^x and everywhere positive, the graph is concave up. Moreover, Figure 7.10 shows that the exponential function has the limits

$$\lim_{x\to-\infty} e^x = 0 \qquad \text{and} \qquad \lim_{x\to\infty} e^x = \infty.$$

From the first of these limits we see that the x-axis is a horizontal asymptote of the graph $y = e^x$.

Laws of Exponents

Even though e^x is defined in a seemingly roundabout way as $\ln^{-1} x$, it obeys the familiar laws of exponents from algebra. The following Theorem 3 shows us that these laws are consequences of the definitions of $\ln x$ and e^x.

THEOREM 3 For all numbers x, x_1, and x_2, the natural exponential e^x obeys the following laws:

1. $e^{x_1}e^{x_2} = e^{x_1+x_2}$ **2.** $e^{-x} = \dfrac{1}{e^x}$

3. $\dfrac{e^{x_1}}{e^{x_2}} = e^{x_1-x_2}$ **4.** $(e^{x_1})^r = e^{rx_1}$, if r is rational

Proof of Law 1 Let $y_1 = e^{x_1}$ and $y_2 = e^{x_2}$. Then

$x_1 = \ln y_1$ and $x_2 = \ln y_2$ $\quad$ Inverse equations

$x_1 + x_2 = \ln y_1 + \ln y_2$

$= \ln y_1y_2$ $\quad$ Product Rule for logarithms

$e^{x_1+x_2} = e^{\ln y_1y_2}$ $\quad$ Exponentiate.

$= y_1y_2$ $\quad$ $e^{\ln u} = u$

$= e^{x_1}e^{x_2}$. ■

Proof of Law 4 Let $y = (e^{x_1})^r$. Then

$\ln y = \ln (e^{x_1})^r$

$= r\ln (e^{x_1})$ $\quad$ Power Rule for logarithms, rational r

$= rx_1$ $\quad$ $\ln e^u = u$ with $u = x_1$

Transcendental Numbers and Transcendental Functions

Numbers that are solutions of polynomial equations with rational coefficients are called **algebraic**: -2 is algebraic because it satisfies the equation $x + 2 = 0$, and $\sqrt{3}$ is algebraic because it satisfies the equation $x^2 - 3 = 0$. Numbers such as e and π that are not algebraic are called **transcendental**.

We call a function $y = f(x)$ algebraic if it satisfies an equation of the form

$$P_n y^n + \cdots + P_1 y + P_0 = 0$$

in which the P's are polynomials in x with rational coefficients. The function $y = 1/\sqrt{x+1}$ is algebraic because it satisfies the equation $(x + 1)y^2 - 1 = 0$. Here the polynomials are $P_2 = x + 1$, $P_1 = 0$, and $P_0 = -1$. Functions that are not algebraic are called transcendental.

Thus, exponentiating each side,

$$y = e^{rx_1}. \qquad e^{\ln y} = y$$

■

Laws 2 and 3 follow from Law 1. Like the Power Rule for logarithms, Law 4 holds for all real numbers r.

The General Exponential Function a^x

Since $a = e^{\ln a}$ for any positive number a, we can express a^x as $(e^{\ln a})^x = e^{x \ln a}$. We therefore use the function e^x to define the other exponential functions, which allow us to raise *any* positive number to an irrational exponent.

DEFINITION For any numbers $a > 0$ and x, the **exponential function with base a** is

$$a^x = e^{x \ln a}.$$

When $a = e$, the definition gives $a^x = e^{x \ln a} = e^{x \ln e} = e^{x \cdot 1} = e^x$.

Theorem 3 is also valid for a^x, the exponential function with base a. For example,

$$\begin{aligned} a^{x_1} \cdot a^{x_2} &= e^{x_1 \ln a} \cdot e^{x_2 \ln a} && \text{Definition of } a^x \\ &= e^{x_1 \ln a + x_2 \ln a} && \text{Law 1} \\ &= e^{(x_1 + x_2) \ln a} && \text{Factor } \ln a \\ &= a^{x_1 + x_2}. && \text{Definition of } a^x \end{aligned}$$

In particular, $a^n \cdot a^{-1} = a^{n-1}$ for any real number n.

Proof of the Power Rule (General Version)

The definition of the general exponential function enables us to make sense of raising any positive number to a real power n, rational or irrational. That is, we can define the power function $y = x^n$ for any exponent n.

DEFINITION For any $x > 0$ and for any real number n,

$$x^n = e^{n \ln x}.$$

Because the logarithm and exponential functions are inverses of each other, the definition gives

$$\ln x^n = n \ln x, \quad \text{for all real numbers } n.$$

That is, the rule for taking the natural logarithm of a power of x holds for *all* real exponents n, not just for rational exponents as previously stated in Theorem 2.

The definition of the power function also enables us to establish the derivative Power Rule for any real power n, as stated in Section 3.3.

General Power Rule for Derivatives
For $x > 0$ and any real number n,

$$\frac{d}{dx} x^n = n x^{n-1}.$$

If $x \leq 0$, then the formula holds whenever the derivative, x^n, and x^{n-1} all exist.

Proof Differentiating x^n with respect to x gives

$$\begin{aligned}
\frac{d}{dx}x^n &= \frac{d}{dx}e^{n\ln x} && \text{Definition of } x^n,\ x > 0\\
&= e^{n\ln x}\cdot\frac{d}{dx}(n\ln x) && \text{Chain Rule for } e^u, \text{ Eq. (2)}\\
&= x^n\cdot\frac{n}{x} && \text{Definition and derivative of } \ln x\\
&= nx^{n-1}. && x^n\cdot x^{-1} = x^{n-1}
\end{aligned}$$

In short, whenever $x > 0$,

$$\frac{d}{dx}x^n = nx^{n-1}.$$

For $x < 0$, if $y = x^n$, y', and x^{n-1} all exist, then

$$\ln|y| = \ln|x|^n = n\ln|x|.$$

Using implicit differentiation (which *assumes* the existence of the derivative y') and Equation (3) in Section 7.2, we have

$$\frac{y'}{y} = \frac{n}{x}.$$

Solving for the derivative,

$$y' = n\frac{y}{x} = n\frac{x^n}{x} = nx^{n-1}.$$

It can be shown directly from the definition of the derivative that the derivative equals 0 when $x = 0$ and $n \geq 1$. This completes the proof of the general version of the Power Rule for all values of x. ■

EXAMPLE 5 Differentiate $f(x) = x^x$, $x > 0$.

Solution We cannot apply the power rule here because the exponent is the *variable* x rather than being a constant value n (rational or irrational). However, from the definition of the general exponential function we note that $f(x) = x^x = e^{x\ln x}$, and differentiation gives

$$\begin{aligned}
f'(x) &= \frac{d}{dx}(e^{x\ln x})\\
&= e^{x\ln x}\frac{d}{dx}(x\ln x) && \text{Eq. (2) with } u = x\ln x\\
&= e^{x\ln x}\left(\ln x + x\cdot\frac{1}{x}\right) && \text{Product Rule}\\
&= x^x(\ln x + 1). && x > 0
\end{aligned}$$

■

The Number e Expressed as a Limit

We have defined the number e as the number for which $\ln e = 1$, or equivalently, the value $\exp(1)$. We see that e is an important constant for the logarithmic and exponential functions, but what is its numerical value? The next theorem shows one way to calculate e as a limit.

THEOREM 4—The Number e as a Limit
The number e can be calculated as the limit

$$e = \lim_{x\to 0}(1 + x)^{1/x}.$$

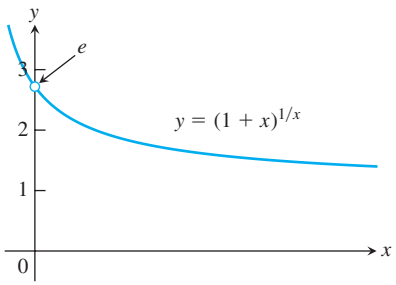


FIGURE 7.12 The number e is the limit of the function graphed here as $x \to 0$.

Proof If $f(x) = \ln x$, then $f'(x) = 1/x$, so $f'(1) = 1$. But, by the definition of derivative,

$$f'(1) = \lim_{h\to 0} \frac{f(1+h) - f(1)}{h} = \lim_{x\to 0} \frac{f(1+x) - f(1)}{x}$$

$$= \lim_{x\to 0} \frac{\ln(1+x) - \ln 1}{x} = \lim_{x\to 0} \frac{1}{x}\ln(1+x) \qquad \ln 1 = 0$$

$$= \lim_{x\to 0} \ln(1+x)^{1/x} = \ln\left[\lim_{x\to 0}(1+x)^{1/x}\right]. \qquad \begin{array}{l}\ln \text{ is continuous;}\\ \text{use Theorem 10}\\ \text{in Chapter 2.}\end{array}$$

Because $f'(1) = 1$, we have

$$\ln\left[\lim_{x\to 0}(1+x)^{1/x}\right] = 1$$

Therefore, exponentiating both sides we get (see Figure 7.12)

$$\lim_{x\to 0}(1+x)^{1/x} = e. \qquad \blacksquare$$

Approximating the limit in Theorem 4 by taking x very small gives approximations to e. Its value is $e \approx 2.718281828459045$ to 15 decimal places as noted before.

The Derivative of a^u

To find this derivative, we start with the defining equation $a^x = e^{x \ln a}$. Then we have

$$\frac{d}{dx}a^x = \frac{d}{dx}e^{x\ln a} = e^{x\ln a}\cdot\frac{d}{dx}(x\ln a) \qquad \frac{d}{dx}e^u = e^u\frac{du}{dx}$$

$$= a^x \ln a.$$

We now see why e^x is the exponential function preferred in calculus. If $a = e$, then $\ln a = 1$ and the derivative of a^x simplifies to

$$\frac{d}{dx}e^x = e^x \ln e = e^x. \qquad \ln e = 1$$

With the Chain Rule, we get the following form for the derivative of the general exponential function.

If $a > 0$ and u is a differentiable function of x, then a^u is a differentiable function of x and

$$\frac{d}{dx}a^u = a^u \ln a \frac{du}{dx}. \tag{3}$$

The integral equivalent of this last result gives the general antiderivative

$$\int a^u\, du = \frac{a^u}{\ln a} + C. \tag{4}$$

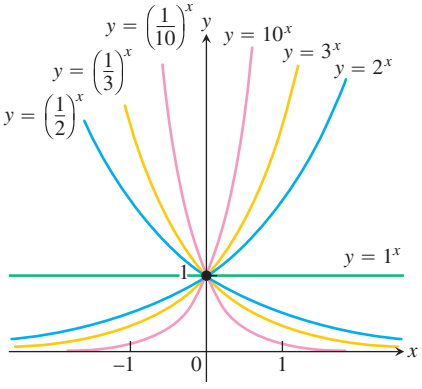


FIGURE 7.13 Exponential functions decrease if $0 < a < 1$ and increase if $a > 1$. As $x \to \infty$, we have $a^x \to 0$ if $0 < a < 1$ and $a^x \to \infty$ if $a > 1$. As $x \to -\infty$, we have $a^x \to \infty$ if $0 < a < 1$ and $a^x \to 0$ if $a > 1$.

From Equation (3) with $u = x$, we see that the derivative of a^x is positive if $\ln a > 0$, or $a > 1$, and negative if $\ln a < 0$, or $0 < a < 1$. Thus, a^x is an increasing function of x if $a > 1$ and a decreasing function of x if $0 < a < 1$. In each case, a^x is one-to-one. The second derivative

$$\frac{d^2}{dx^2}(a^x) = \frac{d}{dx}(a^x \ln a) = (\ln a)^2\, a^x$$

is positive for all x, so the graph of a^x is concave up on every interval of the real line. Figure 7.13 displays the graphs of several exponential functions.

EXAMPLE 6 We find derivatives and integrals using Equations (3) and (4).

(a) $\dfrac{d}{dx}3^x = 3^x \ln 3$ Eq. (3) with $a = 3$, $u = x$

(b) $\dfrac{d}{dx}3^{-x} = 3^{-x}(\ln 3)\dfrac{d}{dx}(-x) = -3^{-x}\ln 3$ Eq. (3) with $a = 3$, $u = -x$

(c) $\dfrac{d}{dx}3^{\sin x} = 3^{\sin x}(\ln 3)\dfrac{d}{dx}(\sin x) = 3^{\sin x}(\ln 3)\cos x$ Eq. (3) with $a = 3$, $u = \sin x$

(d) $\displaystyle\int 2^x\,dx = \frac{2^x}{\ln 2} + C$ Eq. (4) with $a = 2$, $u = x$

(e) $\displaystyle\int 2^{\sin x}\cos x\,dx = \int 2^u\,du = \frac{2^u}{\ln 2} + C$ $u = \sin x$, $du = \cos x\,dx$, and Eq. (4)

$\displaystyle= \frac{2^{\sin x}}{\ln 2} + C$ u replaced by $\sin x$ ■

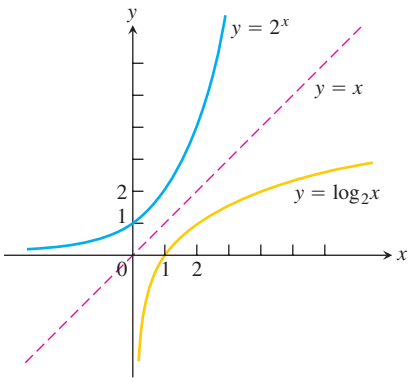


FIGURE 7.14 The graph of 2^x and its inverse, $\log_2 x$.

Logarithms with Base a

If a is any positive number other than 1, the function a^x is one-to-one and has a nonzero derivative at every point. It therefore has a differentiable inverse. We call the inverse the **logarithm of x with base a** and denote it by $\log_a x$.

DEFINITION For any positive number $a \neq 1$,

$\log_a x$ is the inverse function of a^x.

The graph of $y = \log_a x$ can be obtained by reflecting the graph of $y = a^x$ across the 45° line $y = x$ (Figure 7.14). When $a = e$, we have $\log_e x$ = inverse of $e^x = \ln x$. (The function $\log_{10} x$ is sometimes written simply as $\log x$ and is called the **common logarithm** of x.) Since $\log_a x$ and a^x are inverses of one another, composing them in either order gives the identity function.

Inverse Equations for a^x and $\log_a x$

$$a^{\log_a x} = x \qquad (x > 0)$$
$$\log_a(a^x) = x \qquad (\text{all } x)$$

The function $\log_a x$ is actually just a numerical multiple of $\ln x$. To see this, we let $y = \log_a x$ and then take the natural logarithm of both sides of the equivalent equation $a^y = x$ to obtain $y \ln a = \ln x$. Solving for y gives

$$\log_a x = \frac{\ln x}{\ln a}. \tag{5}$$

The algebraic rules satisfied by $\log_a x$ are the same as the ones for $\ln x$. These rules, given in Table 7.2, can be proved using Equation (5) by dividing the corresponding rules for the natural logarithm function by $\ln a$. For example,

$\ln xy = \ln x + \ln y$ Rule 1 for natural logarithms . . .

$\dfrac{\ln xy}{\ln a} = \dfrac{\ln x}{\ln a} + \dfrac{\ln y}{\ln a}$ . . . divided by $\ln a$ . . .

$\log_a xy = \log_a x + \log_a y.$ . . . gives Rule 1 for base a logarithms.

TABLE 7.2 Rules for base a logarithms

For any numbers $x > 0$ and $y > 0$,

1. *Product Rule*: $\log_a xy = \log_a x + \log_a y$
2. *Quotient Rule*: $\log_a \dfrac{x}{y} = \log_a x - \log_a y$
3. *Reciprocal Rule*: $\log_a \dfrac{1}{y} = -\log_a y$
4. *Power Rule*: $\log_a x^y = y \log_a x$

Derivatives and Integrals Involving $\log_a x$

To find derivatives or integrals involving base a logarithms, we convert them to natural logarithms. If u is a positive differentiable function of x, then

$$\frac{d}{dx}(\log_a u) = \frac{d}{dx}\left(\frac{\ln u}{\ln a}\right) = \frac{1}{\ln a}\frac{d}{dx}(\ln u) = \frac{1}{\ln a}\cdot\frac{1}{u}\frac{du}{dx}.$$

$$\boxed{\frac{d}{dx}(\log_a u) = \frac{1}{\ln a}\cdot\frac{1}{u}\frac{du}{dx}}$$

EXAMPLE 7

(a) $$\frac{d}{dx}\log_{10}(3x+1) = \frac{1}{\ln 10}\cdot\frac{1}{3x+1}\frac{d}{dx}(3x+1) = \frac{3}{(\ln 10)(3x+1)}$$

(b) $$\int \frac{\log_2 x}{x}dx = \frac{1}{\ln 2}\int \frac{\ln x}{x}dx \qquad \log_2 x = \frac{\ln x}{\ln 2}$$

$$= \frac{1}{\ln 2}\int u\,du \qquad u = \ln x,\quad du = \frac{1}{x}dx$$

$$= \frac{1}{\ln 2}\frac{u^2}{2} + C = \frac{1}{\ln 2}\frac{(\ln x)^2}{2} + C = \frac{(\ln x)^2}{2\ln 2} + C$$

EXERCISES 7.3

Solving Exponential Equations

In Exercises 1–4, solve for t.

1. **a.** $e^{-0.3t} = 27$ **b.** $e^{kt} = \frac{1}{2}$ **c.** $e^{(\ln 0.2)t} = 0.4$

2. **a.** $e^{-0.01t} = 1000$ **b.** $e^{kt} = \frac{1}{10}$ **c.** $e^{(\ln 2)t} = \frac{1}{2}$

3. $e^{\sqrt{t}} = x^2$ **4.** $e^{(x^2)}e^{(2x+1)} = e^t$

5. $e^{2t} - 3e^t = 0$ **6.** $e^{-2t} + 6 = 5e^{-t}$

Finding Derivatives

In Exercises 7–26, find the derivative of y with respect to x, t, or θ, as appropriate.

7. $y = e^{-5x}$ **8.** $y = e^{2x/3}$

9. $y = e^{5-7x}$ **10.** $y = e^{(4\sqrt{x}+x^2)}$

11. $y = xe^x - e^x$ **12.** $y = (1+2x)e^{-2x}$

13. $y = (x^2 - 2x + 2)e^x$ **14.** $y = (9x^2 - 6x + 2)e^{3x}$

15. $y = e^\theta(\sin\theta + \cos\theta)$ **16.** $y = \ln(3\theta e^{-\theta})$

17. $y = \cos(e^{-\theta^2})$ **18.** $y = \theta^3 e^{-2\theta}\cos 5\theta$

19. $y = \ln(3te^{-t})$ **20.** $y = \ln(2e^{-t}\sin t)$

21. $y = \ln\left(\dfrac{e^\theta}{1+e^\theta}\right)$ **22.** $y = \ln\left(\dfrac{\sqrt{\theta}}{1+\sqrt{\theta}}\right)$

23. $y = e^{(\cos t + \ln t)}$ **24.** $y = e^{\sin t}(\ln t^2 + 1)$

25. $y = \displaystyle\int_0^{\ln x} \sin e^t\,dt$ **26.** $y = \displaystyle\int_{e^{4\sqrt{x}}}^{e^{2x}} \ln t\,dt$

In Exercises 27–32, find dy/dx.

27. $\ln y = e^y \sin x$ **28.** $\ln xy = e^{x+y}$

29. $e^{2x} = \sin(x+3y)$ **30.** $\tan y = e^x + \ln x$

31. $3 + \sin y = y - x^3$ **32.** $\ln y = xe^y - 2$

Finding Integrals

Evaluate the integrals in Exercises 33–54.

33. $\displaystyle\int (e^{3x} + 5e^{-x})\,dx$ **34.** $\displaystyle\int (2e^x - 3e^{-2x})\,dx$

35. $\displaystyle\int_{\ln 2}^{\ln 3} e^x\,dx$ **36.** $\displaystyle\int_{-\ln 2}^{0} e^{-x}\,dx$

37. $\displaystyle\int 8e^{(x+1)}\,dx$ **38.** $\displaystyle\int 2e^{(2x-1)}\,dx$

39. $\displaystyle\int_{\ln 4}^{\ln 9} e^{x/2}\,dx$ **40.** $\displaystyle\int_0^{\ln 16} e^{x/4}\,dx$

41. $\displaystyle\int \frac{e^{\sqrt{r}}}{\sqrt{r}}dr$ **42.** $\displaystyle\int \frac{e^{-\sqrt{r}}}{\sqrt{r}}dr$

43. $\displaystyle\int 2t\,e^{-t^2}\,dt$ **44.** $\displaystyle\int t^3 e^{(t^4)}\,dt$

45. $\displaystyle\int \frac{e^{1/x}}{x^2}dx$ **46.** $\displaystyle\int \frac{e^{-1/x^2}}{x^3}dx$

47. $\displaystyle\int_0^{\pi/4} (1 + e^{\tan\theta})\sec^2\theta\,d\theta$ **48.** $\displaystyle\int_{\pi/4}^{\pi/2} (1 + e^{\cot\theta})\csc^2\theta\,d\theta$

49. $\int e^{\sec \pi t} \sec \pi t \tan \pi t \, dt$

50. $\int e^{\csc(\pi + t)} \csc(\pi + t) \cot(\pi + t) \, dt$

51. $\int_{\ln(\pi/6)}^{\ln(\pi/2)} 2e^v \cos e^v \, dv$

52. $\int_0^{\sqrt{\ln \pi}} 2x \, e^{x^2} \cos(e^{x^2}) \, dx$

53. $\int \frac{e^r}{1 + e^r} dr$

54. $\int \frac{dx}{1 + e^x}$

Initial Value Problems

Solve the initial value problems in Exercises 55–58.

55. $\frac{dy}{dt} = e^t \sin(e^t - 2), \quad y(\ln 2) = 0$

56. $\frac{dy}{dt} = e^{-t} \sec^2(\pi e^{-t}), \quad y(\ln 4) = 2/\pi$

57. $\frac{d^2y}{dx^2} = 2e^{-x}, \quad y(0) = 1 \quad \text{and} \quad y'(0) = 0$

58. $\frac{d^2y}{dt^2} = 1 - e^{2t}, \quad y(1) = -1 \quad \text{and} \quad y'(1) = 0$

Differentiation

In Exercises 59–86, find the derivative of y with respect to the given independent variable.

59. $y = 2^x$

60. $y = 3^{-x}$

61. $y = 5^{\sqrt{s}}$

62. $y = 2^{(s^2)}$

63. $y = x^\pi$

64. $y = t^{1-e}$

65. $y = (\cos \theta)^{\sqrt{2}}$

66. $y = (\ln \theta)^\pi$

67. $y = 7^{\sec \theta} \ln 7$

68. $y = 3^{\tan \theta} \ln 3$

69. $y = 2^{\sin 3t}$

70. $y = 5^{-\cos 2t}$

71. $y = \log_2 5\theta$

72. $y = \log_3(1 + \theta \ln 3)$

73. $y = \log_4 x + \log_4 x^2$

74. $y = \log_{25} e^x - \log_5 \sqrt{x}$

75. $y = x^3 \log_{10} x$

76. $y = \log_3 r \cdot \log_9 r$

77. $y = \log_3\left(\left(\frac{x+1}{x-1}\right)^{\ln 3}\right)$

78. $y = \log_5 \sqrt{\left(\frac{7x}{3x+2}\right)^{\ln 5}}$

79. $y = \theta \sin(\log_7 \theta)$

80. $y = \log_7\left(\frac{\sin \theta \cos \theta}{e^\theta 2^\theta}\right)$

81. $y = \log_{10} e^x$

82. $y = \frac{\theta 5^\theta}{2 - \log_5 \theta}$

83. $y = 3^{\log_2 t}$

84. $y = 3 \log_8(\log_2 t)$

85. $y = \log_2(8t^{\ln 2})$

86. $y = t \log_3\left(e^{(\sin t)(\ln 3)}\right)$

Integration

Evaluate the integrals in Exercises 87–96.

87. $\int 5^x \, dx$

88. $\int \frac{3^x}{3 - 3^x} dx$

89. $\int_0^1 2^{-\theta} \, d\theta$

90. $\int_{-2}^0 5^{-\theta} \, d\theta$

91. $\int_1^{\sqrt{2}} x 2^{(x^2)} \, dx$

92. $\int_1^4 \frac{2^{\sqrt{x}}}{\sqrt{x}} dx$

93. $\int_0^{\pi/2} 7^{\cos t} \sin t \, dt$

94. $\int_0^{\pi/4} \left(\frac{1}{3}\right)^{\tan t} \sec^2 t \, dt$

95. $\int_2^4 x^{2x}(1 + \ln x) \, dx$

96. $\int \frac{x 2^{x^2}}{1 + 2^{x^2}} dx$

Evaluate the integrals in Exercises 97–110.

97. $\int 3x^{\sqrt{3}} \, dx$

98. $\int x^{\sqrt{2}-1} \, dx$

99. $\int_0^3 (\sqrt{2} + 1)x^{\sqrt{2}} \, dx$

100. $\int_1^e x^{(\ln 2)-1} \, dx$

101. $\int \frac{\log_{10} x}{x} dx$

102. $\int_1^4 \frac{\log_2 x}{x} dx$

103. $\int_1^4 \frac{\ln 2 \log_2 x}{x} dx$

104. $\int_1^e \frac{2 \ln 10 \log_{10} x}{x} dx$

105. $\int_0^2 \frac{\log_2(x+2)}{x+2} dx$

106. $\int_{1/10}^{10} \frac{\log_{10}(10x)}{x} dx$

107. $\int_0^9 \frac{2 \log_{10}(x+1)}{x+1} dx$

108. $\int_2^3 \frac{2 \log_2(x-1)}{x-1} dx$

109. $\int \frac{dx}{x \log_{10} x}$

110. $\int \frac{dx}{x(\log_8 x)^2}$

Evaluate the integrals in Exercises 111–114.

111. $\int_1^{\ln x} \frac{1}{t} dt, \quad x > 1$

112. $\int_1^{e^x} \frac{1}{t} dt$

113. $\int_1^{1/x} \frac{1}{t} dt, \quad x > 0$

114. $\frac{1}{\ln a} \int_1^x \frac{1}{t} dt, \quad x > 0$

Logarithmic Differentiation

In Exercises 115–122, use logarithmic differentiation to find the derivative of y with respect to the given independent variable.

115. $y = (x + 1)^x$

116. $y = x^2 + x^{2x}$

117. $y = \left(\sqrt{t}\right)^t$

118. $y = t^{\sqrt{t}}$

119. $y = (\sin x)^x$

120. $y = x^{\sin x}$

121. $y = \sin x^x$

122. $y = (\ln x)^{\ln x}$

123. $y^x = x^3 y$

124. $x^{\sin y} = \ln y$

125. $x = y^{xy}$

126. $e^y = y^{\ln x}$

For Exercises 127 and 128 find a function f satisfying each equation.

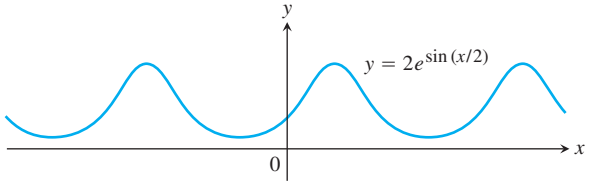
127. $\int_2^x \sqrt{f(t)} \, dt = x \ln x$

128. $f(x) = e^2 + \int_1^x f(t) \, dt$

Theory and Applications

129. Find the absolute maximum and minimum values of $f(x) = e^x - 2x$ on $[0, 1]$.

130. Where does the periodic function $f(x) = 2e^{\sin(x/2)}$ take on its extreme values and what are these values?



131. Let $f(x) = xe^{-x}$.

a. Find all absolute extreme values for f.

b. Find all inflection points for f.

132. Let $f(x) = \dfrac{e^x}{1 + e^{2x}}$.

a. Find all absolute extreme values for f.

b. Find all inflection points for f.

133. Find the absolute maximum value of $f(x) = x^2 \ln (1/x)$ and say where it is assumed.

T **134.** Graph $f(x) = (x - 3)^2 e^x$ and its first derivative together. Comment on the behavior of f in relation to the signs and values of f'. Identify significant points on the graphs with calculus, as necessary.

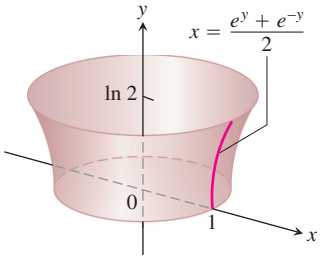
135. Find the area of the "triangular" region in the first quadrant that is bounded above by the curve $y = e^{2x}$, below by the curve $y = e^x$, and on the right by the line $x = \ln 3$.

136. Find the area of the "triangular" region in the first quadrant that is bounded above by the curve $y = e^{x/2}$, below by the curve $y = e^{-x/2}$, and on the right by the line $x = 2 \ln 2$.

137. Find a curve through the origin in the xy-plane whose length from $x = 0$ to $x = 1$ is

$$L = \int_0^1 \sqrt{1 + \frac{1}{4} e^x}\, dx.$$

138. Find the area of the surface generated by revolving the curve $x = (e^y + e^{-y})/2$, $0 \le y \le \ln 2$, about the y-axis.



In Exercises 139–142, find the length of each curve.

139. $y = \dfrac{1}{2}(e^x + e^{-x})$ from $x = 0$ to $x = 1$

140. $y = \ln (e^x - 1) - \ln (e^x + 1)$ from $x = \ln 2$ to $x = \ln 3$

141. $y = \ln (\cos x)$ from $x = 0$ to $x = \pi/4$

142. $y = \ln (\csc x)$ from $x = \pi/6$ to $x = \pi/4$

143. **a.** Show that $\int \ln x\, dx = x \ln x - x + C$.

b. Find the average value of $\ln x$ over $[1, e]$.

144. Find the average value of $f(x) = 1/x$ on $[1, 2]$.

145. The linearization of e^x at $x = 0$

a. Derive the linear approximation $e^x \approx 1 + x$ at $x = 0$.

T **b.** Estimate to five decimal places the magnitude of the error involved in replacing e^x by $1 + x$ on the interval $[0, 0.2]$.

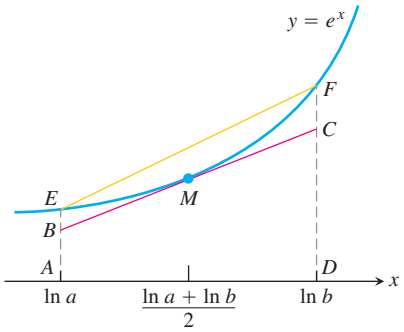
T **c.** Graph e^x and $1 + x$ together for $-2 \le x \le 2$. Use different colors, if available. On what intervals does the approximation appear to overestimate e^x? Underestimate e^x?

146. The geometric, logarithmic, and arithmetic mean inequality

a. Show that the graph of e^x is concave up over every interval of x-values.

b. Show, by reference to the accompanying figure, that if $0 < a < b$, then

$$e^{(\ln a + \ln b)/2} \cdot (\ln b - \ln a) < \int_{\ln a}^{\ln b} e^x\, dx < \frac{e^{\ln a} + e^{\ln b}}{2} \cdot (\ln b - \ln a).$$



NOT TO SCALE

c. Use the inequality in part (b) to conclude that

$$\sqrt{ab} < \frac{b - a}{\ln b - \ln a} < \frac{a + b}{2}.$$

This inequality says that the geometric mean of two positive numbers is less than their logarithmic mean, which in turn is less than their arithmetic mean.

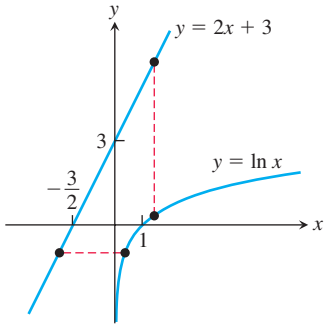
147. Find the area of the region between the curve $y = 2x/(1 + x^2)$ and the interval $-2 \le x \le 2$ of the x-axis.

148. Find the area of the region between the curve $y = 2^{1-x}$ and the interval $-1 \le x \le 1$ of the x-axis.

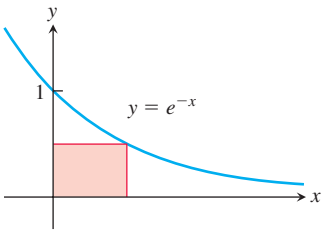
149. Consider the accompanying graphs of $y = 2x + 3$ and $y = \ln x$. Determine the

a. minimum vertical distance

b. minimum horizontal distance between these graphs.



150. Determine the dimensions of the inscribed rectangle of maximum area.



T **151.** The equation $x^2 = 2^x$ has three solutions: $x = 2, x = 4$, and one other. Estimate the third solution as accurately as you can by graphing.

T **152.** Could $x^{\ln 2}$ possibly be the same as $2^{\ln x}$ for $x > 0$? Graph the two functions and explain what you see.

153. The linearization of 2^x

a. Find the linearization of $f(x) = 2^x$ at $x = 0$. Then round its coefficients to two decimal places.

T **b.** Graph the linearization and function together for $-3 \le x \le 3$ and $-1 \le x \le 1$.

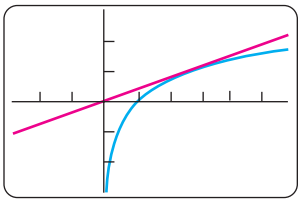
154. The linearization of $\log_3 x$

a. Find the linearization of $f(x) = \log_3 x$ at $x = 3$. Then round its coefficients to two decimal places.

T **b.** Graph the linearization and function together in the windows $0 \le x \le 8$ and $2 \le x \le 4$.

T **155. Which is bigger, π^e or e^π?** Calculators have taken some of the mystery out of this once-challenging question. (Go ahead and check; you will see that it is a very close call.) You can answer the question without a calculator, though.

a. Find an equation for the line through the origin tangent to the graph of $y = \ln x$.



[–3, 6] by [–3, 3]

b. Give an argument based on the graphs of $y = \ln x$ and the tangent line to explain why $\ln x < x/e$ for all positive $x \ne e$.

c. Show that $\ln (x^e) < x$ for all positive $x \ne e$.

d. Conclude that $x^e < e^x$ for all positive $x \ne e$.

e. So which is bigger, π^e or e^π?

T **156. A decimal representation of e** Find e to as many decimal places as your calculator allows by solving the equation $\ln x = 1$ using Newton's method in Section 4.6.

7.4 Exponential Change and Separable Differential Equations

Exponential functions increase or decrease very rapidly with changes in the independent variable. They describe growth or decay in many natural and human-made situations. The variety of models based on these functions partly accounts for their importance.

Exponential Change

In modeling many real-world situations, a quantity y increases or decreases at a rate proportional to its size at a given time t. Examples of such quantities include the size of a population, the amount of a decaying radioactive material, and the temperature difference between a hot object and its surrounding medium. Such quantities are said to undergo **exponential change**.

If the amount present at time $t = 0$ is called y_0, then we can find y as a function of t by solving the following initial value problem:

$$\text{Differential equation:} \quad \frac{dy}{dt} = ky \tag{1a}$$

$$\text{Initial condition:} \quad y = y_0 \quad \text{when} \quad t = 0. \tag{1b}$$

If y is positive and increasing, then k is positive, and we use Equation (1a) to say that the rate of growth is proportional to what has already been accumulated. If y is positive and decreasing, then k is negative, and we use Equation (1a) to say that the rate of decay is proportional to the amount still left.

We see right away that the constant function $y = 0$ is a solution of Equation (1a) if $y_0 = 0$. To find the nonzero solutions, we divide Equation (1a) by y:

$$\begin{aligned}
\frac{1}{y}\cdot\frac{dy}{dt} &= k && y \neq 0\\
\int \frac{1}{y}\frac{dy}{dt}\,dt &= \int k\,dt && \text{Integrate with respect to } t;\ \textstyle\int (1/u)\,du = \ln|u| + C.\\
\ln|y| &= kt + C\\
|y| &= e^{kt+C} && \text{Exponentiate.}\\
|y| &= e^{C}\cdot e^{kt} && e^{a+b} = e^{a}\cdot e^{b}\\
y &= \pm e^{C}e^{kt} && \text{If } |y| = r, \text{ then } y = \pm r.\\
y &= Ae^{kt}. && A \text{ is a shorter name for } \pm e^{C}.
\end{aligned}$$

By allowing A to take on the value 0 in addition to all possible values $\pm e^C$, we can include the solution $y = 0$ in the formula.

We find the value of A for the initial value problem by solving for A when $y = y_0$ and $t = 0$:

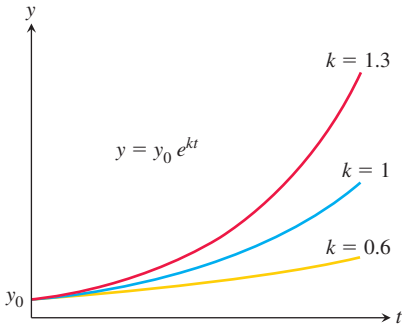
$$y_0 = Ae^{k\cdot 0} = A.$$

The solution of the initial value problem

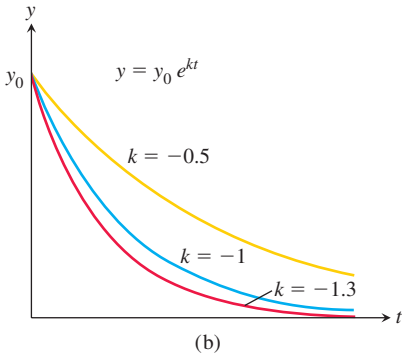
$$\frac{dy}{dt} = ky, \qquad y(0) = y_0$$

is

$$y = y_0e^{kt}. \tag{2}$$



(a)



(b)

FIGURE 7.15 Graphs of (a) exponential growth and (b) exponential decay. As $|k|$ increases, the growth ($k > 0$) or decay ($k < 0$) intensifies.

Quantities changing in this way are said to undergo **exponential growth** if $k > 0$ and **exponential decay** if $k < 0$. The number k is called the **rate constant** of the change. (See Figure 7.15.)

The derivation of Equation (2) shows also that the only functions that are their own derivatives ($k = 1$) are constant multiples of the exponential function.

Before presenting several examples of exponential change, let us consider the process we used to derive it.

Separable Differential Equations

Exponential change is modeled by a differential equation of the form $dy/dx = ky$, where k is a nonzero constant. More generally, suppose we have a differential equation of the form

$$\frac{dy}{dx} = f(x,y), \tag{3}$$

where f is a function of *both* the independent and dependent variables. A **solution** of the equation is a differentiable function $y = y(x)$ defined on an interval of x-values (perhaps infinite) such that

$$\frac{d}{dx}y(x) = f(x, y(x))$$

on that interval. That is, when $y(x)$ and its derivative $y'(x)$ are substituted into the differential equation, the resulting equation is true for all x in the solution interval. The **general solution** is a solution $y(x)$ that contains all possible solutions and it always contains an arbitrary constant.

Equation (3) is **separable** if f can be expressed as a product of a function of x and a function of y. The differential equation then has the form

$$\frac{dy}{dx} = g(x)\,h(y). \qquad \begin{matrix} g \text{ is a function of } x; \\ h \text{ is a function of } y. \end{matrix}$$

Then collect all y terms with dy and all x terms with dx:

$$\frac{1}{h(y)}\,dy = g(x)\,dx.$$

Now we simply integrate both sides of this equation:

$$\int \frac{1}{h(y)}\,dy = \int g(x)\,dx. \tag{4}$$

After completing the integrations, we obtain the solution y defined implicitly as a function of x.

The justification that we can integrate both sides in Equation (4) in this way is based on the Substitution Rule (Section 5.5):

$$\begin{aligned} \int \frac{1}{h(y)}\,dy &= \int \frac{1}{h(y(x))}\frac{dy}{dx}\,dx \\ &= \int \frac{1}{h(y(x))}h(y(x))\,g(x)\,dx \qquad \frac{dy}{dx} = h(y(x))\,g(x) \\ &= \int g(x)\,dx. \end{aligned}$$

EXAMPLE 1 Solve the differential equation

$$\frac{dy}{dx} = (1 + y)\,e^x, \quad y > -1.$$

Solution Since $1 + y$ is never zero for $y > -1$, we can solve the equation by separating the variables.

$$\begin{aligned} \frac{dy}{dx} &= (1 + y)\,e^x \\ dy &= (1 + y)\,e^x\,dx && \text{Treat } dy/dx \text{ as a quotient of differentials and multiply both sides by } dx. \\ \frac{dy}{1 + y} &= e^x\,dx && \text{Divide by } (1 + y). \\ \int \frac{dy}{1 + y} &= \int e^x\,dx && \text{Integrate both sides.} \\ \ln(1 + y) &= e^x + C && C \text{ represents the combined constants of integration.} \end{aligned}$$

The last equation gives y as an implicit function of x. ■

EXAMPLE 2 Solve the equation $y(x + 1)\dfrac{dy}{dx} = x(y^2 + 1)$.

Solution We change to differential form, separate the variables, and integrate:

$$y(x + 1)\,dy = x(y^2 + 1)\,dx$$

$$\frac{y\,dy}{y^2 + 1} = \frac{x\,dx}{x + 1} \qquad x \neq -1$$

$$\int \frac{y\,dy}{1 + y^2} = \int \left(1 - \frac{1}{x + 1}\right) dx \qquad \text{Divide } x \text{ by } x + 1.$$

$$\frac{1}{2}\ln(1 + y^2) = x - \ln|x + 1| + C.$$

The last equation gives the solution y as an implicit function of x. ■

The initial value problem

$$\frac{dy}{dt} = ky, \qquad y(0) = y_0$$

involves a separable differential equation, and the solution $y = y_0 e^{kt}$ expresses exponential change. We now present several examples of such change.

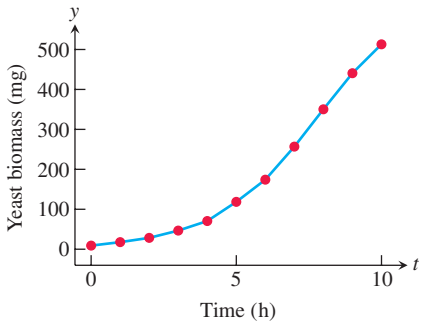


FIGURE 7.16 Graph of the growth of a yeast population over a 10-hour period, based on the data in Example 3.

Time (h)	Yeast biomass (mg)
0	9.6
1	18.3
2	29.0
3	47.2
4	71.1
5	119.1
6	174.6
7	257.3
8	350.7
9	441.0
10	513.3

Unlimited Population Growth

Strictly speaking, the number of individuals in a population (of people, plants, animals, or bacteria, for example) is a discontinuous function of time because it takes on discrete values. However, when the number of individuals becomes large enough, the population can be approximated by a continuous function. Differentiability of the approximating function is another reasonable hypothesis in many settings, allowing for the use of calculus to model and predict population sizes.

If we assume that the proportion of reproducing individuals remains constant and assume a constant fertility, then at any instant t the birth rate is proportional to the number $y(t)$ of individuals present. Let's assume, too, that the death rate of the population is stable and proportional to $y(t)$. If, further, we neglect departures and arrivals, the growth rate dy/dt is the birth rate minus the death rate, which is the difference of the two proportionalities under our assumptions. In other words, $dy/dt = ky$ so that $y = y_0 e^{kt}$, where y_0 is the size of the population at time $t = 0$. As with all kinds of growth, there may be limitations imposed by the surrounding environment, but we will not go into these here. (We treat one model imposing such limitations in Section 16.4.) When k is positive, the proportionality $dy/dt = ky$ models *unlimited population growth.* (See Figure 7.16.)

EXAMPLE 3 The biomass of a yeast culture in an experiment is initially 29 grams. After 30 minutes the mass is 37 grams. Assuming that the equation for unlimited population growth gives a good model for the growth of the yeast when the mass is below 100 grams, how long will its take for the mass to double from its initial value?

Solution Let $y(t)$ be the yeast biomass after t minutes. We use the exponential growth model $dy/dt = ky$ for unlimited population growth, with solution $y = y_0 e^{kt}$.

We have $y_0 = y(0) = 29$. We are also told that

$$y(30) = 29e^{k(30)} = 37.$$

Solving this equation for k, we find

$$e^{k(30)} = \frac{37}{29}$$

$$30k = \ln\left(\frac{37}{29}\right)$$

$$k = \frac{1}{30}\ln\left(\frac{37}{29}\right) \approx 0.008118.$$

Then the mass of the yeast in grams after t minutes is given by the equation

$$y = 29e^{(0.008118)t}.$$

To solve the problem we find the time t for which $y(t) = 58$, which is twice the initial amount.

$$29e^{(0.008118)t} = 58$$

$$(0.008118)t = \ln\left(\frac{58}{29}\right)$$

$$t = \frac{\ln 2}{0.008118} \approx 85.38$$

It takes about 85 minutes for the yeast population to double. ■

In the next example we model the number of people within a given population who are infected by a disease which is being eradicated from the population. Here the constant of proportionality k is negative, and the model describes an exponentially decaying number of infected individuals.

EXAMPLE 4 One model for the way diseases die out when properly treated assumes that the rate dy/dt at which the number of infected people changes is proportional to the number y. The number of people cured is proportional to the number y that are infected with the disease. Suppose that in the course of any given year the number of cases of a disease is reduced by 20%. If there are 10,000 cases today, how many years will it take to reduce the number to 1000?

Solution We use the equation $y = y_0e^{kt}$. There are three things to find: the value of y_0, the value of k, and the time t when $y = 1000$.

The value of y_0. We are free to count time beginning anywhere we want. If we count from today, then $y = 10{,}000$ when $t = 0$, so $y_0 = 10{,}000$. Our equation is now

$$y = 10{,}000e^{kt}. \tag{5}$$

The value of k. When $t = 1$ year, the number of cases will be 80% of its present value, or 8000. Hence,

$$8000 = 10{,}000e^{k(1)} \qquad \text{Eq. (5) with } t = 1 \text{ and } y = 8000$$

$$e^k = 0.8$$

$$\ln(e^k) = \ln 0.8 \qquad \text{Logs of both sides}$$

$$k = \ln 0.8 < 0. \qquad \ln 0.8 \approx -0.223$$

At any given time t,

$$y = 10{,}000e^{(\ln 0.8)t}. \tag{6}$$

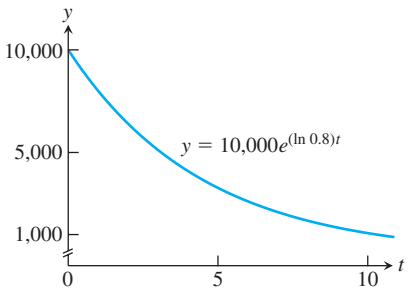


FIGURE 7.17 A graph of the number of people infected by a disease exhibits exponential decay (Example 4).

The value of t that makes $y = 1000$. We set y equal to 1000 in Equation (6) and solve for t:

$$\begin{aligned} 1000 &= 10{,}000e^{(\ln 0.8)t} \\ e^{(\ln 0.8)t} &= 0.1 \\ (\ln 0.8)t &= \ln 0.1 \qquad \text{Logs of both sides} \\ t &= \frac{\ln 0.1}{\ln 0.8} \approx 10.32 \text{ years}. \end{aligned}$$

It will take a little more than 10 years to reduce the number of cases to 1000. (See Figure 7.17.) ■

Radioactivity

Some atoms are unstable and can spontaneously emit mass or radiation. This process is called **radioactive decay**, and an element whose atoms go spontaneously through this process is called **radioactive**. Sometimes when an atom emits some of its mass through this process of radioactivity, the remainder of the atom re-forms to make an atom of some new element. For example, radioactive carbon-14 decays into nitrogen; radium, through a number of intermediate radioactive steps, decays into lead.

Experiments have shown that at any given time the rate at which a radioactive element decays (as measured by the number of nuclei that change per unit time) is approximately proportional to the number of radioactive nuclei present. Thus, the decay of a radioactive element is described by the equation $dy/dt = -ky, k > 0$. It is conventional to use $-k$, with $k > 0$, to emphasize that y is decreasing. If y_0 is the number of radioactive nuclei present at time zero, the number still present at any later time t will be

$$y = y_0 e^{-kt}, \qquad k > 0.$$

For radon-222 gas, t is measured in days and $k = 0.18$. For radium-226, which used to be painted on watch dials to make them glow at night (a dangerous practice), t is measured in years and $k = 4.3 \times 10^{-4}$.

The **half-life** of a radioactive element is the time expected to pass until half of the radioactive nuclei present in a sample decay. It is an interesting fact that the half-life is a constant that does not depend on the number of radioactive nuclei initially present in the sample, but only on the radioactive substance.

To compute the half-life, let y_0 be the number of radioactive nuclei initially present in the sample. Then the number y present at any later time t will be $y = y_0 e^{-kt}$. We seek the value of t at which the number of radioactive nuclei present equals half the original number:

$$\begin{aligned} y_0 e^{-kt} &= \frac{1}{2} y_0 \\ e^{-kt} &= \frac{1}{2} \\ -kt &= \ln \frac{1}{2} = -\ln 2 \qquad \text{Reciprocal Rule for logarithms} \\ t &= \frac{\ln 2}{k}. \end{aligned}$$

This value of t is the half-life of the element. It depends only on the value of k; the number y_0 does not have any effect.

$$\text{Half-life} = \frac{\ln 2}{k} \qquad (7)$$

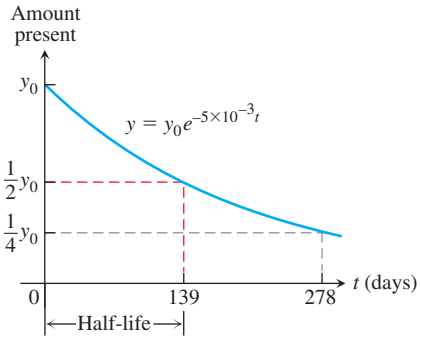


FIGURE 7.18 Amount of polonium-210 present at time t, where y_0 represents the number of radioactive atoms initially present.

The effective radioactive lifetime of polonium-210 is so short that we measure it in days rather than years. The number of radioactive atoms remaining after t days in a sample that starts with y_0 radioactive atoms is

$$y = y_0e^{-5\times10^{-3}\,t}.$$

The element's half-life is

$$\text{Half-life} = \frac{\ln 2}{k} \qquad \text{Eq. (7)}$$

$$= \frac{\ln 2}{5 \times 10^{-3}} \qquad \text{The } k \text{ from polonium's decay equation}$$

$$\approx 139 \text{ days.}$$

This means that after 139 days, $1/2$ of y_0 radioactive atoms remain; after another 139 days (278 days altogether) half of those remain, or $1/4$ of y_0 radioactive atoms remain, and so on (see Figure 7.18).

Carbon-14 dating uses the half-life of 5730 years.

EXAMPLE 5 The decay of radioactive elements can sometimes be used to date events from Earth's past. In a living organism, the ratio of radioactive carbon, carbon-14, to ordinary carbon stays fairly constant during the lifetime of the organism, being approximately equal to the ratio in the organism's atmosphere at the time. After the organism's death, however, no new carbon is ingested, and the proportion of carbon-14 in the organism's remains decreases as the carbon-14 decays.

Scientists who do carbon-14 dating often use a figure of 5730 years for its half-life. Find the age of a sample in which 10% of the radioactive nuclei originally present have decayed.

Solution We use the decay equation $y = y_0e^{-kt}$. There are two things to find: the value of k and the value of t when y is $0.9y_0$ (90% of the radioactive nuclei are still present). That is, find t when $y_0e^{-kt} = 0.9y_0$, or $e^{-kt} = 0.9$.

The value of k. We use the half-life Equation (7):

$$k = \frac{\ln 2}{\text{half-life}} = \frac{\ln 2}{5730} \qquad (\text{about } 1.2 \times 10^{-4}).$$

The value of t that makes $e^{-kt} = 0.9$.

$$e^{-kt} = 0.9$$

$$e^{-(\ln 2/5730)t} = 0.9$$

$$-\frac{\ln 2}{5730}t = \ln 0.9 \qquad \text{Logs of both sides}$$

$$t = -\frac{5730 \ln 0.9}{\ln 2} \approx 871 \text{ years}$$

The sample is about 871 years old. ■

Heat Transfer: Newton's Law of Cooling

Hot soup left in a tin cup cools to the temperature of the surrounding air. A hot silver bar immersed in a large tub of water cools to the temperature of the surrounding water. In situations like these, the rate at which an object's temperature is changing at any given time is roughly proportional to the difference between its temperature and the temperature of the surrounding medium. This observation is called *Newton's Law of Cooling*, although it applies to warming as well.

If H is the temperature of the object at time t and H_S is the constant surrounding temperature, then the differential equation is

$$\frac{dH}{dt} = -k(H - H_S). \tag{8}$$

If we substitute y for $(H - H_S)$, then

$$\begin{aligned}
\frac{dy}{dt} &= \frac{d}{dt}(H - H_S) = \frac{dH}{dt} - \frac{d}{dt}(H_S) \\
&= \frac{dH}{dt} - 0 && H_S \text{ is a constant.} \\
&= \frac{dH}{dt} \\
&= -k(H - H_S) && \text{Eq. (8)} \\
&= -ky. && H - H_S = y
\end{aligned}$$

We know that the solution of the equation $dy/dt = -ky$ is $y = y_0e^{-kt}$, where $y(0) = y_0$. Substituting $(H - H_S)$ for y, this says that

$$H - H_S = (H_0 - H_S)e^{-kt}, \tag{9}$$

where H_0 is the temperature at $t = 0$. This equation is the solution to Newton's Law of Cooling.

EXAMPLE 6 A hard-boiled egg at 98°C is put in a sink of 18°C water. After 5 min, the egg's temperature is 38°C. Assuming that the water has not warmed appreciably, how much longer will it take the egg to reach 20°C?

Solution We find how long it would take the egg to cool from 98°C to 20°C and subtract the 5 min that have already elapsed. Using Equation (9) with $H_S = 18$ and $H_0 = 98$, the egg's temperature t min after it is put in the sink is

$$H = 18 + (98 - 18)e^{-kt} = 18 + 80e^{-kt}.$$

To find k, we use the information that $H = 38$ when $t = 5$:

$$\begin{aligned}
38 &= 18 + 80e^{-5k} \\
e^{-5k} &= \frac{1}{4} \\
-5k &= \ln\frac{1}{4} = -\ln 4 \\
k &= \frac{1}{5}\ln 4 = 0.2 \ln 4 \qquad \text{(about 0.28).}
\end{aligned}$$

The egg's temperature at time t is $H = 18 + 80e^{-(0.2 \ln 4)t}$. Now find the time t when $H = 20$:

$$\begin{aligned}
20 &= 18 + 80e^{-(0.2\ln 4)t} \\
80e^{-(0.2\ln 4)t} &= 2 \\
e^{-(0.2\ln 4)t} &= \frac{1}{40} \\
-(0.2\ln 4)t &= \ln\frac{1}{40} = -\ln 40 \\
t &= \frac{\ln 40}{0.2\ln 4} \approx 13 \text{ min.}
\end{aligned}$$

The egg's temperature will reach 20°C about 13 min after it is put in the water to cool. Since it took 5 min to reach 38°C, it will take about 8 min more to reach 20°C. ■

EXERCISES 7.4

Verifying Solutions

In Exercises 1–4, show that each function $y = f(x)$ is a solution of the accompanying differential equation.

1. $2y' + 3y = e^{-x}$

a. $y = e^{-x}$ **b.** $y = e^{-x} + e^{-(3/2)x}$

c. $y = e^{-x} + Ce^{-(3/2)x}$

2. $y' = y^2$

a. $y = -\dfrac{1}{x}$ **b.** $y = -\dfrac{1}{x+3}$

c. $y = -\dfrac{1}{x+C}$

3. $y = \dfrac{1}{x}\displaystyle\int_1^x \frac{e^t}{t}\,dt, \quad x^2y' + xy = e^x$

4. $y = \dfrac{1}{\sqrt{1+x^4}}\displaystyle\int_1^x \sqrt{1+t^4}\,dt, \quad y' + \dfrac{2x^3}{1+x^4}y = 1$

Initial Value Problems

In Exercises 5–8, show that each function is a solution of the given initial value problem.

	Differential equation	Initial equation	Solution candidate
5.	$y' + y = \dfrac{2}{1+4e^{2x}}$	$y(-\ln 2) = \dfrac{\pi}{2}$	$y = e^{-x}\tan^{-1}(2e^x)$
6.	$y' = e^{-x^2} - 2xy$	$y(2) = 0$	$y = (x-2)e^{-x^2}$
7.	$xy' + y = -\sin x$, $x > 0$	$y\left(\dfrac{\pi}{2}\right) = 0$	$y = \dfrac{\cos x}{x}$
8.	$x^2y' = xy - y^2$, $x > 1$	$y(e) = e$	$y = \dfrac{x}{\ln x}$

Separable Differential Equations

Solve the differential equations in Exercises 9–22.

9. $2\sqrt{xy}\,\dfrac{dy}{dx} = 1, \quad x, y > 0$

10. $\dfrac{dy}{dx} = x^2\sqrt{y}, \quad y > 0$

11. $\dfrac{dy}{dx} = e^{x-y}$

12. $\dfrac{dy}{dx} = 3x^2\,e^{-y}$

13. $\dfrac{dy}{dx} = \sqrt{y}\cos^2\sqrt{y}$

14. $\sqrt{2xy}\,\dfrac{dy}{dx} = 1$

15. $\sqrt{x}\,\dfrac{dy}{dx} = e^{y+\sqrt{x}}, \quad x > 0$

16. $(\sec x)\dfrac{dy}{dx} = e^{y+\sin x}$

17. $\dfrac{dy}{dx} = 2x\sqrt{1-y^2}, \quad -1 < y < 1$

18. $\dfrac{dy}{dx} = \dfrac{e^{2x-y}}{e^{x+y}}$

19. $y^2\,\dfrac{dy}{dx} = 3x^2y^3 - 6x^2$

20. $\dfrac{dy}{dx} = xy + 3x - 2y - 6$

21. $\dfrac{1}{x}\dfrac{dy}{dx} = ye^{x^2} + 2\sqrt{y}\,e^{x^2}$

22. $\dfrac{dy}{dx} = e^{x-y} + e^x + e^{-y} + 1$

Applications and Examples

The answers to most of the following exercises are in terms of logarithms and exponentials. A calculator can be helpful, enabling you to express the answers in decimal form.

23. Human evolution continues The analysis of tooth shrinkage by C. Loring Brace and colleagues at the University of Michigan's Museum of Anthropology indicates that human tooth size is continuing to decrease and that the evolutionary process has not yet come to a halt some 30,000 years ago. In northern Europeans, for example, tooth size reduction now has a rate of 1% per 1000 years.

a. If t represents time in years and y represents tooth size, use the condition that $y = 0.99y_0$ when $t = 1000$ to find the value of k in the equation $y = y_0e^{kt}$. Then use this value of k to answer the following questions.

b. In about how many years will human teeth be 90% of their present size?

c. What will be our descendants' tooth size 20,000 years from now (as a percentage of our present tooth size)?

24. Atmospheric pressure The earth's atmospheric pressure p is often modeled by assuming that the rate dp/dh at which p changes with the altitude h above sea level is proportional to p. Suppose that the pressure at sea level is 1013 hectopascals and that the pressure at an altitude of 20 km is 90 hectopascals.

a. Solve the initial value problem

Differential equation: $dp/dh = kp$ (k a constant)
Initial condition: $p = p_0$ when $h = 0$

to express p in terms of h. Determine the values of p_0 and k from the given altitude-pressure data.

b. What is the atmospheric pressure at $h = 50$ km?

c. At what altitude does the pressure equal 900 hectopascals?

25. First-order chemical reactions In some chemical reactions, the rate at which the amount of a substance changes with time is proportional to the amount present. For the change of δ-glucono lactone into gluconic acid, for example,

$$\frac{dy}{dt} = -0.6y$$

when t is measured in hours. If there are 100 grams of δ-glucono lactone present when $t = 0$, how many grams will be left after the first hour?

26. The inversion of sugar The processing of raw sugar has a step called "inversion" that changes the sugar's molecular structure. Once the process has begun, the rate of change of the amount of raw sugar is proportional to the amount of raw sugar remaining. If 1000 kg of raw sugar reduces to 800 kg of raw sugar during the first 10 hours, how much raw sugar will remain after another 14 hours?

27. Working underwater The intensity $L(x)$ of light x meters beneath the surface of the ocean satisfies the differential equation

$$\frac{dL}{dx} = -kL.$$

As a diver, you know from experience that diving to 6 meters in the Caribbean Sea cuts the intensity in half. You cannot work without artificial light when the intensity falls below one-tenth of the surface value. About how deep can you expect to work without artificial light?

28. Voltage in a discharging capacitor Suppose that electricity is draining from a capacitor at a rate that is proportional to the voltage V across its terminals and that, if t is measured in seconds,

$$\frac{dV}{dt} = -\frac{1}{40}V.$$

Solve this equation for V, using V_0 to denote the value of V when $t = 0$. How long will it take the voltage to drop to 10% of its original value?

29. Cholera bacteria Suppose that the bacteria in a colony can grow unchecked, by the law of exponential change. The colony starts with 1 bacterium and doubles every half-hour. How many bacteria will the colony contain at the end of 24 hours? (Under favorable laboratory conditions, the number of cholera bacteria can double every 30 min. In an infected person, many bacteria are destroyed, but this example helps explain why a person who feels well in the morning may be dangerously ill by evening.)

30. Growth of bacteria A colony of bacteria is grown under ideal conditions in a laboratory so that the population increases exponentially with time. At the end of 3 hours there are 10,000 bacteria. At the end of 5 hours there are 40,000. How many bacteria were present initially?

31. The incidence of a disease (*Continuation of Example 4.*) Suppose that in any given year the number of cases can be reduced by 25% instead of 20%.

a. How long will it take to reduce the number of cases to 1000?

b. How long will it take to eradicate the disease, that is, reduce the number of cases to less than 1?

32. Drug concentration An antibiotic is administered intravenously into the bloodstream at a constant rate r. As the drug flows through the patient's system and acts on the infection that is present, it is removed from the bloodstream at a rate proportional to the amount in the bloodstream at that time. Since the amount of blood in the patient is constant, this means that the concentration $y = y(t)$ of the antibiotic in the bloodstream can be modeled by the differential equation

$$\frac{dy}{dt} = r - ky, \quad k > 0 \text{ and constant.}$$

a. If $y(0) = y_0$, find the concentration $y(t)$ at any time t.

b. Assume that $y_0 < (r/k)$ and find $\lim_{y\to\infty} y(t)$. Sketch the solution curve for the concentration.

33. Endangered species Biologists consider a species of animal or plant to be endangered if it is expected to become extinct within 20 years. If a certain species of wildlife is counted to have 1147 members at the present time, and the population has been steadily declining exponentially at an annual rate averaging 39% over the past 7 years, do you think the species is endangered? Explain your answer.

34. The U.S. population The U.S. Census Bureau keeps a running clock totaling the U.S. population. On September 20, 2012, the total was increasing at the rate of 1 person every 12 s. The population figure for 8:11 P.M. EST on that day was 314,419,198.

a. Assuming exponential growth at a constant rate, find the rate constant for the population's growth (people per 365-day year).

b. At this rate, what will the U.S. population be at 8:11 P.M. EST on September 20, 2019?

35. Oil depletion Suppose the amount of oil pumped from one of the canyon wells in Whittier, California, decreases at the continuous rate of 10% per year. When will the well's output fall to one-fifth of its present value?

36. Continuous price discounting To encourage buyers to place 100-unit orders, your firm's sales department applies a continuous discount that makes the unit price a function $p(x)$ of the number of units x ordered. The discount decreases the price at the rate of \$0.01 per unit ordered. The price per unit for a 100-unit order is $p(100) = \$20.09$.

a. Find $p(x)$ by solving the following initial value problem:

$$\text{Differential equation:} \quad \frac{dp}{dx} = -\frac{1}{100}p$$

$$\text{Initial condition:} \quad p(100) = 20.09.$$

b. Find the unit price $p(10)$ for a 10-unit order and the unit price $p(90)$ for a 90-unit order.

c. The sales department has asked you to find out if it is discounting so much that the firm's revenue, $r(x) = x \cdot p(x)$, will actually be less for a 100-unit order than, say, for a 90-unit order. Reassure them by showing that r has its maximum value at $x = 100$.

d. Graph the revenue function $r(x) = xp(x)$ for $0 \le x \le 200$.

37. Plutonium-239 The half-life of the plutonium isotope is 24,360 years. If 10 g of plutonium is released into the atmosphere by a nuclear accident, how many years will it take for 80% of the isotope to decay?

38. Polonium-210 The half-life of polonium is 139 days, but your sample will not be useful to you after 95% of the radioactive nuclei present on the day the sample arrives has disintegrated. For about how many days after the sample arrives will you be able to use the polonium?

39. The mean life of a radioactive nucleus Physicists using the radioactivity equation $y = y_0e^{-kt}$ call the number $1/k$ the *mean life* of a radioactive nucleus. The mean life of a radon nucleus is about $1/0.18 = 5.6$ days. The mean life of a carbon-14 nucleus is more than 8000 years. Show that 95% of the radioactive nuclei originally present in a sample will disintegrate within three mean lifetimes, i.e., by time $t = 3/k$. Thus, the mean life of a nucleus gives a quick way to estimate how long the radioactivity of a sample will last.

40. Californium-252 What costs \$60 million per gram and can be used to treat brain cancer, analyze coal for its sulfur content, and detect explosives in luggage? The answer is californium-252, a radioactive isotope so rare that only 8 g of it have been made in the Western world since its discovery by Glenn Seaborg in 1950. The half-life of the isotope is 2.645 years—long enough for a useful service life and short enough to have a high radioactivity per unit mass. One microgram of the isotope releases 170 million neutrons per minute.

a. What is the value of k in the decay equation for this isotope?

b. What is the isotope's mean life? (See Exercise 39.)

c. How long will it take 95% of a sample's radioactive nuclei to disintegrate?

41. Cooling soup Suppose that a cup of soup cooled from 90°C to 60°C after 10 min in a room whose temperature was 20°C. Use Newton's Law of Cooling to answer the following questions.

a. How much longer would it take the soup to cool to 35°C?

b. Instead of being left to stand in the room, the cup of 90°C soup is put in a freezer whose temperature is −15°C. How long will it take the soup to cool from 90°C to 35°C?

42. A beam of unknown temperature An aluminum beam was brought from the outside cold into a machine shop where the temperature was held at 18°C. After 10 min, the beam warmed to 2°C and after another 10 min it was 10°C. Use Newton's Law of Cooling to estimate the beam's initial temperature.

43. Surrounding medium of unknown temperature A pan of warm water (46°C) was put in a refrigerator. Ten minutes later, the water's temperature was 39°C; 10 min after that, it was 33°C. Use Newton's Law of Cooling to estimate how cold the refrigerator was.

44. Silver cooling in air The temperature of an ingot of silver is 60°C above room temperature right now. Twenty minutes ago, it was 70°C above room temperature. How far above room temperature will the silver be

a. 15 min from now?

b. 2 hours from now?

c. When will the silver be 10°C above room temperature?

45. The age of Crater Lake The charcoal from a tree killed in the volcanic eruption that formed Crater Lake in Oregon contained 44.5% of the carbon-14 found in living matter. About how old is Crater Lake?

46. The sensitivity of carbon-14 dating to measurement To see the effect of a relatively small error in the estimate of the amount of carbon-14 in a sample being dated, consider this hypothetical situation:

a. A bone fragment found in central Illinois in the year 2000 contains 17% of its original carbon-14 content. Estimate the year the animal died.

b. Repeat part (a), assuming 18% instead of 17%.

c. Repeat part (a), assuming 16% instead of 17%.

47. Carbon-14 The oldest known frozen human mummy, discovered in the Schnalstal glacier of the Italian Alps in 1991 and called *Otzi*, was found wearing straw shoes and a leather coat with goat fur, and holding a copper ax and stone dagger. It was estimated that Otzi died 5000 years before he was discovered in the melting glacier. How much of the original carbon-14 remained in Otzi at the time of his discovery?

48. Art forgery A painting attributed to Vermeer (1632–1675), which should contain no more than 96.2% of its original carbon-14, contains 99.5% instead. About how old is the forgery?

49. Lascaux Cave paintings Prehistoric cave paintings of animals were found in the Lascaux Cave in France in 1940. Scientific analysis revealed that only 15% of the original carbon-14 in the paintings remained. What is an estimate of the age of the paintings?

50. Incan mummy The frozen remains of a young Incan woman were discovered by archeologist Johan Reinhard on Mt. Ampato in Peru during an expedition in 1995.

a. How much of the original carbon-14 was present if the estimated age of the "Ice Maiden" was 500 years?

b. If a 1% error can occur in the carbon-14 measurement, what is the oldest possible age for the Ice Maiden?

7.5 Indeterminate Forms and L'Hôpital's Rule

HISTORICAL BIOGRAPHY
Guillaume François Antoine de l'Hôpital (1661–1704)
www.goo.gl/nMJIKA
Johann Bernoulli (1667–1748)
www.goo.gl/70BgHS

Expressions such as "$0/0$" and "∞/∞" look something like ordinary numbers. We say that they have the *form* of a number. But values cannot be assigned to them in a way that is consistent with the usual rules to add and multiply numbers. We call these expressions "indeterminate forms." Although they are not numbers, these indeterminate forms play a useful role in summarizing the limiting behavior of a function.

John (Johann) Bernoulli discovered a rule using derivatives to calculate limits of fractions whose numerators and denominators both approach zero or ∞. The rule is known today as **l'Hôpital's Rule**, after Guillaume de l'Hôpital, a French nobleman who wrote the earliest introductory differential calculus text, where the rule first appeared in print.

Indeterminate Form 0/0

If we want to know how the function

$$F(x) = \frac{3x - \sin x}{x}$$

behaves *near* $x = 0$ (where it is undefined), we can examine the limit of $F(x)$ as $x \to 0$. We cannot apply the Quotient Rule for limits (Theorem 1 of Chapter 2) because the limit of the denominator is 0. Moreover, in this case, *both* the numerator and denominator

approach 0, and $0/0$ is undefined. Such limits may or may not exist in general, but the limit does exist for the function $F(x)$ under discussion by applying l'Hôpital's Rule, as we will see in Example 1d.

If the continuous functions $f(x)$ and $g(x)$ are both zero at $x = a$, then

$$\lim_{x \to a} \frac{f(x)}{g(x)}$$

cannot be found by substituting $x = a$. The substitution produces $0/0$, a meaningless expression, which we cannot evaluate. We use $0/0$ as a notation for an expression that does not have a numerical value, known as an **indeterminate form**. Other meaningless expressions often occur, such as ∞/∞, $\infty \cdot 0$, $\infty - \infty$, 0^0, and 1^∞, which cannot be evaluated in a consistent way; these are called indeterminate forms as well. Sometimes, but not always, limits that lead to indeterminate forms may be found by cancelation, rearrangement of terms, or other algebraic manipulations. This was our experience in Chapter 2. It took considerable analysis in Section 2.4 to find $\lim_{x \to 0} (\sin x)/x$. But we have had success with the limit

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a},$$

from which we calculate derivatives and which produces the indeterminant form $0/0$ if we attempt to substitute $x = a$. L'Hôpital's Rule enables us to draw on our success with derivatives to evaluate limits for which substitution leads to indeterminate forms.

THEOREM 5—L'Hôpital's Rule

Suppose that $f(a) = g(a) = 0$, that f and g are differentiable on an open interval I containing a, and that $g'(x) \neq 0$ on I if $x \neq a$. Then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

assuming that the limit on the right side of this equation exists.

We give a proof of Theorem 5 at the end of this section.

Caution

To apply l'Hôpital's Rule to f/g, divide the derivative of f by the derivative of g. Do not fall into the trap of taking the derivative of f/g. The quotient to use is f'/g', not $(f/g)'$.

EXAMPLE 1 The following limits involve $0/0$ indeterminate forms, so we apply l'Hôpital's Rule. In some cases, it must be applied repeatedly.

(a) $$\lim_{x \to 0} \frac{3x - \sin x}{x} = \lim_{x \to 0} \frac{3 - \cos x}{1} = \left. \frac{3 - \cos x}{1} \right|_{x=0} = 2$$

(b) $$\lim_{x \to 0} \frac{\sqrt{1 + x} - 1}{x} = \lim_{x \to 0} \frac{\frac{1}{2\sqrt{1 + x}}}{1} = \frac{1}{2}$$

(c) $$\lim_{x \to 0} \frac{\sqrt{1 + x} - 1 - x/2}{x^2}$$ $\frac{0}{0}$; apply l'Hôpital's Rule.

$$= \lim_{x \to 0} \frac{(1/2)(1 + x)^{-1/2} - 1/2}{2x}$$ Still $\frac{0}{0}$; apply l'Hôpital's Rule again.

$$= \lim_{x \to 0} \frac{-(1/4)(1 + x)^{-3/2}}{2} = -\frac{1}{8}$$ Not $\frac{0}{0}$; limit is found.

(d) $\lim_{x\to 0} \dfrac{x - \sin x}{x^3}$ $\quad \frac{0}{0}$; apply l'Hôpital's Rule.

$= \lim_{x\to 0} \dfrac{1 - \cos x}{3x^2}$ Still $\frac{0}{0}$; apply l'Hôpital's Rule again.

$= \lim_{x\to 0} \dfrac{\sin x}{6x}$ Still $\frac{0}{0}$; apply l'Hôpital's Rule again.

$= \lim_{x\to 0} \dfrac{\cos x}{6} = \dfrac{1}{6}$ Not $\frac{0}{0}$; limit is found. ■

Here is a summary of the procedure we followed in Example 1.

Using L'Hôpital's Rule

To find

$$\lim_{x\to a} \frac{f(x)}{g(x)}$$

by l'Hôpital's Rule, we continue to differentiate f and g, so long as we still get the form $0/0$ at $x = a$. But as soon as one or the other of these derivatives is different from zero at $x = a$ we stop differentiating. L'Hôpital's Rule does not apply when either the numerator or denominator has a finite nonzero limit.

EXAMPLE 2 Be careful to apply l'Hôpital's Rule correctly:

$$\lim_{x\to 0} \frac{1 - \cos x}{x + x^2} \qquad \frac{0}{0}$$

$$= \lim_{x\to 0} \frac{\sin x}{1 + 2x} \qquad \text{Not } \frac{0}{0}$$

It is tempting to try to apply l'Hôpital's Rule again, which would result in

$$\lim_{x\to 0} \frac{\cos x}{2} = \frac{1}{2},$$

but this is not the correct limit. L'Hôpital's Rule can be applied only to limits that give indeterminate forms, and $\lim_{x\to 0}(\sin x)/(1 + 2x)$ does not give an indeterminate form. Instead, this limit is $0/1 = 0$, and the correct answer for the original limit is 0. ■

L'Hôpital's Rule applies to one-sided limits as well.

EXAMPLE 3 In this example the one-sided limits are different.

(a) $\lim_{x\to 0^+} \dfrac{\sin x}{x^2}$ $\quad \frac{0}{0}$

$= \lim_{x\to 0^+} \dfrac{\cos x}{2x} = \infty$ Positive for $x > 0$

(b) $\lim_{x\to 0^-} \dfrac{\sin x}{x^2}$ $\quad \frac{0}{0}$

$= \lim_{x\to 0^-} \dfrac{\cos x}{2x} = -\infty$ Negative for $x < 0$ ■

Recall that ∞ and $+\infty$ mean the same thing.

Indeterminate Forms ∞/∞, $\infty \cdot 0$, $\infty - \infty$

Sometimes when we try to evaluate a limit as $x \to a$ by substituting $x = a$ we get an indeterminant form like ∞/∞, $\infty \cdot 0$, or $\infty - \infty$, instead of $0/0$. We first consider the form ∞/∞.

More advanced treatments of calculus prove that l'Hôpital's Rule applies to the indeterminate form ∞/∞, as well as to $0/0$. If $f(x) \to \pm\infty$ and $g(x) \to \pm\infty$ as $x \to a$, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists. In the notation $x \to a$, a may be either finite or infinite. Moreover, $x \to a$ may be replaced by the one-sided limits $x \to a^+$ or $x \to a^-$.

EXAMPLE 4 Find the limits of these ∞/∞ forms:

(a) $\lim_{x\to\pi/2} \dfrac{\sec x}{1 + \tan x}$ **(b)** $\lim_{x\to\infty} \dfrac{\ln x}{2\sqrt{x}}$ **(c)** $\lim_{x\to\infty} \dfrac{e^x}{x^2}$.

Solution

(a) The numerator and denominator are discontinuous at $x = \pi/2$, so we investigate the one-sided limits there. To apply l'Hôpital's Rule, we can choose I to be any open interval with $x = \pi/2$ as an endpoint.

$$\lim_{x\to(\pi/2)^-} \frac{\sec x}{1 + \tan x} \qquad \frac{\infty}{\infty} \text{ from the left so we apply l'Hôpital's Rule.}$$

$$= \lim_{x\to(\pi/2)^-} \frac{\sec x \tan x}{\sec^2 x} = \lim_{x\to(\pi/2)^-} \sin x = 1$$

The right-hand limit is 1 also, with $(-\infty)/(-\infty)$ as the indeterminate form. Therefore, the two-sided limit is equal to 1.

(b) $\lim_{x\to\infty} \dfrac{\ln x}{2\sqrt{x}} = \lim_{x\to\infty} \dfrac{1/x}{1/\sqrt{x}} = \lim_{x\to\infty} \dfrac{1}{\sqrt{x}} = 0$ $\qquad \dfrac{1/x}{1/\sqrt{x}} = \dfrac{\sqrt{x}}{x} = \dfrac{1}{\sqrt{x}}$

(c) $\lim_{x\to\infty} \dfrac{e^x}{x^2} = \lim_{x\to\infty} \dfrac{e^x}{2x} = \lim_{x\to\infty} \dfrac{e^x}{2} = \infty$ ■

Next we turn our attention to the indeterminate forms $\infty \cdot 0$ and $\infty - \infty$. Sometimes these forms can be handled by using algebra to convert them to a $0/0$ or ∞/∞ form. Here again we do not mean to suggest that $\infty \cdot 0$ or $\infty - \infty$ is a number. They are only notations for functional behaviors when considering limits. Here are examples of how we might work with these indeterminate forms.

EXAMPLE 5 Find the limits of these $\infty \cdot 0$ forms:

(a) $\lim_{x\to\infty} \left(x \sin \dfrac{1}{x}\right)$ **(b)** $\lim_{x\to 0^+} \sqrt{x} \ln x$

Solution

(a) $\lim_{x\to\infty} \left(x \sin \dfrac{1}{x}\right) = \lim_{h\to 0^+} \left(\dfrac{1}{h} \sin h\right) = \lim_{h\to 0^+} \dfrac{\sin h}{h} = 1$ $\qquad \infty \cdot 0$; let $h = 1/x$.

(b) $\lim_{x\to 0^+} \sqrt{x} \ln x = \lim_{x\to 0^+} \dfrac{\ln x}{1/\sqrt{x}}$ $\qquad \infty \cdot 0$ converted to ∞/∞

$$= \lim_{x\to 0^+} \frac{1/x}{-1/2x^{3/2}} \qquad \text{l'Hôpital's Rule applied}$$

$$= \lim_{x\to 0^+} \left(-2\sqrt{x}\right) = 0$$ ■

EXAMPLE 6 Find the limit of this $\infty - \infty$ form:

$$\lim_{x\to 0} \left(\frac{1}{\sin x} - \frac{1}{x}\right).$$

Solution If $x \to 0^+$, then $\sin x \to 0^+$ and

$$\frac{1}{\sin x} - \frac{1}{x} \to \infty - \infty.$$

Similarly, if $x \to 0^-$, then $\sin x \to 0^-$ and

$$\frac{1}{\sin x} - \frac{1}{x} \to -\infty - (-\infty) = -\infty + \infty.$$

Neither form reveals what happens in the limit. To find out, we first combine the fractions:

$$\frac{1}{\sin x} - \frac{1}{x} = \frac{x - \sin x}{x \sin x}. \qquad \text{Common denominator is } x \sin x.$$

Then we apply l'Hôpital's Rule to the result:

$$\lim_{x \to 0}\left(\frac{1}{\sin x} - \frac{1}{x}\right) = \lim_{x \to 0} \frac{x - \sin x}{x \sin x} \qquad \frac{0}{0}$$

$$= \lim_{x \to 0} \frac{1 - \cos x}{\sin x + x \cos x} \qquad \text{Still } \frac{0}{0}$$

$$= \lim_{x \to 0} \frac{\sin x}{2\cos x - x \sin x} = \frac{0}{2} = 0.$$ ■

Indeterminate Powers

Limits that lead to the indeterminate forms 1^∞, 0^0, and ∞^0 can sometimes be handled by first taking the logarithm of the function. We use l'Hôpital's Rule to find the limit of the logarithm expression and then exponentiate the result to find the original function limit. This procedure is justified by the continuity of the exponential function and Theorem 10 in Section 2.5, and it is formulated as follows. (The formula is also valid for one-sided limits.)

If $\lim_{x \to a} \ln f(x) = L$, then

$$\lim_{x \to a} f(x) = \lim_{x \to a} e^{\ln f(x)} = e^L.$$

Here a may be either finite or infinite.

EXAMPLE 7 Apply l'Hôpital's Rule to show that $\lim_{x \to 0^+} (1 + x)^{1/x} = e$.

Solution The limit leads to the indeterminate form 1^∞. We let $f(x) = (1 + x)^{1/x}$ and find $\lim_{x \to 0^+} \ln f(x)$. Since

$$\ln f(x) = \ln (1 + x)^{1/x} = \frac{1}{x} \ln (1 + x),$$

l'Hôpital's Rule now applies to give

$$\lim_{x \to 0^+} \ln f(x) = \lim_{x \to 0^+} \frac{\ln (1 + x)}{x} \qquad \frac{0}{0}$$

$$= \lim_{x \to 0^+} \frac{\dfrac{1}{1 + x}}{1} \qquad \text{l'Hôpital's Rule applied}$$

$$= \frac{1}{1} = 1.$$

Therefore, $\lim_{x \to 0^+} (1 + x)^{1/x} = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} e^{\ln f(x)} = e^1 = e.$ ■

EXAMPLE 8 Find $\lim_{x\to\infty} x^{1/x}$.

Solution The limit leads to the indeterminate form ∞^0. We let $f(x) = x^{1/x}$ and find $\lim_{x\to\infty} \ln f(x)$. Since

$$\ln f(x) = \ln x^{1/x} = \frac{\ln x}{x},$$

l'Hôpital's Rule gives

$$\lim_{x\to\infty} \ln f(x) = \lim_{x\to\infty} \frac{\ln x}{x} \qquad \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty} \frac{1/x}{1} \qquad \text{l'Hôpital's Rule applied}$$

$$= \frac{0}{1} = 0.$$

Therefore $\lim_{x\to\infty} x^{1/x} = \lim_{x\to\infty} f(x) = \lim_{x\to\infty} e^{\ln f(x)} = e^0 = 1.$ ■

Proof of L'Hôpital's Rule

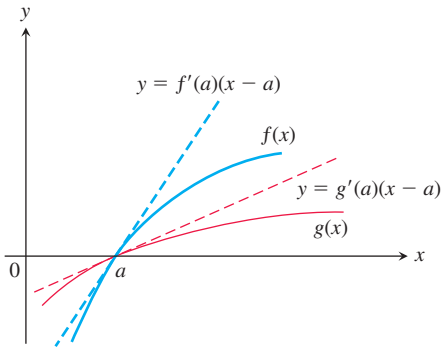


FIGURE 7.19 The two functions in l'Hôpital's Rule, graphed with their linear approximations at $x = a$.

HISTORICAL BIOGRAPHY
Augustin-Louis Cauchy (1789–1857)
www.goo.gl/ORQbxf

Before we prove l'Hôpital's Rule, we consider a special case to provide some geometric insight for its reasonableness. Consider the two functions $f(x)$ and $g(x)$ having *continuous* derivatives and satisfying $f(a) = g(a) = 0$, $g'(a) \neq 0$. The graphs of $f(x)$ and $g(x)$, together with their linearizations $y = f'(a)(x - a)$ and $y = g'(a)(x - a)$, are shown in Figure 7.19. We know that near $x = a$, the linearizations provide good approximations to the functions. In fact,

$$f(x) = f'(a)(x - a) + \varepsilon_1(x - a) \quad \text{and} \quad g(x) = g'(a)(x - a) + \varepsilon_2(x - a)$$

where $\varepsilon_1 \to 0$ and $\varepsilon_2 \to 0$ as $x \to a$. So, as Figure 7.19 suggests,

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(a)(x - a) + \varepsilon_1(x - a)}{g'(a)(x - a) + \varepsilon_2(x - a)}$$

$$= \lim_{x\to a} \frac{f'(a) + \varepsilon_1}{g'(a) + \varepsilon_2} = \frac{f'(a)}{g'(a)} \qquad g'(a) \neq 0$$

$$= \lim_{x\to a} \frac{f'(x)}{g'(x)}, \qquad \text{Continuous derivatives}$$

as asserted by l'Hôpital's Rule. We now proceed to a proof of the rule based on the more general assumptions stated in Theorem 5, which do not require that $g'(a) \neq 0$ and that the two functions have *continuous* derivatives.

The proof of l'Hôpital's Rule is based on Cauchy's Mean Value Theorem, an extension of the Mean Value Theorem that involves two functions instead of one. We prove Cauchy's Theorem first and then show how it leads to l'Hôpital's Rule.

When $g(x) = x$, Theorem 6 is the Mean Value Theorem in Chapter 4.

THEOREM 6—Cauchy's Mean Value Theorem
Suppose functions f and g are continuous on $[a, b]$ and differentiable throughout (a, b) and also suppose $g'(x) \neq 0$ throughout (a, b). Then there exists a number c in (a, b) at which

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Proof We apply the Mean Value Theorem of Section 4.2 twice. First we use it to show that $g(a) \neq g(b)$. For if $g(b)$ did equal $g(a)$, then the Mean Value Theorem would give

$$g'(c) = \frac{g(b) - g(a)}{b - a} = 0$$

for some c between a and b, which cannot happen because $g'(x) \neq 0$ in (a, b).

We next apply the Mean Value Theorem to the function

$$F(x) = f(x) - f(a) - \frac{f(b) - f(a)}{g(b) - g(a)}[\,g(x) - g(a)\,].$$

This function is continuous and differentiable where f and g are, and $F(b) = F(a) = 0$. Therefore, there is a number c between a and b for which $F'(c) = 0$. When expressed in terms of f and g, this equation becomes

$$F'(c) = f'(c) - \frac{f(b) - f(a)}{g(b) - g(a)}[\,g'(c)\,] = 0$$

so that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

■

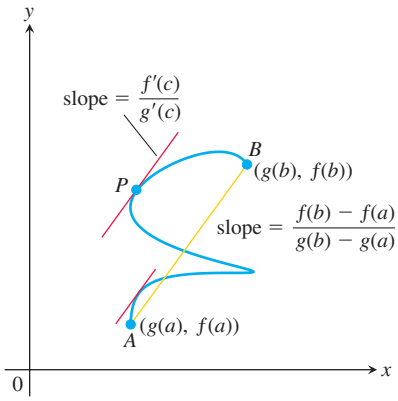


FIGURE 7.20 There is at least one point P on the curve C for which the slope of the tangent line to the curve at P is the same as the slope of the secant line joining the points $A(g(a), f(a))$ and $B(g(b), f(b))$.

Cauchy's Mean Value Theorem has a geometric interpretation for a general winding curve C in the plane joining the two points $A = (g(a), f(a))$ and $B = (g(b), f(b))$. In Chapter 10 you will learn how the curve C can be formulated to show that there is at least one point P on the curve for which the tangent line to the curve at P is parallel to the secant line joining the points A and B. The slope of that tangent line turns out to be the quotient f'/g' evaluated at the number c in the interval (a, b), which is the left-hand side of the equation in Theorem 6. Because the slope of the secant line joining A and B is

$$\frac{f(b) - f(a)}{g(b) - g(a)},$$

the equation in Cauchy's Mean Value Theorem says that the slope of the tangent line equals the slope of the secant line. This geometric interpretation is shown in Figure 7.20. Notice from the figure that it is possible for more than one point on the curve C to have a tangent line that is parallel to the secant line joining A and B.

Proof of l'Hôpital's Rule We first establish the limit equation for the case $x \to a^+$. The method needs almost no change to apply to $x \to a^-$, and the combination of these two cases establishes the result.

Suppose that x lies to the right of a. Then $g'(x) \neq 0$, and we can apply Cauchy's Mean Value Theorem to the closed interval from a to x. This step produces a number c between a and x such that

$$\frac{f'(c)}{g'(c)} = \frac{f(x) - f(a)}{g(x) - g(a)}.$$

But $f(a) = g(a) = 0$, so

$$\frac{f'(c)}{g'(c)} = \frac{f(x)}{g(x)}.$$

As x approaches a, c approaches a because it always lies between a and x. Therefore,

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = \lim_{c \to a^+} \frac{f'(c)}{g'(c)} = \lim_{x \to a^+} \frac{f'(x)}{g'(x)},$$

which establishes l'Hôpital's Rule for the case where x approaches a from above. The case where x approaches a from below is proved by applying Cauchy's Mean Value Theorem to the closed interval $[\,x, a\,]$, $x < a$. ■

EXERCISES 7.5

Finding Limits in Two Ways

In Exercises 1–6, use l'Hôpital's Rule to evaluate the limit. Then evaluate the limit using a method studied in Chapter 2.

1. $\lim_{x\to -2} \dfrac{x+2}{x^2-4}$ **2.** $\lim_{x\to 0} \dfrac{\sin 5x}{x}$ **3.** $\lim_{x\to\infty} \dfrac{5x^2-3x}{7x^2+1}$

4. $\lim_{x\to 1} \dfrac{x^3-1}{4x^3-x-3}$ **5.** $\lim_{x\to 0} \dfrac{1-\cos x}{x^2}$ **6.** $\lim_{x\to\infty} \dfrac{2x^2+3x}{x^3+x+1}$

Applying l'Hôpital's Rule

Use l'Hôpital's rule to find the limits in Exercises 7–50.

7. $\lim_{x\to 2} \dfrac{x-2}{x^2-4}$ **8.** $\lim_{x\to -5} \dfrac{x^2-25}{x+5}$

9. $\lim_{t\to -3} \dfrac{t^3-4t+15}{t^2-t-12}$ **10.** $\lim_{t\to -1} \dfrac{3t^3+3}{4t^3-t+3}$

11. $\lim_{x\to\infty} \dfrac{5x^3-2x}{7x^3+3}$ **12.** $\lim_{x\to\infty} \dfrac{x-8x^2}{12x^2+5x}$

13. $\lim_{t\to 0} \dfrac{\sin t^2}{t}$ **14.** $\lim_{t\to 0} \dfrac{\sin 5t}{2t}$

15. $\lim_{x\to 0} \dfrac{8x^2}{\cos x-1}$ **16.** $\lim_{x\to 0} \dfrac{\sin x-x}{x^3}$

17. $\lim_{\theta\to \pi/2} \dfrac{2\theta-\pi}{\cos(2\pi-\theta)}$ **18.** $\lim_{\theta\to -\pi/3} \dfrac{3\theta+\pi}{\sin(\theta+(\pi/3))}$

19. $\lim_{\theta\to \pi/2} \dfrac{1-\sin\theta}{1+\cos 2\theta}$ **20.** $\lim_{x\to 1} \dfrac{x-1}{\ln x-\sin \pi x}$

21. $\lim_{x\to 0} \dfrac{x^2}{\ln(\sec x)}$ **22.** $\lim_{x\to \pi/2} \dfrac{\ln(\csc x)}{(x-(\pi/2))^2}$

23. $\lim_{t\to 0} \dfrac{t(1-\cos t)}{t-\sin t}$ **24.** $\lim_{t\to 0} \dfrac{t\sin t}{1-\cos t}$

25. $\lim_{x\to(\pi/2)^-} \left(x-\dfrac{\pi}{2}\right)\sec x$ **26.** $\lim_{x\to(\pi/2)^-} \left(\dfrac{\pi}{2}-x\right)\tan x$

27. $\lim_{\theta\to 0} \dfrac{3^{\sin\theta}-1}{\theta}$ **28.** $\lim_{\theta\to 0} \dfrac{(1/2)^\theta-1}{\theta}$

29. $\lim_{x\to 0} \dfrac{x2^x}{2^x-1}$ **30.** $\lim_{x\to 0} \dfrac{3^x-1}{2^x-1}$

31. $\lim_{x\to\infty} \dfrac{\ln(x+1)}{\log_2 x}$ **32.** $\lim_{x\to\infty} \dfrac{\log_2 x}{\log_3(x+3)}$

33. $\lim_{x\to 0^+} \dfrac{\ln(x^2+2x)}{\ln x}$ **34.** $\lim_{x\to 0^+} \dfrac{\ln(e^x-1)}{\ln x}$

35. $\lim_{y\to 0} \dfrac{\sqrt{5y+25}-5}{y}$ **36.** $\lim_{y\to 0} \dfrac{\sqrt{ay+a^2}-a}{y}, \quad a>0$

37. $\lim_{x\to\infty} (\ln 2x-\ln(x+1))$ **38.** $\lim_{x\to 0^+} (\ln x-\ln\sin x)$

39. $\lim_{x\to 0^+} \dfrac{(\ln x)^2}{\ln(\sin x)}$ **40.** $\lim_{x\to 0^+} \left(\dfrac{3x+1}{x}-\dfrac{1}{\sin x}\right)$

41. $\lim_{x\to 1^+} \left(\dfrac{1}{x-1}-\dfrac{1}{\ln x}\right)$ **42.** $\lim_{x\to 0^+} (\csc x-\cot x+\cos x)$

43. $\lim_{\theta\to 0} \dfrac{\cos\theta-1}{e^\theta-\theta-1}$ **44.** $\lim_{h\to 0} \dfrac{e^h-(1+h)}{h^2}$

45. $\lim_{t\to\infty} \dfrac{e^t+t^2}{e^t-t}$ **46.** $\lim_{x\to\infty} x^2e^{-x}$

47. $\lim_{x\to 0} \dfrac{x-\sin x}{x\tan x}$ **48.** $\lim_{x\to 0} \dfrac{(e^x-1)^2}{x\sin x}$

49. $\lim_{\theta\to 0} \dfrac{\theta-\sin\theta\cos\theta}{\tan\theta-\theta}$ **50.** $\lim_{x\to 0} \dfrac{\sin 3x-3x+x^2}{\sin x\sin 2x}$

Indeterminate Powers and Products

Find the limits in Exercises 51–66.

51. $\lim_{x\to 1^+} x^{1/(1-x)}$ **52.** $\lim_{x\to 1^+} x^{1/(x-1)}$ **53.** $\lim_{x\to\infty} (\ln x)^{1/x}$

54. $\lim_{x\to e^+} (\ln x)^{1/(x-e)}$ **55.** $\lim_{x\to 0^+} x^{-1/\ln x}$ **56.** $\lim_{x\to\infty} x^{1/\ln x}$

57. $\lim_{x\to\infty} (1+2x)^{1/(2\ln x)}$ **58.** $\lim_{x\to 0} (e^x+x)^{1/x}$ **59.** $\lim_{x\to 0^+} x^x$

60. $\lim_{x\to 0^+} \left(1+\dfrac{1}{x}\right)^x$ **61.** $\lim_{x\to\infty} \left(\dfrac{x+2}{x-1}\right)^x$ **62.** $\lim_{x\to\infty} \left(\dfrac{x^2+1}{x+2}\right)^{1/x}$

63. $\lim_{x\to 0^+} x^2\ln x$ **64.** $\lim_{x\to 0^+} x(\ln x)^2$

65. $\lim_{x\to 0^+} x\tan\left(\dfrac{\pi}{2}-x\right)$ **66.** $\lim_{x\to 0^+} \sin x\cdot\ln x$

Theory and Applications

L'Hôpital's Rule does not help with the limits in Exercises 67–74. Try it—you just keep on cycling. Find the limits some other way.

67. $\lim_{x\to\infty} \dfrac{\sqrt{9x+1}}{\sqrt{x+1}}$ **68.** $\lim_{x\to 0^+} \dfrac{\sqrt{x}}{\sqrt{\sin x}}$ **69.** $\lim_{x\to(\pi/2)^-} \dfrac{\sec x}{\tan x}$

70. $\lim_{x\to 0^+} \dfrac{\cot x}{\csc x}$ **71.** $\lim_{x\to\infty} \dfrac{2^x-3^x}{3^x+4^x}$ **72.** $\lim_{x\to -\infty} \dfrac{2^x+4^x}{5^x-2^x}$

73. $\lim_{x\to\infty} \dfrac{e^{x^2}}{xe^x}$ **74.** $\lim_{x\to 0^+} \dfrac{x}{e^{-1/x}}$

75. Which one is correct, and which one is wrong? Give reasons for your answers.

a. $\lim_{x\to 3} \dfrac{x-3}{x^2-3} = \lim_{x\to 3} \dfrac{1}{2x} = \dfrac{1}{6}$ **b.** $\lim_{x\to 3} \dfrac{x-3}{x^2-3} = \dfrac{0}{6} = 0$

76. Which one is correct, and which one is wrong? Give reasons for your answers.

a. $\lim_{x\to 0} \dfrac{x^2-2x}{x^2-\sin x} = \lim_{x\to 0} \dfrac{2x-2}{2x-\cos x}$

$= \lim_{x\to 0} \dfrac{2}{2+\sin x} = \dfrac{2}{2+0} = 1$

b. $\lim_{x\to 0} \dfrac{x^2-2x}{x^2-\sin x} = \lim_{x\to 0} \dfrac{2x-2}{2x-\cos x} = \dfrac{-2}{0-1} = 2$

77. Only one of these calculations is correct. Which one? Why are the others wrong? Give reasons for your answers.

a. $\lim_{x\to 0^+} x \ln x = 0 \cdot (-\infty) = 0$

b. $\lim_{x\to 0^+} x \ln x = 0 \cdot (-\infty) = -\infty$

c. $\lim_{x\to 0^+} x \ln x = \lim_{x\to 0^+} \dfrac{\ln x}{(1/x)} = \dfrac{-\infty}{\infty} = -1$

d. $\lim_{x\to 0^+} x \ln x = \lim_{x\to 0^+} \dfrac{\ln x}{(1/x)}$

$$= \lim_{x\to 0^+} \frac{(1/x)}{(-1/x^2)} = \lim_{x\to 0^+} (-x) = 0$$

78. Find all values of c that satisfy the conclusion of Cauchy's Mean Value Theorem for the given functions and interval.

a. $f(x) = x, \quad g(x) = x^2, \quad (a, b) = (-2, 0)$

b. $f(x) = x, \quad g(x) = x^2, \quad (a, b)$ arbitrary

c. $f(x) = x^3/3 - 4x, \quad g(x) = x^2, \quad (a, b) = (0, 3)$

79. Continuous extension Find a value of c that makes the function

$$f(x) = \begin{cases} \dfrac{9x - 3\sin 3x}{5x^3}, & x \neq 0 \\ c, & x = 0 \end{cases}$$

continuous at $x = 0$. Explain why your value of c works.

80. For what values of a and b is

$$\lim_{x\to 0}\left(\frac{\tan 2x}{x^3} + \frac{a}{x^2} + \frac{\sin bx}{x}\right) = 0?$$

T **81. $\infty - \infty$ Form**

a. Estimate the value of

$$\lim_{x\to\infty}\left(x - \sqrt{x^2 + x}\right)$$

by graphing $f(x) = x - \sqrt{x^2 + x}$ over a suitably large interval of x-values.

b. Now confirm your estimate by finding the limit with l'Hôpital's Rule. As the first step, multiply $f(x)$ by the fraction $\left(x + \sqrt{x^2 + x}\right)/\left(x + \sqrt{x^2 + x}\right)$ and simplify the new numerator.

82. Find $\lim_{x\to\infty}\left(\sqrt{x^2 + 1} - \sqrt{x}\right)$.

T **83. 0/0 Form** Estimate the value of

$$\lim_{x\to 1} \frac{2x^2 - (3x + 1)\sqrt{x} + 2}{x - 1}$$

by graphing. Then confirm your estimate with l'Hôpital's Rule.

84. This exercise explores the difference between the limit

$$\lim_{x\to\infty}\left(1 + \frac{1}{x^2}\right)^x$$

and the limit

$$\lim_{x\to\infty}\left(1 + \frac{1}{x}\right)^x = e.$$

a. Use l'Hôpital's Rule to show that

$$\lim_{x\to\infty}\left(1 + \frac{1}{x}\right)^x = e.$$

T **b.** Graph

$$f(x) = \left(1 + \frac{1}{x^2}\right)^x \quad \text{and} \quad g(x) = \left(1 + \frac{1}{x}\right)^x$$

together for $x \geq 0$. How does the behavior of f compare with that of g? Estimate the value of $\lim_{x\to\infty} f(x)$.

c. Confirm your estimate of $\lim_{x\to\infty} f(x)$ by calculating it with l'Hôpital's Rule.

85. Show that

$$\lim_{k\to\infty}\left(1 + \frac{r}{k}\right)^k = e^r.$$

86. Given that $x > 0$, find the maximum value, if any, of

a. $x^{1/x}$ **b.** x^{1/x^2} **c.** x^{1/x^n} (n a positive integer)

d. Show that $\lim_{x\to\infty} x^{1/x^n} = 1$ for every positive integer n.

87. Use limits to find horizontal asymptotes for each function.

a. $y = x \tan\left(\dfrac{1}{x}\right)$ **b.** $y = \dfrac{3x + e^{2x}}{2x + e^{3x}}$

88. Find $f'(0)$ for $f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0. \end{cases}$

T **89. The continuous extension of $(\sin x)^x$ to $[0, \pi]$**

a. Graph $f(x) = (\sin x)^x$ on the interval $0 \leq x \leq \pi$. What value would you assign to f to make it continuous at $x = 0$?

b. Verify your conclusion in part (a) by finding $\lim_{x\to 0^+} f(x)$ with l'Hôpital's Rule.

c. Returning to the graph, estimate the maximum value of f on $[0, \pi]$. About where is max f taken on?

d. Sharpen your estimate in part (c) by graphing f' in the same window to see where its graph crosses the x-axis. To simplify your work, you might want to delete the exponential factor from the expression for f' and graph just the factor that has a zero.

T **90. The function $(\sin x)^{\tan x}$** (*Continuation of Exercise 89.*)

a. Graph $f(x) = (\sin x)^{\tan x}$ on the interval $-7 \leq x \leq 7$. How do you account for the gaps in the graph? How wide are the gaps?

b. Now graph f on the interval $0 \leq x \leq \pi$. The function is not defined at $x = \pi/2$, but the graph has no break at this point. What is going on? What value does the graph appear to give for f at $x = \pi/2$? (*Hint:* Use l'Hôpital's Rule to find $\lim f$ as $x \to (\pi/2)^-$ and $x \to (\pi/2)^+$.)

c. Continuing with the graphs in part (b), find max f and min f as accurately as you can and estimate the values of x at which they are taken on.

7.6 Inverse Trigonometric Functions

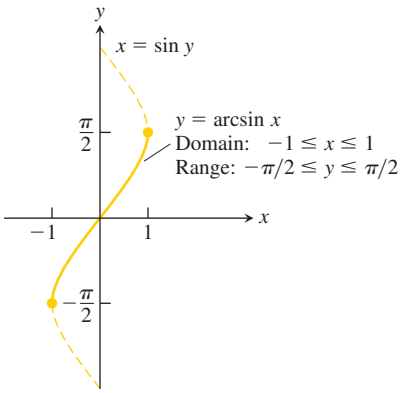


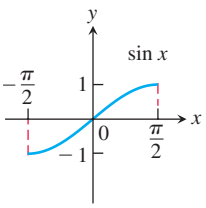
FIGURE 7.21 The graph of $y = \arcsin x$.

Inverse trigonometric functions arise when we want to calculate angles from side measurements in triangles. They also provide useful antiderivatives and appear frequently in the solutions of differential equations. This section shows how these functions are defined, graphed, and evaluated, how their derivatives are computed, and why they appear as important antiderivatives.

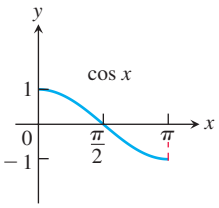
Defining the Inverse Trigonometric Functions

The six basic trigonometric functions are not one-to-one (since their values repeat periodically). However, we can restrict their domains to intervals on which they are one-to-one. The sine function increases from -1 at $x = -\pi/2$ to $+1$ at $x = \pi/2$. By restricting its domain to the interval $[-\pi/2, \pi/2]$ we make it one-to-one, so that it has an inverse which is called arcsin x (Figure 7.21). Similar domain restrictions can be applied to all six trigonometric functions.

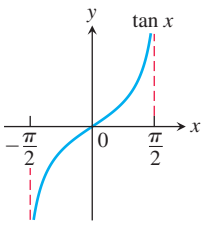
Domain restrictions that make the trigonometric functions one-to-one



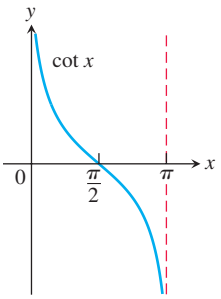
$y = \sin x$
Domain: $[-\pi/2, \pi/2]$
Range: $[-1, 1]$



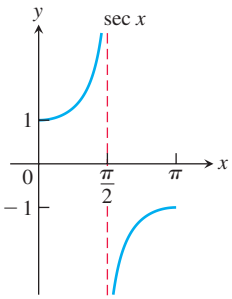
$y = \cos x$
Domain: $[0, \pi]$
Range: $[-1, 1]$



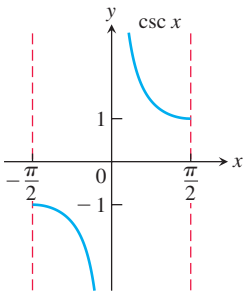
$y = \tan x$
Domain: $(-\pi/2, \pi/2)$
Range: $(-\infty, \infty)$



$y = \cot x$
Domain: $(0, \pi)$
Range: $(-\infty, \infty)$



$y = \sec x$
Domain: $[0, \pi/2) \cup (\pi/2, \pi]$
Range: $(-\infty, -1] \cup [1, \infty)$



$y = \csc x$
Domain: $[-\pi/2, 0) \cup (0, \pi/2]$
Range: $(-\infty, -1] \cup [1, \infty)$

Since these restricted functions are now one-to-one, they have inverses, which we denote by

$$y = \sin^{-1} x \quad \text{or} \quad y = \arcsin x, \qquad y = \cos^{-1} x \quad \text{or} \quad y = \arccos x$$
$$y = \tan^{-1} x \quad \text{or} \quad y = \arctan x, \qquad y = \cot^{-1} x \quad \text{or} \quad y = \operatorname{arccot} x$$
$$y = \sec^{-1} x \quad \text{or} \quad y = \operatorname{arcsec} x, \qquad y = \csc^{-1} x \quad \text{or} \quad y = \operatorname{arccsc} x$$

These equations are read "y equals the arcsine of x" or "y equals arcsin x" and so on.

Caution The -1 in the expressions for the inverse means "inverse." It does *not* mean reciprocal. For example, the *reciprocal* of $\sin x$ is $(\sin x)^{-1} = 1/\sin x = \csc x$. ●

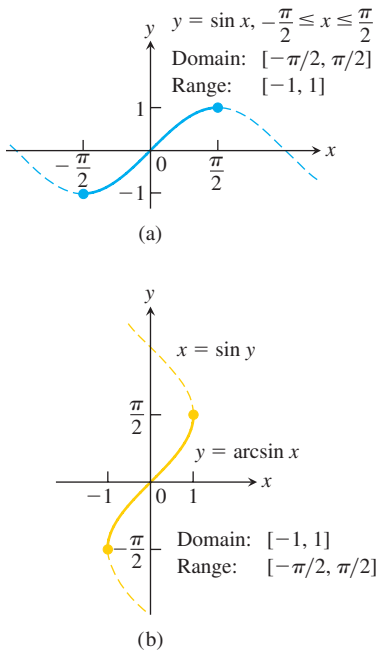


FIGURE 7.22 The graphs of (a) $y = \sin x, -\pi/2 \le x \le \pi/2$, and (b) its inverse, $y = \arcsin x$. The graph of arcsin x, obtained by reflection across the line $y = x$, is a portion of the curve $x = \sin y$.

The graphs of the six inverse trigonometric functions are obtained by reflecting the graphs of the restricted trigonometric functions through the line $y = x$. Figure 7.22b shows the graph of $y = \arcsin x$ and Figure 7.23 shows the graphs of all six functions. We now take a closer look at these functions.

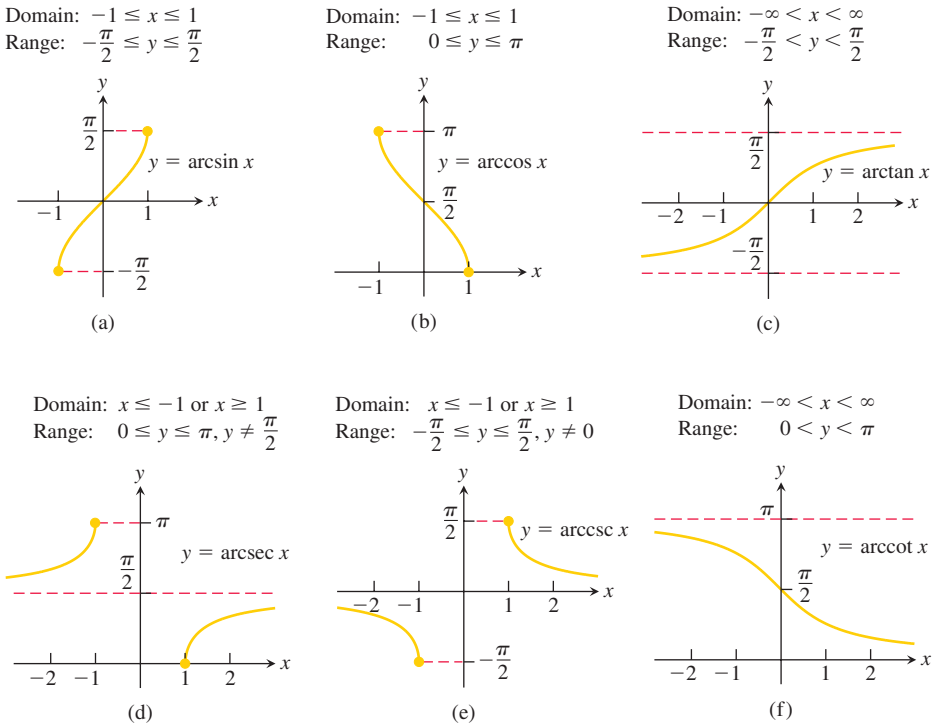


FIGURE 7.23 Graphs of the six basic inverse trigonometric functions.

The Arcsine and Arccosine Functions

We define the arcsine and arccosine as functions whose values are angles (measured in radians) that belong to restricted domains of the sine and cosine functions.

DEFINITION

y = arcsin x is the number in $[-\pi/2, \pi/2]$ for which $\sin y = x$.

y = arccos x is the number in $[0, \pi]$ for which $\cos y = x$.

The graph of $y = \arcsin x$ (Figure 7.22b) is symmetric about the origin (it lies along the graph of $x = \sin y$). The arcsine is therefore an odd function:

$$\arcsin(-x) = -\arcsin x. \tag{1}$$

The graph of $y = \arccos x$ (Figure 7.24b) has no such symmetry.

EXAMPLE 1 Evaluate **(a)** $\arcsin\left(\frac{\sqrt{3}}{2}\right)$ and **(b)** $\arccos\left(-\frac{1}{2}\right)$.

Solution

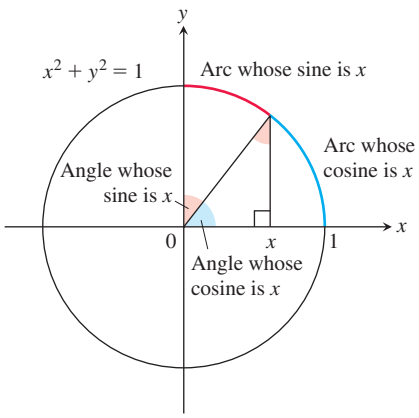
(a) We see that

$$\arcsin\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$$

because $\sin(\pi/3) = \sqrt{3}/2$ and $\pi/3$ belongs to the range $[-\pi/2, \pi/2]$ of the arcsine function. See Figure 7.25a.

The "Arc" in Arcsine and Arccosine

For a unit circle and radian angles, the arc length equation $s = r\theta$ becomes $s = \theta$, so central angles and the arcs they subtend have the same measure. If $x = \sin y$, then, in addition to being the angle whose sine is x, y is also the length of the arc on the unit circle that subtends an angle whose sine is x. So we call y "the arc whose sine is x."



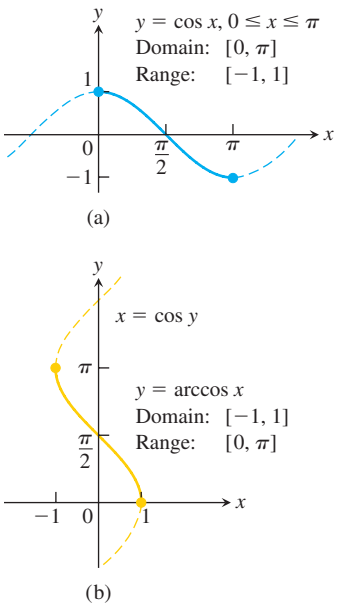


FIGURE 7.24 The graphs of (a) $y = \cos x$, $0 \le x \le \pi$, and (b) its inverse, $y = \arccos x$. The graph of $\arccos x$, obtained by reflection across the line $y = x$, is a portion of the curve $x = \cos y$.

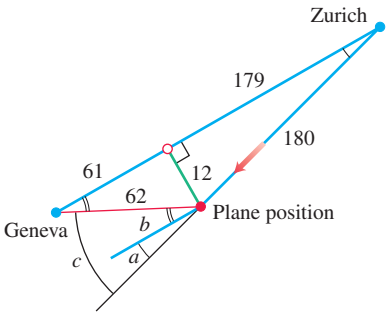


FIGURE 7.26 Diagram for drift correction (Example 2), with distances rounded to the nearest kilometers (drawing not to scale).

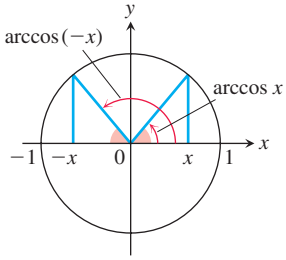


FIGURE 7.27 $\arccos x$ and $\arccos(-x)$ are supplementary angles (so their sum is π).

(b) We have

$$\arccos\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$$

because $\cos(2\pi/3) = -1/2$ and $2\pi/3$ belongs to the range $[0, \pi]$ of the arccosine function. See Figure 7.25b. ■

Using the same procedure illustrated in Example 1, we can create the following table of common values for the arcsine and arccosine functions.

x	$\arcsin x$	$\arccos x$
$\sqrt{3}/2$	$\pi/3$	$\pi/6$
$\sqrt{2}/2$	$\pi/4$	$\pi/4$
$1/2$	$\pi/6$	$\pi/3$
$-1/2$	$-\pi/6$	$2\pi/3$
$-\sqrt{2}/2$	$-\pi/4$	$3\pi/4$
$-\sqrt{3}/2$	$-\pi/3$	$5\pi/6$

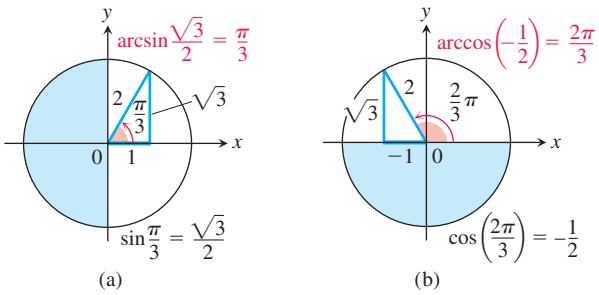


FIGURE 7.25 Values of the arcsine and arccosine functions (Example 1).

EXAMPLE 2 During a 240 km airplane flight from Zurich to Geneva, after flying 180 km the navigator determines that the plane is 12 km off course, as shown in Figure 7.26. Find the angle a for a course parallel to the original correct course, the angle b, and the drift correction angle $c = a + b$.

Solution From the Pythagorean theorem and given information, we compute an approximate hypothetical flight distance of 179 km, had the plane been flying along the original correct course (see Figure 7.26). Knowing the flight distance from Zurich to Geneva, we next calculate the remaining leg of the original course to be 61 km. Applying the Pythagorean theorem again then gives an approximate distance of 62 km from the position of the plane to Geneva. Finally, from Figure 7.26, we see that $180 \sin a = 12$ and $62 \sin b = 12$, so

$$a = \sin^{-1}\frac{12}{180} \approx 0.067 \text{ radian} \approx 3.8°$$

$$b = \sin^{-1}\frac{12}{62} \approx 0.195 \text{ radian} \approx 11.2°$$

$$c = a + b \approx 15°.$$ ■

Identities Involving Arcsine and Arccosine

As we can see from Figure 7.27, the arccosine of x satisfies the identity

$$\arccos x + \arccos(-x) = \pi, \tag{2}$$

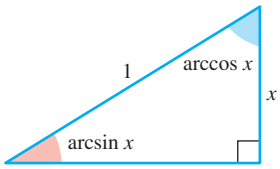


FIGURE 7.28 arcsin x and arccos x are complementary angles (so their sum is $\pi/2$).

or

$$\arccos(-x) = \pi - \arccos x. \tag{3}$$

Also, we can see from the triangle in Figure 7.28 that for $x > 0$,

$$\arcsin x + \arccos x = \pi/2. \tag{4}$$

Equation (4) holds for the other values of x in $[-1, 1]$ as well, but we cannot conclude this from the triangle in Figure 7.28. It is, however, a consequence of Equations (1) and (3) (Exercise 119).

Inverses of tan x, cot x, sec x, and csc x

The graphs of these four basic inverse trigonometric functions were shown in Figure 7.23. We obtain these graphs by reflecting the graphs of the restricted trigonometric functions through the line $y = x$. The arctangent of x is a radian angle whose tangent is x. The arccotangent of x is an angle whose cotangent is x, and so forth. The angles belong to the restricted domains of the tangent, cotangent, secant, and cosecant functions.

DEFINITIONS

$y =$ **arctan** x is the number in $(-\pi/2, \pi/2)$ for which $\tan y = x$.
$y =$ **arccot** x is the number in $(0, \pi)$ for which $\cot y = x$.
$y =$ **arcsec** x is the number in $[0, \pi/2) \cup (\pi/2, \pi]$ for which $\sec y = x$.
$y =$ **arccsc** x is the number in $[-\pi/2, 0) \cup (0, \pi/2]$ for which $\csc y = x$.

We use open or half-open intervals to avoid values for which the tangent, cotangent, secant, and cosecant functions are undefined. (See Figure 7.23.)

Just as the inverse sine and the inverse cosine functions are often written as arcsin x and arccos x instead of $\sin^{-1} x$ and $\cos^{-1} x$, we denote the other inverse trigonometric functions both by $\tan^{-1} x$, $\cot^{-1} x$, $\sec^{-1} x$ and $\csc^{-1} x$, and by arctan x, arccot x, arcsec x, and arccsc x.

The graph of $y = \arctan x$ is symmetric about the origin because it is a branch of the graph $x = \tan y$ that is symmetric about the origin (Figure 7.23c). Algebraically this means that

$$\arctan(-x) = -\arctan x;$$

the arctangent is an odd function. The graph of $y = \operatorname{arccot} x$ has no such symmetry (Figure 7.23d). Notice from Figure 7.23a that the graph of the arctangent function has two horizontal asymptotes: one at $y = \pi/2$ and the other at $y = -\pi/2$.

The inverses of the restricted forms of sec x and csc x are chosen to be the functions graphed in Figures 7.23e and 7.24f.

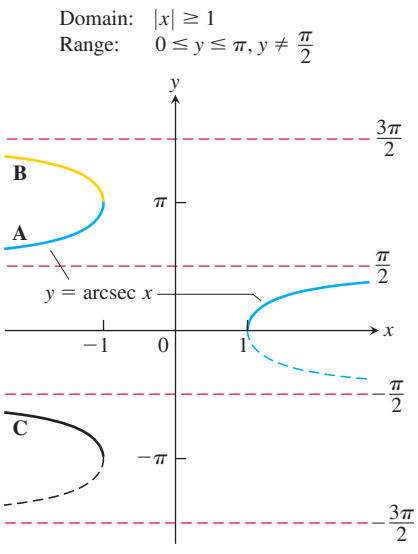


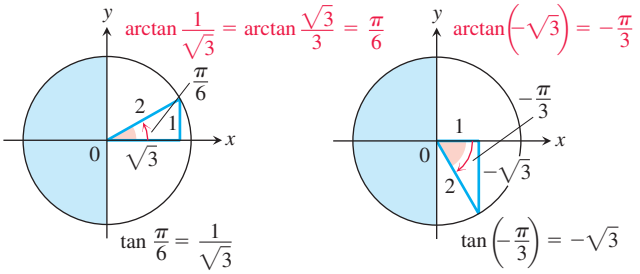
FIGURE 7.29 There are several logical choices for the left-hand branch of $y = \operatorname{arcsec} x$. With choice **A**, $\operatorname{arcsec} x = \arccos(1/x)$, a useful identity employed by many calculators.

Caution There is no general agreement about how to define arcsec x for negative values of x. We chose angles in the second quadrant between $\pi/2$ and π. This choice makes $\operatorname{arcsec} x = \arccos(1/x)$. It also makes arcsec x an increasing function on each interval of its domain. Some tables choose arcsec x to lie in $[-\pi, -\pi/2)$ for $x < 0$ and some texts choose it to lie in $[\pi, 3\pi/2)$ (Figure 7.29). These choices simplify the formula for the derivative (our formula needs absolute value signs) but fail to satisfy the computational equation $\operatorname{arcsec} x = \arccos(1/x)$. From this, we can derive the identity

$$\operatorname{arcsec} x = \arccos\left(\frac{1}{x}\right) = \frac{\pi}{2} - \arcsin\left(\frac{1}{x}\right) \tag{5}$$

by applying Equation (4). ●

EXAMPLE 3 The accompanying figures show two values of arctan x.



x	arctan x
$\sqrt{3}$	$\pi/3$
1	$\pi/4$
$\sqrt{3}/3$	$\pi/6$
$-\sqrt{3}/3$	$-\pi/6$
-1	$-\pi/4$
$-\sqrt{3}$	$-\pi/3$

The angles come from the first and fourth quadrants because the range of arctan x is $(-\pi/2, \pi/2)$.

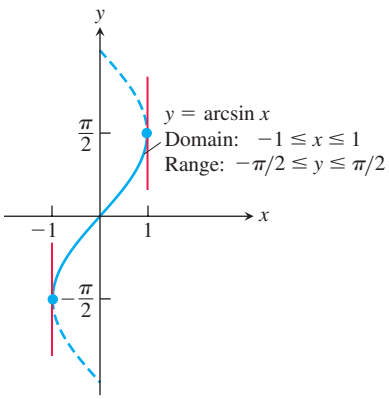


FIGURE 7.30 The graph of $y = \arcsin x$ has vertical tangents at $x = -1$ and $x = 1$.

The Derivative of $y = \arcsin u$

We know that the function $x = \sin y$ is differentiable in the interval $-\pi/2 < y < \pi/2$ and that its derivative, the cosine, is positive there. Theorem 1 in Section 7.1 therefore assures us that the inverse function $y = \arcsin x$ is differentiable throughout the interval $-1 < x < 1$. We cannot expect it to be differentiable at $x = 1$ or $x = -1$ because the tangents to the graph are vertical at these points (see Figure 7.30).

We find the derivative of $y = \arcsin x$ by applying Theorem 1 with $f(x) = \sin x$ and $f^{-1}(x) = \arcsin x$:

$$
\begin{aligned}
(f^{-1})'(x) &= \frac{1}{f'(f^{-1}(x))} && \text{Theorem 1} \\
&= \frac{1}{\cos(\arcsin x)} && f'(u) = \cos u \\
&= \frac{1}{\sqrt{1 - \sin^2(\arcsin x)}} && \cos u = \sqrt{1 - \sin^2 u} \\
&= \frac{1}{\sqrt{1 - x^2}}. && \sin(\arcsin x) = x
\end{aligned}
$$

If u is a differentiable function of x with $|u| < 1$, we apply the Chain Rule to get the general formula

$$\frac{d}{dx}(\arcsin u) = \frac{1}{\sqrt{1 - u^2}}\frac{du}{dx}, \qquad |u| < 1.$$

EXAMPLE 4 Using the Chain Rule, we calculate the derivative

$$\frac{d}{dx}(\arcsin x^2) = \frac{1}{\sqrt{1-(x^2)^2}} \cdot \frac{d}{dx}(x^2) = \frac{2x}{\sqrt{1-x^4}}.$$

The Derivative of $y = \arctan u$

We find the derivative of $y = \arctan x$ by applying Theorem 1 with $f(x) = \tan x$ and $f^{-1}(x) = \arctan x$. Theorem 1 can be applied because the derivative of $\tan x$ is positive for $-\pi/2 < x < \pi/2$:

$$\begin{aligned}
(f^{-1})'(x) &= \frac{1}{f'(f^{-1}(x))} && \text{Theorem 1} \\
&= \frac{1}{\sec^2(\arctan x)} && f'(u) = \sec^2 u \\
&= \frac{1}{1+\tan^2(\arctan x)} && \sec^2 u = 1 + \tan^2 u \\
&= \frac{1}{1+x^2}. && \tan(\arctan x) = x
\end{aligned}$$

The derivative is defined for all real numbers. If u is a differentiable function of x, we get the Chain Rule form:

$$\frac{d}{dx}(\arctan u) = \frac{1}{1+u^2}\frac{du}{dx}.$$

The Derivative of $y = \text{arcsec } u$

Since the derivative of $\sec x$ is positive for $0 < x < \pi/2$ and $\pi/2 < x < \pi$, Theorem 1 says that the inverse function $y = \text{arcsec } x$ is differentiable. Instead of applying the formula in Theorem 1 directly, we find the derivative of $y = \text{arcsec } x$, $|x| > 1$, using implicit differentiation and the Chain Rule as follows:

$$\begin{aligned}
y &= \text{arcsec } x \\
\sec y &= x && \text{Inverse function relationship} \\
\frac{d}{dx}(\sec y) &= \frac{d}{dx}x && \text{Differentiate both sides.} \\
\sec y \tan y \frac{dy}{dx} &= 1 && \text{Chain Rule} \\
\frac{dy}{dx} &= \frac{1}{\sec y \tan y}. && \text{Since } |x| > 1, y \text{ lies in } (0, \pi/2) \cup (\pi/2, \pi) \text{ and } \sec y \tan y \neq 0.
\end{aligned}$$

To express the result in terms of x, we use the relationships

$$\sec y = x \qquad \text{and} \qquad \tan y = \pm\sqrt{\sec^2 y - 1} = \pm\sqrt{x^2-1}$$

to get

$$\frac{dy}{dx} = \pm\frac{1}{x\sqrt{x^2-1}}.$$

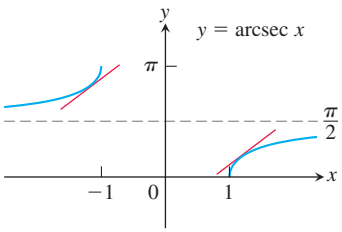


FIGURE 7.31 The slope of the curve $y = \text{arcsec } x$ is positive for both $x < -1$ and $x > 1$.

Can we do anything about the $\pm$ sign? A glance at Figure 7.31 shows that the slope of the graph $y = \text{arcsec } x$ is always positive. Thus,

$$\frac{d}{dx}\text{arcsec } x = \begin{cases} +\dfrac{1}{x\sqrt{x^2-1}} & \text{if } x > 1 \\ -\dfrac{1}{x\sqrt{x^2-1}} & \text{if } x < -1. \end{cases}$$

With the absolute value symbol, we can write a single expression that eliminates the "$\pm$" ambiguity:

$$\frac{d}{dx}\text{arcsec } x = \frac{1}{|x|\sqrt{x^2-1}}.$$

If u is a differentiable function of x with $|u| > 1$, we have the formula

$$\frac{d}{dx}(\text{arcsec } u) = \frac{1}{|u|\sqrt{u^2-1}}\frac{du}{dx}, \qquad |u| > 1.$$

EXAMPLE 5 Using the Chain Rule and derivative of the arcsecant function, we find

$$\begin{aligned} \frac{d}{dx}\text{arcsec } (5x^4) &= \frac{1}{|5x^4|\sqrt{(5x^4)^2-1}}\frac{d}{dx}(5x^4) \\ &= \frac{1}{5x^4\sqrt{25x^8-1}}(20x^3) && 5x^4 > 1 > 0 \\ &= \frac{4}{x\sqrt{25x^8-1}}. \end{aligned}$$

Derivatives of the Other Three Inverse Trigonometric Functions

We could use the same techniques to find the derivatives of the other three inverse trigonometric functions—arccosine, arccotangent, and arccosecant—but there is an easier way, thanks to the following identities.

Inverse Function–Inverse Cofunction Identities

$$\begin{aligned} \arccos x &= \pi/2 - \arcsin x \\ \text{arccot } x &= \pi/2 - \arctan x \\ \text{arccsc } x &= \pi/2 - \text{arcsec } x \end{aligned}$$

We saw the first of these identities in Equation (4). The others are derived in a similar way. It follows easily that the derivatives of the inverse cofunctions are the negatives of the derivatives of the corresponding inverse functions. For example, the derivative of $\cos^{-1} x$ is calculated as follows:

$$\begin{aligned} \frac{d}{dx}(\arccos x) &= \frac{d}{dx}\left(\frac{\pi}{2} - \arcsin x\right) && \text{Identity} \\ &= -\frac{d}{dx}(\arcsin x) \\ &= -\frac{1}{\sqrt{1-x^2}}. && \text{Derivative of arcsine} \end{aligned}$$

The derivatives of the inverse trigonometric functions are summarized in Table 7.3.

TABLE 7.3 Derivatives of the inverse trigonometric functions

1. $\dfrac{d(\arcsin u)}{dx} = \dfrac{1}{\sqrt{1-u^2}}\dfrac{du}{dx}, \quad |u| < 1$
2. $\dfrac{d(\arccos u)}{dx} = -\dfrac{1}{\sqrt{1-u^2}}\dfrac{du}{dx}, \quad |u| < 1$
3. $\dfrac{d(\arctan u)}{dx} = \dfrac{1}{1+u^2}\dfrac{du}{dx}$
4. $\dfrac{d(\operatorname{arccot} u)}{dx} = -\dfrac{1}{1+u^2}\dfrac{du}{dx}$
5. $\dfrac{d(\operatorname{arcsec} u)}{dx} = \dfrac{1}{|u|\sqrt{u^2-1}}\dfrac{du}{dx}, \quad |u| > 1$
6. $\dfrac{d(\operatorname{arccsc} u)}{dx} = -\dfrac{1}{|u|\sqrt{u^2-1}}\dfrac{du}{dx}, \quad |u| > 1$

Integration Formulas

The derivative formulas in Table 7.3 yield three useful integration formulas in Table 7.4. The formulas are readily verified by differentiating the functions on the right-hand sides. Since two notations are regularly used to represent the inverse sine function, we state these formulas using both notations, $\sin^{-1} x$ as well as $\arcsin x$, and similarly for the other inverse trigonometric functions.

TABLE 7.4 Integrals evaluated with inverse trigonometric functions

The following formulas hold for any constant $a > 0$.

1. $\displaystyle\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$ (Valid for $u^2 < a^2$)
2. $\displaystyle\int \frac{du}{a^2+u^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$ (Valid for all u)
3. $\displaystyle\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{u}{a}\right| + C$ (Valid for $|u| > a > 0$)

The derivative formulas in Table 7.3 have $a = 1$, but in most integrations $a \neq 1$, and the formulas in Table 7.4 are more useful.

EXAMPLE 6 These examples illustrate how we use Table 7.4.

(a)
$$\int_{\sqrt{2}/2}^{\sqrt{3}/2} \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x \Big]_{\sqrt{2}/2}^{\sqrt{3}/2} \qquad a = 1,\ u = x \text{ in Table 7.4, Formula 1}$$
$$= \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$$

(b)
$$\int \frac{dx}{\sqrt{3-4x^2}} = \frac{1}{2}\int \frac{du}{\sqrt{a^2-u^2}} \qquad a = \sqrt{3},\ u = 2x, \text{ and } du/2 = dx$$
$$= \frac{1}{2}\sin^{-1}\left(\frac{u}{a}\right) + C \qquad \text{Table 7.4, Formula 1}$$
$$= \frac{1}{2}\sin^{-1}\left(\frac{2x}{\sqrt{3}}\right) + C$$

(c)
$$\int \frac{dx}{\sqrt{e^{2x}-6}} = \int \frac{du/u}{\sqrt{u^2-a^2}} \qquad u = e^x,\ du = e^x,\ dx = du/e^x = du/u,\ a = \sqrt{6}$$
$$= \int \frac{du}{u\sqrt{u^2-a^2}}$$
$$= \frac{1}{a}\sec^{-1}\left|\frac{u}{a}\right| + C \qquad \text{Table 7.4, Formula 3}$$
$$= \frac{1}{\sqrt{6}}\sec^{-1}\left(\frac{e^x}{\sqrt{6}}\right) + C$$

EXAMPLE 7 Evaluate

(a) $\displaystyle\int \frac{dx}{\sqrt{4x - x^2}}$ **(b)** $\displaystyle\int \frac{dx}{4x^2 + 4x + 2}$

Solution

(a) The expression $\sqrt{4x - x^2}$ does not match any of the formulas in Table 7.4, so we first rewrite $4x - x^2$ by completing the square:

$$4x - x^2 = -(x^2 - 4x) = -(x^2 - 4x + 4) + 4 = 4 - (x - 2)^2.$$

Then we substitute $a = 2$, $u = x - 2$, and $du = dx$ to get

$$\int \frac{dx}{\sqrt{4x - x^2}} = \int \frac{dx}{\sqrt{4 - (x - 2)^2}}$$

$$= \int \frac{du}{\sqrt{a^2 - u^2}} \qquad a = 2,\ u = x - 2, \text{ and } du = dx$$

$$= \sin^{-1}\left(\frac{u}{a}\right) + C \qquad \text{Table 7.4, Formula 1}$$

$$= \sin^{-1}\left(\frac{x - 2}{2}\right) + C$$

(b) We complete the square on the binomial $4x^2 + 4x$:

$$4x^2 + 4x + 2 = 4(x^2 + x) + 2 = 4\left(x^2 + x + \frac{1}{4}\right) + 2 - \frac{4}{4}$$

$$= 4\left(x + \frac{1}{2}\right)^2 + 1 = (2x + 1)^2 + 1.$$

Then,

$$\int \frac{dx}{4x^2 + 4x + 2} = \int \frac{dx}{(2x + 1)^2 + 1} = \frac{1}{2}\int \frac{du}{u^2 + a^2} \qquad a = 1,\ u = 2x + 1, \text{ and } du/2 = dx$$

$$= \frac{1}{2} \cdot \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C \qquad \text{Table 7.4, Formula 2}$$

$$= \frac{1}{2}\tan^{-1}(2x + 1) + C \qquad a = 1,\ u = 2x + 1$$

EXERCISES 7.6

Common Values

Use reference triangles like those in Examples 1 and 3 to find the angles in Exercises 1–8.

1. a. $\tan^{-1} 1$ **b.** $\arctan\left(-\sqrt{3}\right)$ **c.** $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

2. a. $\arctan(-1)$ **b.** $\tan^{-1}\sqrt{3}$ **c.** $\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

3. a. $\sin^{-1}\left(\frac{-1}{2}\right)$ **b.** $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ **c.** $\arcsin\left(\frac{-\sqrt{3}}{2}\right)$

4. a. $\sin^{-1}\left(\frac{1}{2}\right)$ **b.** $\arcsin\left(\frac{-1}{\sqrt{2}}\right)$ **c.** $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

5. a. $\cos^{-1}\left(\frac{1}{2}\right)$ **b.** $\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$ **c.** $\arccos\left(\frac{\sqrt{3}}{2}\right)$

6. a. $\csc^{-1}\sqrt{2}$ **b.** $\operatorname{arccsc}\left(\frac{-2}{\sqrt{3}}\right)$ **c.** $\csc^{-1} 2$

7. a. $\operatorname{arcsec}\left(-\sqrt{2}\right)$ **b.** $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ **c.** $\sec^{-1}(-2)$

8. a. $\cot^{-1}(-1)$ **b.** $\operatorname{arccot}\left(\sqrt{3}\right)$ **c.** $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

Evaluations

Find the values in Exercises 9–12.

9. $\sin\left(\cos^{-1}\left(\frac{\sqrt{2}}{2}\right)\right)$

10. $\sec\left(\arccos\frac{1}{2}\right)$

11. $\tan\left(\arcsin\left(-\frac{1}{2}\right)\right)$

12. $\cot\left(\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)$

Limits

Find the limits in Exercises 13–20. (If in doubt, look at the function's graph.)

13. $\lim_{x\to 1^-} \sin^{-1} x$

14. $\lim_{x\to -1^+} \cos^{-1} x$

15. $\lim_{x\to\infty} \tan^{-1} x$

16. $\lim_{x\to -\infty} \tan^{-1} x$

17. $\lim_{x\to\infty} \sec^{-1} x$

18. $\lim_{x\to -\infty} \sec^{-1} x$

19. $\lim_{x\to\infty} \csc^{-1} x$

20. $\lim_{x\to -\infty} \csc^{-1} x$

Finding Derivatives

In Exercises 21–42, find the derivative of y with respect to the appropriate variable.

21. $y = \cos^{-1}(x^2)$

22. $y = \cos^{-1}(1/x)$

23. $y = \arcsin\sqrt{2}\,t$

24. $y = \sin^{-1}(1 - t)$

25. $y = \sec^{-1}(2s + 1)$

26. $y = \sec^{-1} 5s$

27. $y = \csc^{-1}(x^2 + 1),\ x > 0$

28. $y = \csc^{-1}\frac{x}{2}$

29. $y = \sec^{-1}\frac{1}{t},\ 0 < t < 1$

30. $y = \arcsin\frac{3}{t^2}$

31. $y = \operatorname{arccot}\sqrt{t}$

32. $y = \cot^{-1}\sqrt{t-1}$

33. $y = \ln(\tan^{-1} x)$

34. $y = \tan^{-1}(\ln x)$

35. $y = \csc^{-1}(e^t)$

36. $y = \arccos(e^{-t})$

37. $y = s\sqrt{1 - s^2} + \cos^{-1} s$

38. $y = \sqrt{s^2 - 1} - \sec^{-1} s$

39. $y = \tan^{-1}\sqrt{x^2 - 1} + \csc^{-1} x,\quad x > 1$

40. $y = \cot^{-1}\frac{1}{x} - \tan^{-1} x$

41. $y = x\arcsin x + \sqrt{1 - x^2}$

42. $y = \ln(x^2 + 4) - x\tan^{-1}\left(\frac{x}{2}\right)$

For problems 43–46 use implicit differentiation to find $\frac{dy}{dx}$ at the given point P.

43. $3\tan^{-1} x + \sin^{-1} y = \frac{\pi}{4};\quad P(1,-1)$

44. $\sin^{-1}(x + y) + \cos^{-1}(x - y) = \frac{5\pi}{6};\quad P\left(0,\frac{1}{2}\right)$

45. $y\cos^{-1}(xy) = \frac{-3\sqrt{2}}{4}\pi;\quad P\left(\frac{1}{2},-\sqrt{2}\right)$

46. $16(\tan^{-1} 3y)^2 + 9(\tan^{-1} 2x)^2 = 2\pi^2;\quad P\left(\frac{\sqrt{3}}{2},\frac{1}{3}\right)$

Evaluating Integrals

Evaluate the integrals in Exercises 47–70.

47. $\int \frac{dx}{\sqrt{9 - x^2}}$

48. $\int \frac{dx}{\sqrt{1 - 4x^2}}$

49. $\int \frac{dx}{17 + x^2}$

50. $\int \frac{dx}{9 + 3x^2}$

51. $\int \frac{dx}{x\sqrt{25x^2 - 2}}$

52. $\int \frac{dx}{x\sqrt{5x^2 - 4}}$

53. $\int_0^1 \frac{4\,ds}{\sqrt{4 - s^2}}$

54. $\int_0^{3\sqrt{2/4}} \frac{ds}{\sqrt{9 - 4s^2}}$

55. $\int_0^2 \frac{dt}{8 + 2t^2}$

56. $\int_{-2}^2 \frac{dt}{4 + 3t^2}$

57. $\int_{-1}^{-\sqrt{2/2}} \frac{dy}{y\sqrt{4y^2 - 1}}$

58. $\int_{-2/3}^{-\sqrt{2/3}} \frac{dy}{y\sqrt{9y^2 - 1}}$

59. $\int \frac{3\,dr}{\sqrt{1 - 4(r-1)^2}}$

60. $\int \frac{6\,dr}{\sqrt{4 - (r+1)^2}}$

61. $\int \frac{dx}{2 + (x-1)^2}$

62. $\int \frac{dx}{1 + (3x+1)^2}$

63. $\int \frac{dx}{(2x-1)\sqrt{(2x-1)^2 - 4}}$

64. $\int \frac{dx}{(x+3)\sqrt{(x+3)^2 - 25}}$

65. $\int_{-\pi/2}^{\pi/2} \frac{2\cos\theta\,d\theta}{1 + (\sin\theta)^2}$

66. $\int_{\pi/6}^{\pi/4} \frac{\csc^2 x\,dx}{1 + (\cot x)^2}$

67. $\int_0^{\ln\sqrt{3}} \frac{e^x\,dx}{1 + e^{2x}}$

68. $\int_1^{e^{\pi/4}} \frac{4\,dt}{t(1 + \ln^2 t)}$

69. $\int \frac{y\,dy}{\sqrt{1 - y^4}}$

70. $\int \frac{\sec^2 y\,dy}{\sqrt{1 - \tan^2 y}}$

Evaluate the integrals in Exercises 71–84.

71. $\int \frac{dx}{\sqrt{-x^2 + 4x - 3}}$

72. $\int \frac{dx}{\sqrt{2x - x^2}}$

73. $\int_{-1}^0 \frac{6\,dt}{\sqrt{3 - 2t - t^2}}$

74. $\int_{1/2}^1 \frac{6\,dt}{\sqrt{3 + 4t - 4t^2}}$

75. $\int \frac{dy}{y^2 - 2y + 5}$

76. $\int \frac{dy}{y^2 + 6y + 10}$

77. $\int_1^2 \frac{8\,dx}{x^2 - 2x + 2}$

78. $\int_2^4 \frac{2\,dx}{x^2 - 6x + 10}$

79. $\int \frac{x+4}{x^2+4}\,dx$

80. $\int \frac{t-2}{t^2 - 6t + 10}\,dt$

81. $\displaystyle\int \frac{x^2 + 2x - 1}{x^2 + 9}\,dx$

82. $\displaystyle\int \frac{t^3 - 2t^2 + 3t - 4}{t^2 + 1}\,dt$

83. $\displaystyle\int \frac{dx}{(x + 1)\sqrt{x^2 + 2x}}$

84. $\displaystyle\int \frac{dx}{(x - 2)\sqrt{x^2 - 4x + 3}}$

Evaluate the integrals in Exercises 85–94.

85. $\displaystyle\int \frac{e^{\sin^{-1}x}\,dx}{\sqrt{1 - x^2}}$

86. $\displaystyle\int \frac{e^{\cos^{-1}x}\,dx}{\sqrt{1 - x^2}}$

87. $\displaystyle\int \frac{(\sin^{-1}x)^2\,dx}{\sqrt{1 - x^2}}$

88. $\displaystyle\int \frac{\sqrt{\tan^{-1}x}\,dx}{1 + x^2}$

89. $\displaystyle\int \frac{dy}{(\tan^{-1}y)(1 + y^2)}$

90. $\displaystyle\int \frac{dy}{(\sin^{-1}y)\sqrt{1 - y^2}}$

91. $\displaystyle\int_{\sqrt{2}}^{2} \frac{\sec^2(\sec^{-1}x)\,dx}{x\sqrt{x^2 - 1}}$

92. $\displaystyle\int_{2/\sqrt{3}}^{2} \frac{\cos(\sec^{-1}x)\,dx}{x\sqrt{x^2 - 1}}$

93. $\displaystyle\int \frac{1}{\sqrt{x}(x + 1)\left((\tan^{-1}\sqrt{x})^2 + 9\right)}\,dx$

94. $\displaystyle\int \frac{e^x \sin^{-1}e^x}{\sqrt{1 - e^{2x}}}\,dx$

95. $\displaystyle\int_0^1 \frac{\tan^{-1}x}{1 + x^2}\,dx$

96. $\displaystyle\int_{-\sqrt{3}}^{1/\sqrt{3}} \frac{\cos(\tan^{-1}3x)}{1 + 9x^2}\,dx$

L'Hôpital's Rule

Find the limits in Exercises 97–104.

97. $\displaystyle\lim_{x\to 0} \frac{\sin^{-1}5x}{x}$

98. $\displaystyle\lim_{x\to 1^+} \frac{\sqrt{x^2 - 1}}{\sec^{-1}x}$

99. $\displaystyle\lim_{x\to\infty} x\tan^{-1}\frac{2}{x}$

100. $\displaystyle\lim_{x\to 0} \frac{2\tan^{-1}3x^2}{7x^2}$

101. $\displaystyle\lim_{x\to 0} \frac{\tan^{-1}x^2}{x\sin^{-1}x}$

102. $\displaystyle\lim_{x\to\infty} \frac{e^x\tan^{-1}e^x}{e^{2x} + x}$

103. $\displaystyle\lim_{x\to 0^+} \frac{(\tan^{-1}\sqrt{x})^2}{x\sqrt{x + 1}}$

104. $\displaystyle\lim_{x\to 0^+} \frac{\sin^{-1}x^2}{(\sin^{-1}x)^2}$

Integration Formulas

Verify the integration formulas in Exercises 105–108.

105. $\displaystyle\int \frac{\tan^{-1}x}{x^2}dx = \ln x - \frac{1}{2}\ln(1 + x^2) - \frac{\tan^{-1}x}{x} + C$

106. $\displaystyle\int x^3\cos^{-1}5x\,dx = \frac{x^4}{4}\cos^{-1}5x + \frac{5}{4}\int \frac{x^4\,dx}{\sqrt{1 - 25x^2}}$

107. $\displaystyle\int (\sin^{-1}x)^2\,dx = x(\sin^{-1}x)^2 - 2x + 2\sqrt{1 - x^2}\sin^{-1}x + C$

108. $\displaystyle\int \ln(a^2 + x^2)\,dx = x\ln(a^2 + x^2) - 2x + 2a\tan^{-1}\frac{x}{a} + C$

Initial Value Problems

Solve the initial value problems in Exercises 109–114.

109. $\displaystyle\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}, \quad y(0) = 0$

110. $\displaystyle\frac{dy}{dx} = \frac{1}{x^2 + 1} - 1, \quad y(0) = 1$

111. $\displaystyle\frac{dy}{dx} = \frac{1}{x\sqrt{x^2 - 1}}, \quad x > 1; \quad y(2) = \pi$

112. $\displaystyle\frac{dy}{dx} = \frac{1}{1 + x^2} - \frac{2}{\sqrt{1 - x^2}}, \quad y(0) = 2$

Applications and Theory

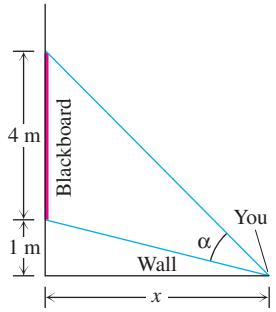
113. You are sitting in a classroom next to the wall looking at the blackboard at the front of the room. The blackboard is 4 m long and starts 1 m from the wall you are sitting next to.

a. Show that your viewing angle is

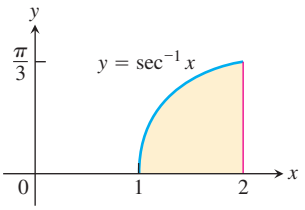
$$\alpha = \cot^{-1}\frac{x}{5} - \cot^{-1}x$$

if you are x m from the front wall.

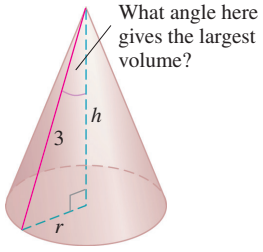
b. Find x so that α is as large as possible.



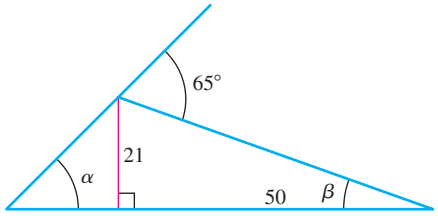
114. The region between the curve $y = \sec^{-1}x$ and the x-axis from $x = 1$ to $x = 2$ (shown here) is revolved about the y-axis to generate a solid. Find the volume of the solid.



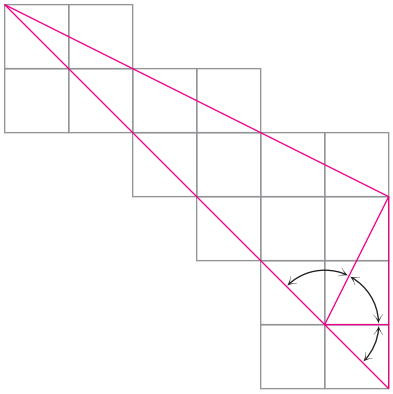
115. The slant height of the cone shown here is 3 m. How large should the indicated angle be to maximize the cone's volume?



116. Find the angle α.

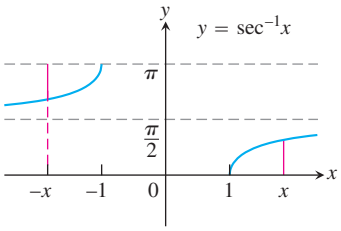


117. Here is an informal proof that $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \pi$. Explain what is going on.



118. Two derivations of the identity $\sec^{-1}(-x) = \pi - \sec^{-1}x$

a. (*Geometric*) Here is a pictorial proof that $\sec^{-1}(-x) = \pi - \sec^{-1}x$. See if you can tell what is going on.



b. (*Algebraic*) Derive the identity $\sec^{-1}(-x) = \pi - \sec^{-1}x$ by combining the following two equations from the text:

$$\cos^{-1}(-x) = \pi - \cos^{-1}x \qquad \text{Eq. (3)}$$
$$\sec^{-1}x = \cos^{-1}(1/x) \qquad \text{Eq. (5)}$$

119. The identity $\sin^{-1}x + \cos^{-1}x = \pi/2$ Figure 7.28 establishes the identity for $0 < x < 1$. To establish it for the rest of $[-1, 1]$, verify by direct calculation that it holds for $x = 1$, 0, and -1. Then, for values of x in $(-1, 0)$, let $x = -a$, $a > 0$, and apply Eqs. (1) and (3) to the sum $\sin^{-1}(-a) + \cos^{-1}(-a)$.

120. Show that the sum $\tan^{-1}x + \tan^{-1}(1/x)$ is constant.

121. Use the identity

$$\csc^{-1}u = \frac{\pi}{2} - \sec^{-1}u$$

to derive the formula for the derivative of $\csc^{-1}u$ in Table 7.3 from the formula for the derivative of $\sec^{-1}u$.

122. Derive the formula

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

for the derivative of $y = \tan^{-1}x$ by differentiating both sides of the equivalent equation $\tan y = x$.

123. Use the Derivative Rule, Theorem 1, to derive

$$\frac{d}{dx}\sec^{-1}x = \frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1.$$

124. Use the identity

$$\cot^{-1}u = \frac{\pi}{2} - \tan^{-1}u$$

to derive the formula for the derivative of $\cot^{-1}u$ in Table 7.3 from the formula for the derivative of $\tan^{-1}u$.

125. What is special about the functions

$$f(x) = \sin^{-1}\frac{x-1}{x+1}, \quad x \geq 0, \quad \text{and} \quad g(x) = 2\tan^{-1}\sqrt{x}?$$

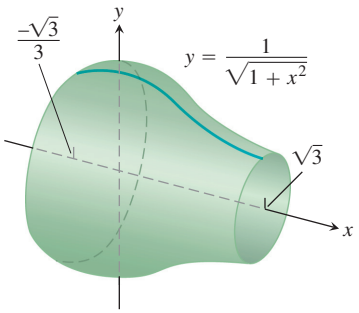
Explain.

126. What is special about the functions

$$f(x) = \sin^{-1}\frac{1}{\sqrt{x^2+1}} \quad \text{and} \quad g(x) = \tan^{-1}\frac{1}{x}?$$

Explain.

127. Find the volume of the solid of revolution shown here.



128. Arc length Find the circumference of a circle of radius r using Eq. (3) in Section 6.3.

Find the volumes of the solids in Exercises 129 and 130.

129. The solid lies between planes perpendicular to the x-axis at $x = -1$ and $x = 1$. The cross-sections perpendicular to the x-axis are

a. circles whose diameters stretch from the curve $y = -1/\sqrt{1+x^2}$ to the curve $y = 1/\sqrt{1+x^2}$.

b. vertical squares whose base edges run from the curve $y = -1/\sqrt{1+x^2}$ to the curve $y = 1/\sqrt{1+x^2}$.

130. The solid lies between planes perpendicular to the x-axis at $x = -\sqrt{2}/2$ and $x = \sqrt{2}/2$. The cross-sections perpendicular to the x-axis are

a. circles whose diameters stretch from the x-axis to the curve $y = 2/\sqrt[4]{1-x^2}$.

b. squares whose diagonals stretch from the x-axis to the curve $y = 2/\sqrt[4]{1-x^2}$.

T **131.** Find the values of the following.

a. $\sec^{-1}1.5$ **b.** $\csc^{-1}(-1.5)$ **c.** $\cot^{-1}2$

T **132.** Find the values of the following.

a. $\sec^{-1}(-3)$ **b.** $\csc^{-1}1.7$ **c.** $\cot^{-1}(-2)$

T In Exercises 133–135, find the domain and range of each composite function. Then graph the compositions on separate screens. Do the graphs make sense in each case? Give reasons for your answers. Comment on any differences you see.

133. a. $y = \tan^{-1}(\tan x)$ **b.** $y = \tan(\tan^{-1}x)$

134. a. $y = \sin^{-1}(\sin x)$ **b.** $y = \sin(\sin^{-1}x)$

135. a. $y = \cos^{-1}(\cos x)$ **b.** $y = \cos(\cos^{-1}x)$

T Use your graphing utility for Exercises 136–140.

136. Graph $y = \sec(\sec^{-1}x) = \sec(\cos^{-1}(1/x))$. Explain what you see.

137. Newton's serpentine Graph $y = 4x/(x^2 + 1)$, known as Newton's serpentine. Then graph $y = 2\sin(2\tan^{-1}x)$ in the same graphing window. What do you see? Explain.

138. Graph the rational function $y = (2 - x^2)/x^2$. Then graph $y = \cos(2\sec^{-1}x)$ in the same graphing window. What do you see? Explain.

139. Graph $f(x) = \sin^{-1}x$ together with its first two derivatives. Comment on the behavior of f and the shape of its graph in relation to the signs and values of f' and f''.

140. Graph $f(x) = \tan^{-1}x$ together with its first two derivatives. Comment on the behavior of f and the shape of its graph in relation to the signs and values of f' and f''.

7.7 Hyperbolic Functions

The hyperbolic functions are formed by taking combinations of the two exponential functions e^x and e^{-x}. The hyperbolic functions simplify many mathematical expressions and occur frequently in mathematical and engineering applications.

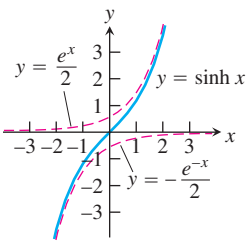
Definitions and Identities

The hyperbolic sine and hyperbolic cosine functions are defined by the equations

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad \text{and} \quad \cosh x = \frac{e^x + e^{-x}}{2}.$$

We pronounce $\sinh x$ as "cinch x," rhyming with "pinch x," and $\cosh x$ as "kosh x," rhyming with "gosh x." From this basic pair, we define the hyperbolic tangent, cotangent, secant, and cosecant functions. The defining equations and graphs of these functions are shown in Table 7.5. We will see that the hyperbolic functions bear many similarities to the trigonometric functions after which they are named.

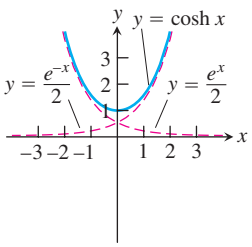
TABLE 7.5 The six basic hyperbolic functions



(a)

Hyperbolic sine:

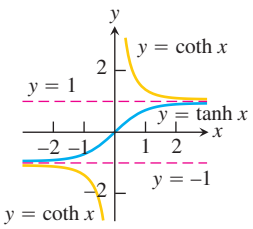
$$\sinh x = \frac{e^x - e^{-x}}{2}$$



(b)

Hyperbolic cosine:

$$\cosh x = \frac{e^x + e^{-x}}{2}$$



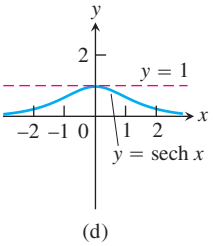
(c)

Hyperbolic tangent:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Hyperbolic cotangent:

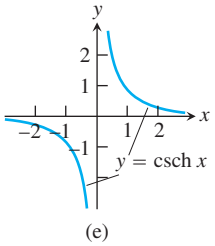
$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$



(d)

Hyperbolic secant:

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$



(e)

Hyperbolic cosecant:

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

TABLE 7.6 Identities for hyperbolic functions

$$\cosh^2 x - \sinh^2 x = 1$$
$$\sinh 2x = 2 \sinh x \cosh x$$
$$\cosh 2x = \cosh^2 x + \sinh^2 x$$
$$\cosh^2 x = \frac{\cosh 2x + 1}{2}$$
$$\sinh^2 x = \frac{\cosh 2x - 1}{2}$$
$$\tanh^2 x = 1 - \operatorname{sech}^2 x$$
$$\coth^2 x = 1 + \operatorname{csch}^2 x$$

TABLE 7.7 Derivatives of hyperbolic functions

$$\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$$
$$\frac{d}{dx}(\cosh u) = \sinh u \frac{du}{dx}$$
$$\frac{d}{dx}(\tanh u) = \operatorname{sech}^2 u \frac{du}{dx}$$
$$\frac{d}{dx}(\coth u) = -\operatorname{csch}^2 u \frac{du}{dx}$$
$$\frac{d}{dx}(\operatorname{sech} u) = -\operatorname{sech} u \tanh u \frac{du}{dx}$$
$$\frac{d}{dx}(\operatorname{csch} u) = -\operatorname{csch} u \coth u \frac{du}{dx}$$

TABLE 7.8 Integral formulas for hyperbolic functions

$$\int \sinh u \, du = \cosh u + C$$
$$\int \cosh u \, du = \sinh u + C$$
$$\int \operatorname{sech}^2 u \, du = \tanh u + C$$
$$\int \operatorname{csch}^2 u \, du = -\coth u + C$$
$$\int \operatorname{sech} u \tanh u \, du = -\operatorname{sech} u + C$$
$$\int \operatorname{csch} u \coth u \, du = -\operatorname{csch} u + C$$

Hyperbolic functions satisfy the identities in Table 7.6. Except for differences in sign, these resemble identities we know for the trigonometric functions. The identities are proved directly from the definitions, as we show here for the second one:

$$\begin{aligned} 2 \sinh x \cosh x &= 2\left(\frac{e^x - e^{-x}}{2}\right)\left(\frac{e^x + e^{-x}}{2}\right) \\ &= \frac{e^{2x} - e^{-2x}}{2} && \text{Simplify.} \\ &= \sinh 2x. && \text{Definition of sinh} \end{aligned}$$

The other identities are obtained similarly, by substituting in the definitions of the hyperbolic functions and using algebra.

For any real number u, we know the point with coordinates $(\cos u, \sin u)$ lies on the unit circle $x^2 + y^2 = 1$. So the trigonometric functions are sometimes called the *circular* functions. Because of the first identity

$$\cosh^2 u - \sinh^2 u = 1,$$

with u substituted for x in Table 7.6, the point having coordinates $(\cosh u, \sinh u)$ lies on the right-hand branch of the hyperbola $x^2 - y^2 = 1$. This is where the *hyperbolic* functions get their names (see Exercise 86).

Hyperbolic functions are useful in finding integrals, which we will see in Chapter 8. They play an important role in science and engineering as well. The hyperbolic cosine describes the shape of a hanging cable or wire that is strung between two points at the same height and hanging freely (see Exercise 83). The shape of the St. Louis Arch is an inverted hyperbolic cosine. The hyperbolic tangent occurs in the formula for the velocity of an ocean wave moving over water having a constant depth, and the inverse hyperbolic tangent describes how relative velocities sum according to Einstein's Law in the Special Theory of Relativity.

Derivatives and Integrals of Hyperbolic Functions

The six hyperbolic functions, being rational combinations of the differentiable functions e^x and e^{-x}, have derivatives at every point at which they are defined (Table 7.7). Again, there are similarities with trigonometric functions.

The derivative formulas are derived from the derivative of e^u:

$$\begin{aligned} \frac{d}{dx}(\sinh u) &= \frac{d}{dx}\left(\frac{e^u - e^{-u}}{2}\right) && \text{Definition of sinh } u \\ &= \frac{e^u \, du/dx + e^{-u} \, du/dx}{2} && \text{Derivative of } e^u \\ &= \cosh u \frac{du}{dx}. && \text{Definition of cosh } u \end{aligned}$$

This gives the first derivative formula. From the definition, we can calculate the derivative of the hyperbolic cosecant function, as follows:

$$\begin{aligned} \frac{d}{dx}(\operatorname{csch} u) &= \frac{d}{dx}\left(\frac{1}{\sinh u}\right) && \text{Definition of csch } u \\ &= -\frac{\cosh u \frac{du}{dx}}{\sinh^2 u} && \text{Quotient Rule for derivatives} \\ &= -\frac{1}{\sinh u}\frac{\cosh u}{\sinh u}\frac{du}{dx} && \text{Rearrange terms.} \\ &= -\operatorname{csch} u \coth u \frac{du}{dx} && \text{Definitions of csch } u \text{ and coth } u \end{aligned}$$

The other formulas in Table 7.7 are obtained similarly.

The derivative formulas lead to the integral formulas in Table 7.8.

EXAMPLE 1 We illustrate the derivative and integral formulas.

(a) $$\frac{d}{dt}\left(\tanh\sqrt{1+t^2}\right) = \operatorname{sech}^2\sqrt{1+t^2}\,\frac{d}{dt}\left(\sqrt{1+t^2}\right)$$
$$= \frac{t}{\sqrt{1+t^2}}\operatorname{sech}^2\sqrt{1+t^2}$$

(b) $$\int \coth 5x\,dx = \int \frac{\cosh 5x}{\sinh 5x}\,dx = \frac{1}{5}\int \frac{du}{u} \qquad u = \sinh 5x,\ du = 5\cosh 5x\,dx$$
$$= \frac{1}{5}\ln|u| + C = \frac{1}{5}\ln|\sinh 5x| + C$$

(c) $$\int_0^1 \sinh^2 x\,dx = \int_0^1 \frac{\cosh 2x - 1}{2}\,dx \qquad \text{Table 7.6}$$
$$= \frac{1}{2}\int_0^1 (\cosh 2x - 1)\,dx = \frac{1}{2}\left[\frac{\sinh 2x}{2} - x\right]_0^1$$
$$= \frac{\sinh 2}{4} - \frac{1}{2} \approx 0.40672 \qquad \text{Evaluate with a calculator.}$$

(d) $$\int_0^{\ln 2} 4e^x \sinh x\,dx = \int_0^{\ln 2} 4e^x\,\frac{e^x - e^{-x}}{2}\,dx = \int_0^{\ln 2} (2e^{2x} - 2)\,dx$$
$$= \left[e^{2x} - 2x\right]_0^{\ln 2} = (e^{2\ln 2} - 2\ln 2) - (1 - 0)$$
$$= 4 - 2\ln 2 - 1 \approx 1.6137$$

Inverse Hyperbolic Functions

The inverses of the six basic hyperbolic functions are very useful in integration (see Chapter 8). Since $d(\sinh x)/dx = \cosh x > 0$, the hyperbolic sine is an increasing function of x. We denote its inverse by

$$y = \sinh^{-1} x.$$

For every value of x in the interval $-\infty < x < \infty$, the value of $y = \sinh^{-1} x$ is the number whose hyperbolic sine is x. The graphs of $y = \sinh x$ and $y = \sinh^{-1} x$ are shown in Figure 7.32a.

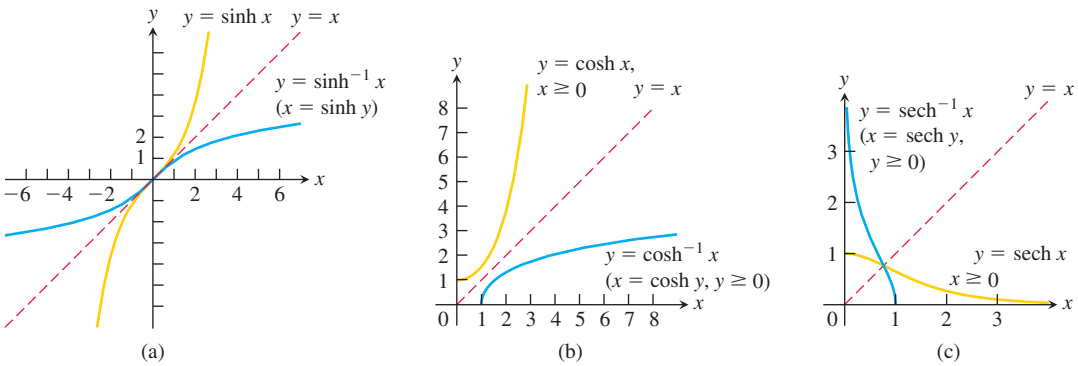


FIGURE 7.32 The graphs of the inverse hyperbolic sine, cosine, and secant of x. Notice the symmetries about the line $y = x$.

The function $y = \cosh x$ is not one-to-one because its graph in Table 7.5 does not pass the horizontal line test. The restricted function $y = \cosh x, x \geq 0$, however, is one-to-one and therefore has an inverse, denoted by

$$y = \cosh^{-1} x.$$

For every value of $x \geq 1$, $y = \cosh^{-1} x$ is the number in the interval $0 \leq y < \infty$ whose hyperbolic cosine is x. The graphs of $y = \cosh x$, $x \geq 0$, and $y = \cosh^{-1} x$ are shown in Figure 7.32b.

Like $y = \cosh x$, the function $y = \operatorname{sech} x = 1/\cosh x$ fails to be one-to-one, but its restriction to nonnegative values of x does have an inverse, denoted by

$$y = \operatorname{sech}^{-1} x.$$

For every value of x in the interval $(0, 1]$, $y = \operatorname{sech}^{-1} x$ is the nonnegative number whose hyperbolic secant is x. The graphs of $y = \operatorname{sech} x$, $x \geq 0$, and $y = \operatorname{sech}^{-1} x$ are shown in Figure 7.32c.

The hyperbolic tangent, cotangent, and cosecant are one-to-one on their domains and therefore have inverses, denoted by

$$y = \tanh^{-1} x, \qquad y = \coth^{-1} x, \qquad y = \operatorname{csch}^{-1} x.$$

These functions are graphed in Figure 7.33.

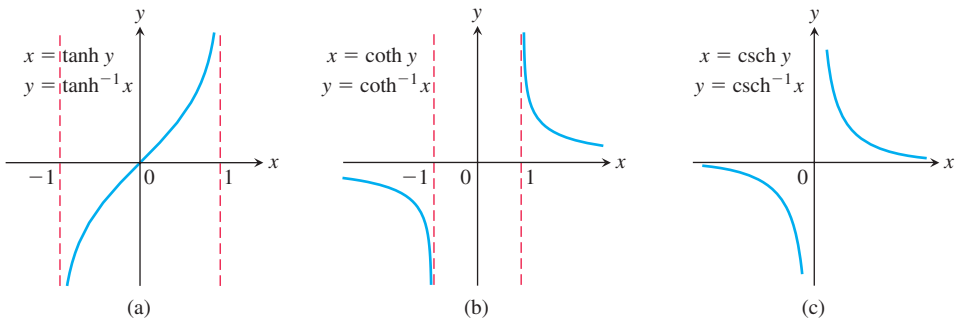


FIGURE 7.33 The graphs of the inverse hyperbolic tangent, cotangent, and cosecant of x.

TABLE 7.9 Identities for inverse hyperbolic functions

$$\operatorname{sech}^{-1} x = \cosh^{-1} \frac{1}{x}$$

$$\operatorname{csch}^{-1} x = \sinh^{-1} \frac{1}{x}$$

$$\coth^{-1} x = \tanh^{-1} \frac{1}{x}$$

Useful Identities

We use the identities in Table 7.9 to calculate the values of $\operatorname{sech}^{-1} x$, $\operatorname{csch}^{-1} x$, and $\coth^{-1} x$ on calculators that give only $\cosh^{-1} x$, $\sinh^{-1} x$, and $\tanh^{-1} x$. These identities are direct consequences of the definitions. For example, if $0 < x \leq 1$, then

$$\operatorname{sech}\left(\cosh^{-1}\left(\frac{1}{x}\right)\right) = \frac{1}{\cosh\left(\cosh^{-1}\left(\frac{1}{x}\right)\right)} = \frac{1}{\left(\frac{1}{x}\right)} = x.$$

We also know that $\operatorname{sech}(\operatorname{sech}^{-1} x) = x$, so because the hyperbolic secant is one-to-one on $(0, 1]$, we have

$$\cosh^{-1}\left(\frac{1}{x}\right) = \operatorname{sech}^{-1} x.$$

Derivatives of Inverse Hyperbolic Functions

An important use of inverse hyperbolic functions lies in antiderivatives that reverse the derivative formulas in Table 7.10.

The restrictions $|u| < 1$ and $|u| > 1$ on the derivative formulas for $\tanh^{-1} u$ and $\coth^{-1} u$ come from the natural restrictions on the values of these functions. (See Figure 7.33a and b.) The distinction between $|u| < 1$ and $|u| > 1$ becomes important when we convert the derivative formulas into integral formulas.

We illustrate how the derivatives of the inverse hyperbolic functions are found in Example 2, where we calculate $d(\cosh^{-1} u)/dx$. The other derivatives are obtained by similar calculations.

TABLE 7.10 Derivatives of inverse hyperbolic functions

$$\frac{d(\sinh^{-1}u)}{dx} = \frac{1}{\sqrt{1+u^2}}\frac{du}{dx}$$

$$\frac{d(\cosh^{-1}u)}{dx} = \frac{1}{\sqrt{u^2-1}}\frac{du}{dx}, \qquad u > 1$$

$$\frac{d(\tanh^{-1}u)}{dx} = \frac{1}{1-u^2}\frac{du}{dx}, \qquad |u| < 1$$

$$\frac{d(\coth^{-1}u)}{dx} = \frac{1}{1-u^2}\frac{du}{dx}, \qquad |u| > 1$$

$$\frac{d(\operatorname{sech}^{-1}u)}{dx} = -\frac{1}{u\sqrt{1-u^2}}\frac{du}{dx}, \qquad 0 < u < 1$$

$$\frac{d(\operatorname{csch}^{-1}u)}{dx} = -\frac{1}{|u|\sqrt{1+u^2}}\frac{du}{dx}, \qquad u \neq 0$$

EXAMPLE 2 Show that if u is a differentiable function of x whose values are greater than 1, then

$$\frac{d}{dx}(\cosh^{-1}u) = \frac{1}{\sqrt{u^2-1}}\frac{du}{dx}.$$

Solution First we find the derivative of $y = \cosh^{-1}x$ for $x > 1$ by applying Theorem 1 of Section 7.1 with $f(x) = \cosh x$ and $f^{-1}(x) = \cosh^{-1}x$. Theorem 1 can be applied because the derivative of $\cosh x$ is positive when $x > 0$.

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \qquad \text{Theorem 1}$$

$$= \frac{1}{\sinh(\cosh^{-1}x)} \qquad f'(u) = \sinh u$$

$$= \frac{1}{\sqrt{\cosh^2(\cosh^{-1}x) - 1}} \qquad \cosh^2 u - \sinh^2 u = 1,\ \sinh u = \sqrt{\cosh^2 u - 1}$$

$$= \frac{1}{\sqrt{x^2-1}} \qquad \cosh(\cosh^{-1}x) = x$$

HISTORICAL BIOGRAPHY
Sonya Kovalevsky
(1850–1891)
www.goo.gl/6TLoDz

The Chain Rule gives the final result:

$$\frac{d}{dx}(\cosh^{-1}u) = \frac{1}{\sqrt{u^2-1}}\frac{du}{dx}.$$

■

With appropriate substitutions, the derivative formulas in Table 7.10 lead to the integration formulas in Table 7.11. Each of the formulas in Table 7.11 can be verified by differentiating the expression on the right-hand side.

EXAMPLE 3 Evaluate

$$\int_0^1 \frac{2\,dx}{\sqrt{3+4x^2}}.$$

TABLE 7.11 Integrals leading to inverse hyperbolic functions

1. $\displaystyle\int \frac{du}{\sqrt{a^2 + u^2}} = \sinh^{-1}\left(\frac{u}{a}\right) + C, \qquad a > 0$

2. $\displaystyle\int \frac{du}{\sqrt{u^2 - a^2}} = \cosh^{-1}\left(\frac{u}{a}\right) + C, \qquad u > a > 0$

3. $\displaystyle\int \frac{du}{a^2 - u^2} = \begin{cases} \frac{1}{a}\tanh^{-1}\left(\frac{u}{a}\right) + C, & u^2 < a^2 \\ \frac{1}{a}\coth^{-1}\left(\frac{u}{a}\right) + C, & u^2 > a^2 \end{cases}$

4. $\displaystyle\int \frac{du}{u\sqrt{a^2 - u^2}} = -\frac{1}{a}\operatorname{sech}^{-1}\left(\frac{u}{a}\right) + C, \qquad 0 < u < a$

5. $\displaystyle\int \frac{du}{u\sqrt{a^2 + u^2}} = -\frac{1}{a}\operatorname{csch}^{-1}\left|\frac{u}{a}\right| + C, \qquad u \neq 0 \text{ and } a > 0$

Solution The indefinite integral is

$$\int \frac{2\,dx}{\sqrt{3 + 4x^2}} = \int \frac{du}{\sqrt{a^2 + u^2}} \qquad u = 2x,\quad du = 2\,dx,\quad a = \sqrt{3}$$

$$= \sinh^{-1}\left(\frac{u}{a}\right) + C \qquad \text{Formula from Table 7.11}$$

$$= \sinh^{-1}\left(\frac{2x}{\sqrt{3}}\right) + C.$$

Therefore,

$$\int_0^1 \frac{2\,dx}{\sqrt{3 + 4x^2}} = \sinh^{-1}\left(\frac{2x}{\sqrt{3}}\right)\Big]_0^1 = \sinh^{-1}\left(\frac{2}{\sqrt{3}}\right) - \sinh^{-1}(0)$$

$$= \sinh^{-1}\left(\frac{2}{\sqrt{3}}\right) - 0 \approx 0.98665.$$

EXERCISES 7.7

Values and Identities

Each of Exercises 1–4 gives a value of $\sinh x$ or $\cosh x$. Use the definitions and the identity $\cosh^2 x - \sinh^2 x = 1$ to find the values of the remaining five hyperbolic functions.

1. $\sinh x = -\frac{3}{4}$

2. $\sinh x = \frac{4}{3}$

3. $\cosh x = \frac{17}{15}, \quad x > 0$

4. $\cosh x = \frac{13}{5}, \quad x > 0$

Rewrite the expressions in Exercises 5–10 in terms of exponentials and simplify the results as much as you can.

5. $2\cosh(\ln x)$

6. $\sinh(2\ln x)$

7. $\cosh 5x + \sinh 5x$

8. $\cosh 3x - \sinh 3x$

9. $(\sinh x + \cosh x)^4$

10. $\ln(\cosh x + \sinh x) + \ln(\cosh x - \sinh x)$

11. Prove the identities

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y,$$
$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y.$$

Then use them to show that

a. $\sinh 2x = 2\sinh x \cosh x.$

b. $\cosh 2x = \cosh^2 x + \sinh^2 x.$

12. Use the definitions of $\cosh x$ and $\sinh x$ to show that

$$\cosh^2 x - \sinh^2 x = 1.$$

Finding Derivatives

In Exercises 13–24, find the derivative of y with respect to the appropriate variable.

13. $y = 6 \sinh \frac{x}{3}$

14. $y = \frac{1}{2}\sinh (2x + 1)$

15. $y = 2\sqrt{t} \tanh \sqrt{t}$

16. $y = t^2 \tanh \frac{1}{t}$

17. $y = \ln (\sinh z)$

18. $y = \ln (\cosh z)$

19. $y = (\operatorname{sech} \theta)(1 - \ln \operatorname{sech} \theta)$

20. $y = (\operatorname{csch} \theta)(1 - \ln \operatorname{csch} \theta)$

21. $y = \ln \cosh v - \frac{1}{2}\tanh^2 v$

22. $y = \ln \sinh v - \frac{1}{2}\coth^2 v$

23. $y = (x^2 + 1)\operatorname{sech} (\ln x)$

(*Hint:* Before differentiating, express in terms of exponentials and simplify.)

24. $y = (4x^2 - 1)\operatorname{csch} (\ln 2x)$

In Exercises 25–36, find the derivative of y with respect to the appropriate variable.

25. $y = \sinh^{-1} \sqrt{x}$

26. $y = \cosh^{-1} 2\sqrt{x + 1}$

27. $y = (1 - \theta)\tanh^{-1} \theta$

28. $y = (\theta^2 + 2\theta)\tanh^{-1}(\theta + 1)$

29. $y = (1 - t)\coth^{-1} \sqrt{t}$

30. $y = (1 - t^2)\coth^{-1} t$

31. $y = \cos^{-1} x - x \operatorname{sech}^{-1} x$

32. $y = \ln x + \sqrt{1 - x^2} \operatorname{sech}^{-1} x$

33. $y = \operatorname{csch}^{-1}\left(\frac{1}{2}\right)^{\theta}$

34. $y = \operatorname{csch}^{-1} 2^{\theta}$

35. $y = \sinh^{-1}(\tan x)$

36. $y = \cosh^{-1}(\sec x), \quad 0 < x < \pi/2$

Integration Formulas

Verify the integration formulas in Exercises 37–40.

37. a. $\displaystyle\int \operatorname{sech} x \, dx = \tan^{-1}(\sinh x) + C$

b. $\displaystyle\int \operatorname{sech} x \, dx = \sin^{-1}(\tanh x) + C$

38. $\displaystyle\int x \operatorname{sech}^{-1} x \, dx = \frac{x^2}{2} \operatorname{sech}^{-1} x - \frac{1}{2}\sqrt{1 - x^2} + C$

39. $\displaystyle\int x \coth^{-1} x \, dx = \frac{x^2 - 1}{2} \coth^{-1} x + \frac{x}{2} + C$

40. $\displaystyle\int \tanh^{-1} x \, dx = x \tanh^{-1} x + \frac{1}{2}\ln (1 - x^2) + C$

Evaluating Integrals

Evaluate the integrals in Exercises 41–60.

41. $\displaystyle\int \sinh 2x \, dx$

42. $\displaystyle\int \sinh \frac{x}{5} dx$

43. $\displaystyle\int 6 \cosh \left(\frac{x}{2} - \ln 3\right) dx$

44. $\displaystyle\int 4 \cosh (3x - \ln 2) \, dx$

45. $\displaystyle\int \tanh \frac{x}{7} dx$

46. $\displaystyle\int \coth \frac{\theta}{\sqrt{3}} d\theta$

47. $\displaystyle\int \operatorname{sech}^2\left(x - \frac{1}{2}\right) dx$

48. $\displaystyle\int \operatorname{csch}^2(5 - x) \, dx$

49. $\displaystyle\int \frac{\operatorname{sech} \sqrt{t} \tanh \sqrt{t} \, dt}{\sqrt{t}}$

50. $\displaystyle\int \frac{\operatorname{csch} (\ln t) \coth (\ln t) \, dt}{t}$

51. $\displaystyle\int_{\ln 2}^{\ln 4} \coth x \, dx$

52. $\displaystyle\int_{0}^{\ln 2} \tanh 2x \, dx$

53. $\displaystyle\int_{-\ln 4}^{-\ln 2} 2e^{\theta} \cosh \theta \, d\theta$

54. $\displaystyle\int_{0}^{\ln 2} 4e^{-\theta} \sinh \theta \, d\theta$

55. $\displaystyle\int_{-\pi/4}^{\pi/4} \cosh (\tan \theta) \sec^2 \theta \, d\theta$

56. $\displaystyle\int_{0}^{\pi/2} 2 \sinh (\sin \theta) \cos \theta \, d\theta$

57. $\displaystyle\int_{1}^{2} \frac{\cosh (\ln t)}{t} dt$

58. $\displaystyle\int_{1}^{4} \frac{8 \cosh \sqrt{x}}{\sqrt{x}} dx$

59. $\displaystyle\int_{-\ln 2}^{0} \cosh^2\left(\frac{x}{2}\right) dx$

60. $\displaystyle\int_{0}^{\ln 10} 4 \sinh^2\left(\frac{x}{2}\right) dx$

Inverse Hyperbolic Functions and Integrals

Since the hyperbolic functions can be written in terms of exponential functions, it is possible to express the inverse hyperbolic functions in terms of logarithms, as shown in the following table.

$$\sinh^{-1} x = \ln\left(x + \sqrt{x^2 + 1}\right), \qquad -\infty < x < \infty$$

$$\cosh^{-1} x = \ln\left(x + \sqrt{x^2 - 1}\right), \qquad x \geq 1$$

$$\tanh^{-1} x = \frac{1}{2}\ln\frac{1 + x}{1 - x}, \qquad |x| < 1$$

$$\operatorname{sech}^{-1} x = \ln\left(\frac{1 + \sqrt{1 - x^2}}{x}\right), \qquad 0 < x \leq 1$$

$$\operatorname{csch}^{-1} x = \ln\left(\frac{1}{x} + \frac{\sqrt{1 + x^2}}{|x|}\right), \qquad x \neq 0$$

$$\coth^{-1} x = \frac{1}{2}\ln\frac{x + 1}{x - 1}, \qquad |x| > 1$$

Use these formulas to express the numbers in Exercises 61–66 in terms of natural logarithms.

61. $\sinh^{-1}(-5/12)$

62. $\cosh^{-1}(5/3)$

63. $\tanh^{-1}(-1/2)$

64. $\coth^{-1}(5/4)$

65. $\operatorname{sech}^{-1}(3/5)$

66. $\operatorname{csch}^{-1}\left(-1/\sqrt{3}\right)$

Evaluate the integrals in Exercises 67–74 in terms of

a. inverse hyperbolic functions.

b. natural logarithms.

67. $\displaystyle\int_{0}^{2\sqrt{3}} \frac{dx}{\sqrt{4 + x^2}}$

68. $\displaystyle\int_{0}^{1/3} \frac{6 \, dx}{\sqrt{1 + 9x^2}}$

69. $\displaystyle\int_{5/4}^{2} \frac{dx}{1 - x^2}$

70. $\displaystyle\int_{0}^{1/2} \frac{dx}{1 - x^2}$

71. $\displaystyle\int_{1/5}^{3/13} \frac{dx}{x\sqrt{1-16x^2}}$ **72.** $\displaystyle\int_{1}^{2} \frac{dx}{x\sqrt{4+x^2}}$

73. $\displaystyle\int_{0}^{\pi} \frac{\cos x\,dx}{\sqrt{1+\sin^2 x}}$ **74.** $\displaystyle\int_{1}^{e} \frac{dx}{x\sqrt{1+(\ln x)^2}}$

Applications and Examples

75. Show that if a function f is defined on an interval symmetric about the origin (so that f is defined at $-x$ whenever it is defined at x), then

$$f(x) = \frac{f(x)+f(-x)}{2} + \frac{f(x)-f(-x)}{2}. \tag{1}$$

Then show that $(f(x)+f(-x))/2$ is even and that $(f(x)-f(-x))/2$ is odd.

76. Derive the formula $\sinh^{-1} x = \ln\left(x+\sqrt{x^2+1}\right)$ for all real x. Explain in your derivation why the plus sign is used with the square root instead of the minus sign.

77. Skydiving If a body of mass m falling from rest under the action of gravity encounters an air resistance proportional to the square of the velocity, then the body's velocity t s into the fall satisfies the differential equation

$$m\frac{dv}{dt} = mg - kv^2,$$

where k is a constant that depends on the body's aerodynamic properties and the density of the air. (We assume that the fall is short enough so that the variation in the air's density will not affect the outcome significantly.)

a. Show that

$$v = \sqrt{\frac{mg}{k}}\tanh\left(\sqrt{\frac{gk}{m}}t\right)$$

satisfies the differential equation and the initial condition that $v = 0$ when $t = 0$.

b. Find the body's *limiting velocity*, $\lim_{t\to\infty} v$.

c. For a 75 kg skydiver ($mg = 735$ N), with time in seconds and distance in meters, a typical value for k is 0.235. What is the diver's limiting velocity?

78. Accelerations whose magnitudes are proportional to displacement Suppose that the position of a body moving along a coordinate line at time t is

a. $s = a\cos kt + b\sin kt$.

b. $s = a\cosh kt + b\sinh kt$.

Show in both cases that the acceleration d^2s/dt^2 is proportional to s but that in the first case it is directed toward the origin, whereas in the second case it is directed away from the origin.

79. Volume A region in the first quadrant is bounded above by the curve $y = \cosh x$, below by the curve $y = \sinh x$, and on the left and right by the y-axis and the line $x = 2$, respectively. Find the volume of the solid generated by revolving the region about the x-axis.

80. Volume The region enclosed by the curve $y = \operatorname{sech} x$, the x-axis, and the lines $x = \pm\ln\sqrt{3}$ is revolved about the x-axis to generate a solid. Find the volume of the solid.

81. Arc length Find the length of the graph of $y = (1/2)\cosh 2x$ from $x = 0$ to $x = \ln\sqrt{5}$.

82. Use the definitions of the hyperbolic functions to find each of the following limits.

a. $\lim_{x\to\infty}\tanh x$ **b.** $\lim_{x\to-\infty}\tanh x$

c. $\lim_{x\to\infty}\sinh x$ **d.** $\lim_{x\to-\infty}\sinh x$

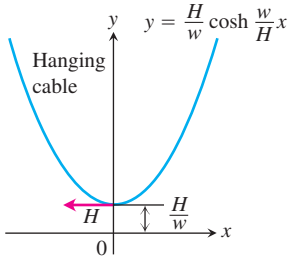
e. $\lim_{x\to\infty}\operatorname{sech} x$ **f.** $\lim_{x\to\infty}\coth x$

g. $\lim_{x\to0^+}\coth x$ **h.** $\lim_{x\to0^-}\coth x$

i. $\lim_{x\to-\infty}\operatorname{csch} x$

83. Hanging cables Imagine a cable, like a telephone line or TV cable, strung from one support to another and hanging freely. The cable's weight per unit length is a constant w and the horizontal tension at its lowest point is a *vector* of length H. If we choose a coordinate system for the plane of the cable in which the x-axis is horizontal, the force of gravity is straight down, the positive y-axis points straight up, and the lowest point of the cable lies at the point $y = H/w$ on the y-axis (see accompanying figure), then it can be shown that the cable lies along the graph of the hyperbolic cosine

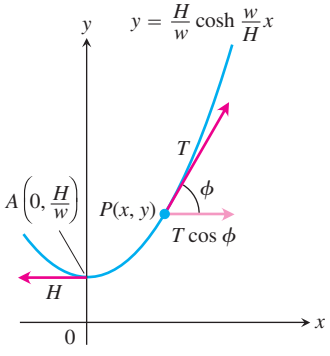
$$y = \frac{H}{w}\cosh\frac{w}{H}x.$$



Such a curve is sometimes called a **chain curve** or a **catenary**, the latter deriving from the Latin *catena*, meaning "chain."

a. Let $P(x, y)$ denote an arbitrary point on the cable. The next accompanying figure displays the tension at P as a vector of length (magnitude) T, as well as the tension H at the lowest point A. Show that the cable's slope at P is

$$\tan\phi = \frac{dy}{dx} = \sinh\frac{w}{H}x.$$



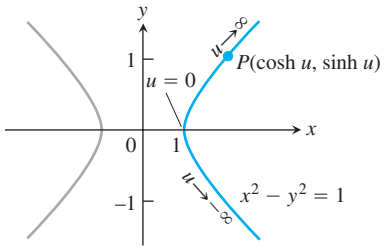
b. Using the result from part (a) and the fact that the horizontal tension at P must equal H (the cable is not moving), show that $T = wy$. Hence, the magnitude of the tension at $P(x, y)$ is exactly equal to the weight of y units of cable.

84. (*Continuation of Exercise 83.*) The length of arc AP in the Exercise 83 figure is $s = (1/a)\sinh ax$, where $a = w/H$. Show that the coordinates of P may be expressed in terms of s as

$$x = \frac{1}{a}\sinh^{-1} as, \qquad y = \sqrt{s^2 + \frac{1}{a^2}}.$$

85. Area Show that the area of the region in the first quadrant enclosed by the curve $y = (1/a)\cosh ax$, the coordinate axes, and the line $x = b$ is the same as the area of a rectangle of height $1/a$ and length s, where s is the length of the curve from $x = 0$ to $x = b$. Draw a figure illustrating this result.

86. The hyperbolic in hyperbolic functions Just as $x = \cos u$ and $y = \sin u$ are identified with points (x, y) on the unit circle, the functions $x = \cosh u$ and $y = \sinh u$ are identified with points (x, y) on the right-hand branch of the unit hyperbola, $x^2 - y^2 = 1$.



Since $\cosh^2 u - \sinh^2 u = 1$, the point $(\cosh u, \sinh u)$ lies on the right-hand branch of the hyperbola $x^2 - y^2 = 1$ for every value of u.

Another analogy between hyperbolic and circular functions is that the variable u in the coordinates $(\cosh u, \sinh u)$ for the points of the right-hand branch of the hyperbola $x^2 - y^2 = 1$ is twice the area of the sector AOP pictured in the accompanying figure. To see why this is so, carry out the following steps.

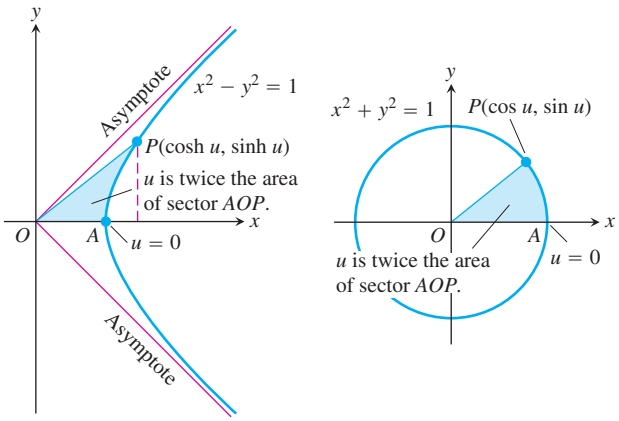
a. Show that the area $A(u)$ of sector AOP is

$$A(u) = \frac{1}{2}\cosh u \sinh u - \int_1^{\cosh u} \sqrt{x^2 - 1}\, dx.$$

b. Differentiate both sides of the equation in part (a) with respect to u to show that

$$A'(u) = \frac{1}{2}.$$

c. Solve this last equation for $A(u)$. What is the value of $A(0)$? What is the value of the constant of integration C in your solution? With C determined, what does your solution say about the relationship of u to $A(u)$?



One of the analogies between hyperbolic and circular functions is revealed by these two diagrams (Exercise 86).

7.8 Relative Rates of Growth

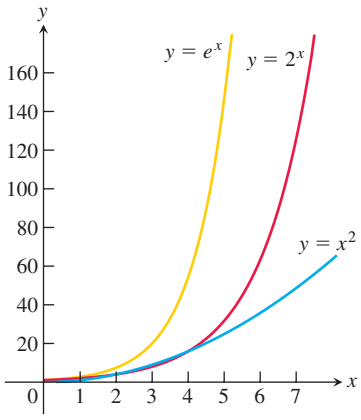


FIGURE 7.34 The graphs of e^x, 2^x, and x^2.

It is often important in mathematics, computer science, and engineering to compare the rates at which functions of x grow as x becomes large. Exponential functions are important in these comparisons because of their very fast growth, and logarithmic functions because of their very slow growth. In this section we introduce the *little-oh* and *big-oh* notation used to describe the results of these comparisons. We restrict our attention to functions whose values eventually become and remain positive as $x \to \infty$.

Growth Rates of Functions

You may have noticed that exponential functions like 2^x and e^x seem to grow more rapidly as x gets large than do polynomials and rational functions. These exponentials certainly grow more rapidly than x itself, and you can see 2^x outgrowing x^2 as x increases in Figure 7.34. In fact, as $x \to \infty$, the functions 2^x and e^x grow faster than any power of x, even $x^{1,000,000}$ (Exercise 19). In contrast, logarithmic functions like $y = \log_2 x$ and $y = \ln x$ grow more slowly as $x \to \infty$ than any positive power of x (Exercise 21).

To get a feeling for how rapidly the values of $y = e^x$ grow with increasing x, think of graphing the function on a large blackboard, with the axes scaled in centimeters. At $x = 1$ cm, the graph is $e^1 \approx 3$ cm above the x-axis. At $x = 6$ cm, the graph is

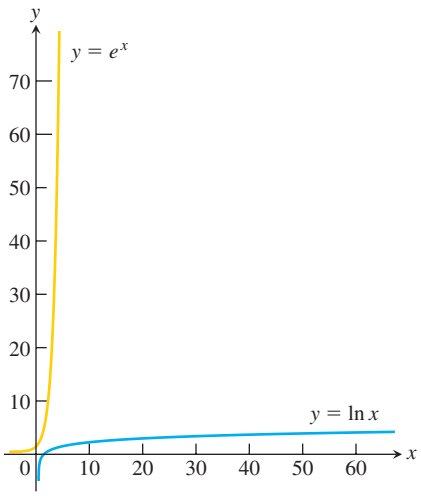


FIGURE 7.35 Scale drawings of the graphs of e^x and $\ln x$.

$e^6 \approx 403$ cm ≈ 4 m high (it is about to go through the ceiling if it hasn't done so already). At $x = 10$ cm, the graph is $e^{10} \approx 22{,}026$ cm ≈ 220 m high, higher than most buildings. At $x = 24$ cm, the graph is more than halfway to the moon, and at $x = 43$ cm from the origin, the graph is high enough to reach past the sun's closest stellar neighbor, the red dwarf star Proxima Centauri. By contrast, with axes scaled in centimeters, you have to go nearly 5 light-years out on the x-axis to find a point where the graph of $y = \ln x$ is even $y = 43$ cm high. See Figure 7.35.

These important comparisons of exponential, polynomial, and logarithmic functions can be made precise by defining what it means for a function $f(x)$ to grow faster than another function $g(x)$ as $x \to \infty$.

DEFINITION Let $f(x)$ and $g(x)$ be positive for x sufficiently large.

1. f **grows faster than** g as $x \to \infty$ if

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \infty$$

or, equivalently, if

$$\lim_{x\to\infty} \frac{g(x)}{f(x)} = 0.$$

We also say that g **grows slower than** f as $x \to \infty$.

2. f and g **grow at the same rate** as $x \to \infty$ if

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L$$

where L is finite and positive.

According to these definitions, $y = 2x$ does not grow faster than $y = x$. The two functions grow at the same rate because

$$\lim_{x\to\infty} \frac{2x}{x} = \lim_{x\to\infty} 2 = 2,$$

which is a finite, positive limit. The reason for this choice of terminology is that we want "f grows faster than g" to mean that for large x-values g is negligible when compared with f.

EXAMPLE 1 We compare the growth rates of several common functions.

(a) e^x grows faster than x^2 as $x \to \infty$ because

$$\underbrace{\lim_{x\to\infty} \frac{e^x}{x^2}}_{\infty/\infty} = \underbrace{\lim_{x\to\infty} \frac{e^x}{2x}}_{\infty/\infty} = \lim_{x\to\infty} \frac{e^x}{2} = \infty.$$

Using l'Hôpital's Rule twice

(b) 3^x grows faster than 2^x as $x \to \infty$ because

$$\lim_{x\to\infty} \frac{3^x}{2^x} = \lim_{x\to\infty} \left(\frac{3}{2}\right)^x = \infty.$$

(c) x^2 grows faster than $\ln x$ as $x \to \infty$ because

$$\lim_{x\to\infty} \frac{x^2}{\ln x} = \lim_{x\to\infty} \frac{2x}{1/x} = \lim_{x\to\infty} 2x^2 = \infty.$$

l'Hôpital's Rule

(d) $\ln x$ grows slower than $x^{1/n}$ as $x \to \infty$ for any positive integer n because

$$\lim_{x\to\infty} \frac{\ln x}{x^{1/n}} = \lim_{x\to\infty} \frac{1/x}{(1/n)x^{(1/n)-1}} \qquad \text{l'Hôpital's Rule}$$

$$= \lim_{x\to\infty} \frac{n}{x^{1/n}} = 0. \qquad n \text{ is constant.}$$

(e) As part (b) suggests, exponential functions with different bases never grow at the same rate as $x \to \infty$. If $a > b > 0$, then a^x grows faster than b^x. Since $(a/b) > 1$,

$$\lim_{x\to\infty} \frac{a^x}{b^x} = \lim_{x\to\infty} \left(\frac{a}{b}\right)^x = \infty.$$

(f) In contrast to exponential functions, logarithmic functions with different bases $a > 1$ and $b > 1$ always grow at the same rate as $x \to \infty$:

$$\lim_{x\to\infty} \frac{\log_a x}{\log_b x} = \lim_{x\to\infty} \frac{\ln x/\ln a}{\ln x/\ln b} = \frac{\ln b}{\ln a}.$$

The limiting ratio is always finite and never zero. ■

If f grows at the same rate as g as $x \to \infty$, and g grows at the same rate as h as $x \to \infty$, then f grows at the same rate as h as $x \to \infty$. The reason is that

$$\lim_{x\to\infty} \frac{f}{g} = L_1 \qquad \text{and} \qquad \lim_{x\to\infty} \frac{g}{h} = L_2$$

together imply

$$\lim_{x\to\infty} \frac{f}{h} = \lim_{x\to\infty} \frac{f}{g} \cdot \frac{g}{h} = L_1 L_2.$$

If L_1 and L_2 are finite and nonzero, then so is $L_1 L_2$.

EXAMPLE 2 Show that $\sqrt{x^2 + 5}$ and $(2\sqrt{x} - 1)^2$ grow at the same rate as $x \to \infty$.

Solution We show that the functions grow at the same rate by showing that they both grow at the same rate as the function $g(x) = x$:

$$\lim_{x\to\infty} \frac{\sqrt{x^2 + 5}}{x} = \lim_{x\to\infty} \sqrt{1 + \frac{5}{x^2}} = 1,$$

$$\lim_{x\to\infty} \frac{(2\sqrt{x} - 1)^2}{x} = \lim_{x\to\infty} \left(\frac{2\sqrt{x} - 1}{\sqrt{x}}\right)^2 = \lim_{x\to\infty} \left(2 - \frac{1}{\sqrt{x}}\right)^2 = 4.$$

■

Order and Oh-Notation

The "little-oh" and "big-oh" notation was invented by number theorists a hundred years ago and is now commonplace in mathematical analysis and computer science. According to this definition, saying $f = o(g)$ as $x \to \infty$ is another way to say that f grows slower than g as $x \to \infty$.

DEFINITION A function f is **of smaller order than** g as $x \to \infty$ if $\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0$. We indicate this by writing $\boldsymbol{f = o(g)}$ ("f is little-oh of g").

EXAMPLE 3 Using little-oh notation.

(a) $\ln x = o(x)$ as $x \to \infty$ because $\lim_{x\to\infty} \frac{\ln x}{x} = 0$

(b) $x^2 = o(x^3 + 1)$ as $x \to \infty$ because $\lim_{x\to\infty} \frac{x^2}{x^3 + 1} = 0$ ■

DEFINITION Let $f(x)$ and $g(x)$ be positive for x sufficiently large. Then f is **of at most the order of** g as $x \to \infty$ if there is a positive integer M for which

$$\frac{f(x)}{g(x)} \le M,$$

for x sufficiently large. We indicate this by writing $\boldsymbol{f = O(g)}$ ("f is big-oh of g").

EXAMPLE 4 Using big-oh notation.

(a) $x + \sin x = O(x)$ as $x \to \infty$ because $\frac{x + \sin x}{x} \le 2$ for x sufficiently large.

(b) $e^x + x^2 = O(e^x)$ as $x \to \infty$ because $\frac{e^x + x^2}{e^x} \to 1$ as $x \to \infty$.

(c) $x = O(e^x)$ as $x \to \infty$ because $\frac{x}{e^x} \to 0$ as $x \to \infty$. ■

If you look at the definitions again, you will see that $f = o(g)$ implies $f = O(g)$ for functions that are positive for all sufficiently large x. Also, if f and g grow at the same rate, then $f = O(g)$ and $g = O(f)$ (Exercise 11).

Sequential vs. Binary Search

Computer scientists often measure the efficiency of an algorithm by counting the number of steps a computer must take to execute the algorithm. There can be significant differences in how efficiently algorithms perform, even if they are designed to accomplish the same task. These differences are often described using big-oh notation. Here is an example.

Webster's International Dictionary lists about 26,000 words that begin with the letter *a*. One way to look up a word, or to learn if it is not there, is to read through the list one word at a time until you either find the word or determine that it is not there. This method, called **sequential search**, makes no particular use of the words' alphabetical arrangement in the list. You are sure to get an answer, but it might take 26,000 steps.

Another way to find the word or to learn it is not there is to go straight to the middle of the list (give or take a few words). If you do not find the word, then go to the middle of the half that contains it and forget about the half that does not. (You know which half contains it because you know the list is ordered alphabetically.) This method, called a **binary search**, eliminates roughly 13,000 words in a single step. If you do not find the word on the second try, then jump to the middle of the half that contains it. Continue this way until you have either found the word or divided the list in half so many times there are no words left. How many times do you have to divide the list to find the word or learn that it is not there? At most 15, because

$$(26{,}000/2^{15}) < 1.$$

That certainly beats a possible 26,000 steps.

For a list of length n, a sequential search algorithm takes on the order of n steps to find a word or determine that it is not in the list. A binary search, as the second algorithm is called, takes on the order of $\log_2 n$ steps. The reason is that if $2^{m-1} < n \le 2^m$, then $m - 1 < \log_2 n \le m$, and the number of bisections required to narrow the list to one word will be at most $m = \lceil \log_2 n \rceil$, the integer ceiling of the number $\log_2 n$.

Big-oh notation provides a compact way to say all this. The number of steps in a sequential search of an ordered list is $O(n)$; the number of steps in a binary search is $O(\log_2 n)$. In our example, there is a big difference between the two (26,000 vs. 15), and the difference can only increase with n because n grows faster than $\log_2 n$ as $n \to \infty$.

EXERCISES 7.8

Comparisons with the Exponential e^x

1. Which of the following functions grow faster than e^x as $x \to \infty$? Which grow at the same rate as e^x? Which grow slower?

a. $x - 3$ **b.** $x^3 + \sin^2 x$
c. $\sqrt{x}$ **d.** 4^x
e. $(3/2)^x$ **f.** $e^{x/2}$
g. $e^x/2$ **h.** $\log_{10} x$

2. Which of the following functions grow faster than e^x as $x \to \infty$? Which grow at the same rate as e^x? Which grow slower?

a. $10x^4 + 30x + 1$ **b.** $x \ln x - x$
c. $\sqrt{1 + x^4}$ **d.** $(5/2)^x$
e. e^{-x} **f.** xe^x
g. $e^{\cos x}$ **h.** e^{x-1}

Comparisons with the Power x^2

3. Which of the following functions grow faster than x^2 as $x \to \infty$? Which grow at the same rate as x^2? Which grow slower?

a. $x^2 + 4x$ **b.** $x^5 - x^2$
c. $\sqrt{x^4 + x^3}$ **d.** $(x + 3)^2$
e. $x \ln x$ **f.** 2^x
g. $x^3 e^{-x}$ **h.** $8x^2$

4. Which of the following functions grow faster than x^2 as $x \to \infty$? Which grow at the same rate as x^2? Which grow slower?

a. $x^2 + \sqrt{x}$ **b.** $10x^2$
c. $x^2 e^{-x}$ **d.** $\log_{10}(x^2)$
e. $x^3 - x^2$ **f.** $(1/10)^x$
g. $(1.1)^x$ **h.** $x^2 + 100x$

Comparisons with the Logarithm $\ln x$

5. Which of the following functions grow faster than $\ln x$ as $x \to \infty$? Which grow at the same rate as $\ln x$? Which grow slower?

a. $\log_3 x$ **b.** $\ln 2x$
c. $\ln \sqrt{x}$ **d.** $\sqrt{x}$
e. x **f.** $5 \ln x$
g. $1/x$ **h.** e^x

6. Which of the following functions grow faster than $\ln x$ as $x \to \infty$? Which grow at the same rate as $\ln x$? Which grow slower?

a. $\log_2(x^2)$ **b.** $\log_{10} 10x$
c. $1/\sqrt{x}$ **d.** $1/x^2$
e. $x - 2 \ln x$ **f.** e^{-x}
g. $\ln(\ln x)$ **h.** $\ln(2x + 5)$

Ordering Functions by Growth Rates

7. Order the following functions from slowest growing to fastest growing as $x \to \infty$.

a. e^x **b.** x^x
c. $(\ln x)^x$ **d.** $e^{x/2}$

8. Order the following functions from slowest growing to fastest growing as $x \to \infty$.

a. 2^x **b.** x^2
c. $(\ln 2)^x$ **d.** e^x

Big-oh and Little-oh; Order

9. True, or false? As $x \to \infty$,

a. $x = o(x)$ **b.** $x = o(x + 5)$
c. $x = O(x + 5)$ **d.** $x = O(2x)$
e. $e^x = o(e^{2x})$ **f.** $x + \ln x = O(x)$
g. $\ln x = o(\ln 2x)$ **h.** $\sqrt{x^2 + 5} = O(x)$

10. True, or false? As $x \to \infty$,

a. $\dfrac{1}{x+3} = O\left(\dfrac{1}{x}\right)$ **b.** $\dfrac{1}{x} + \dfrac{1}{x^2} = O\left(\dfrac{1}{x}\right)$
c. $\dfrac{1}{x} - \dfrac{1}{x^2} = o\left(\dfrac{1}{x}\right)$ **d.** $2 + \cos x = O(2)$
e. $e^x + x = O(e^x)$ **f.** $x \ln x = o(x^2)$
g. $\ln(\ln x) = O(\ln x)$ **h.** $\ln(x) = o(\ln(x^2 + 1))$

11. Show that if positive functions $f(x)$ and $g(x)$ grow at the same rate as $x \to \infty$, then $f = O(g)$ and $g = O(f)$.

12. When is a polynomial $f(x)$ of smaller order than a polynomial $g(x)$ as $x \to \infty$? Give reasons for your answer.

13. When is a polynomial $f(x)$ of at most the order of a polynomial $g(x)$ as $x \to \infty$? Give reasons for your answer.

14. What do the conclusions we drew in Section 2.8 about the limits of rational functions tell us about the relative growth of polynomials as $x \to \infty$?

Other Comparisons

T **15.** Investigate

$$\lim_{x\to\infty} \frac{\ln(x+1)}{\ln x} \quad \text{and} \quad \lim_{x\to\infty} \frac{\ln(x+999)}{\ln x}.$$

Then use l'Hôpital's Rule to explain what you find.

16. (*Continuation of Exercise 15.*) Show that the value of

$$\lim_{x\to\infty} \frac{\ln(x+a)}{\ln x}$$

is the same no matter what value you assign to the constant a. What does this say about the relative rates at which the functions $f(x) = \ln(x + a)$ and $g(x) = \ln x$ grow?

17. Show that $\sqrt{10x+1}$ and $\sqrt{x+1}$ grow at the same rate as $x \to \infty$ by showing that they both grow at the same rate as $\sqrt{x}$ as $x \to \infty$.

18. Show that $\sqrt{x^4+x}$ and $\sqrt{x^4-x^3}$ grow at the same rate as $x \to \infty$ by showing that they both grow at the same rate as x^2 as $x \to \infty$.

19. Show that e^x grows faster as $x \to \infty$ than x^n for any positive integer n, even $x^{1{,}000{,}000}$. (*Hint:* What is the nth derivative of x^n?)

20. The function e^x outgrows any polynomial Show that e^x grows faster as $x \to \infty$ than any polynomial

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0.$$

21. a. Show that $\ln x$ grows slower as $x \to \infty$ than $x^{1/n}$ for any positive integer n, even $x^{1/1{,}000{,}000}$.

T **b.** Although the values of $x^{1/1{,}000{,}000}$ eventually overtake the values of $\ln x$, you have to go way out on the x-axis before this happens. Find a value of x greater than 1 for which $x^{1/1{,}000{,}000} > \ln x$. You might start by observing that when $x > 1$ the equation $\ln x = x^{1/1{,}000{,}000}$ is equivalent to the equation $\ln(\ln x) = (\ln x)/1{,}000{,}000$.

T **c.** Even $x^{1/10}$ takes a long time to overtake $\ln x$. Experiment with a calculator to find the value of x at which the graphs of $x^{1/10}$ and $\ln x$ cross, or, equivalently, at which $\ln x = 10 \ln(\ln x)$. Bracket the crossing point between powers of 10 and then close in by successive halving.

T **d.** (*Continuation of part* (*c*).) The value of x at which $\ln x = 10 \ln(\ln x)$ is too far out for some graphers and root finders to identify. Try it on the equipment available to you and see what happens.

22. The function $\ln x$ grows slower than any polynomial Show that $\ln x$ grows slower as $x \to \infty$ than any nonconstant polynomial.

Algorithms and Searches

23. a. Suppose you have three different algorithms for solving the same problem and each algorithm takes a number of steps that is of the order of one of the functions listed here:

$$n \log_2 n, \quad n^{3/2}, \quad n(\log_2 n)^2.$$

Which of the algorithms is the most efficient in the long run? Give reasons for your answer.

T **b.** Graph the functions in part (a) together to get a sense of how rapidly each one grows.

24. Repeat Exercise 23 for the functions

$$n, \quad \sqrt{n}\log_2 n, \quad (\log_2 n)^2.$$

T **25.** Suppose you are looking for an item in an ordered list one million items long. How many steps might it take to find that item with a sequential search? A binary search?

T **26.** You are looking for an item in an ordered list 450,000 items long (the length of *Webster's Third New International Dictionary*). How many steps might it take to find the item with a sequential search? A binary search?

CHAPTER 7 Questions to Guide Your Review

1. What functions have inverses? How do you know if two functions f and g are inverses of one another? Give examples of functions that are (are not) inverses of one another.

2. How are the domains, ranges, and graphs of functions and their inverses related? Give an example.

3. How can you sometimes express the inverse of a function of x as a function of x?

4. Under what circumstances can you be sure that the inverse of a function f is differentiable? How are the derivatives of f and f^{-1} related?

5. What is the natural logarithm function? What are its domain, range, and derivative? What arithmetic properties does it have? Comment on its graph.

6. What is logarithmic differentiation? Give an example.

7. What integrals lead to logarithms? Give examples. What are the integrals of $\tan x$, $\cot x$, $\sec x$, and $\csc x$?

8. How is the exponential function e^x defined? What are its domain, range, and derivative? What laws of exponents does it obey? Comment on its graph.

9. How are the functions a^x and $\log_a x$ defined? Are there any restrictions on a? How is the graph of $\log_a x$ related to the graph of $\ln x$? What truth is there in the statement that there is really only one exponential function and one logarithmic function?

10. How do you solve separable first-order differential equations?

11. What is the law of exponential change? How can it be derived from an initial value problem? What are some of the applications of the law?

12. Describe l'Hôpital's Rule. How do you know when to use the rule and when to stop? Give an example.

13. How can you sometimes handle limits that lead to indeterminate forms ∞/∞, $\infty \cdot 0$, and $\infty - \infty$? Give examples.

14. How can you sometimes handle limits that lead to indeterminate forms 1^∞, 0^0, and ∞^∞? Give examples.

15. How are the inverse trigonometric functions defined? How can you sometimes use right triangles to find values of these functions? Give examples.

16. What are the derivatives of the inverse trigonometric functions? How do the domains of the derivatives compare with the domains of the functions?
17. What integrals lead to inverse trigonometric functions? How do substitution and completing the square broaden the application of these integrals?
18. What are the six basic hyperbolic functions? Comment on their domains, ranges, and graphs. What are some of the identities relating them?
19. What are the derivatives of the six basic hyperbolic functions? What are the corresponding integral formulas? What similarities do you see here with the six basic trigonometric functions?
20. How are the inverse hyperbolic functions defined? Comment on their domains, ranges, and graphs. How can you find values of $\operatorname{sech}^{-1} x$, $\operatorname{csch}^{-1} x$, and $\coth^{-1} x$ using a calculator's keys for $\cosh^{-1} x$, $\sinh^{-1} x$, and $\tanh^{-1} x$?
21. What integrals lead naturally to inverse hyperbolic functions?
22. How do you compare the growth rates of positive functions as $x \to \infty$?
23. What roles do the functions e^x and $\ln x$ play in growth comparisons?
24. Describe big-oh and little-oh notation. Give examples.
25. Which is more efficient—a sequential search or a binary search? Explain.

CHAPTER 7 Practice Exercises

Finding Derivatives

In Exercises 1–24, find the derivative of y with respect to the appropriate variable.

1. $y = 10e^{-x/5}$
2. $y = \sqrt{2}e^{\sqrt{2}x}$
3. $y = \frac{1}{4}xe^{4x} - \frac{1}{16}e^{4x}$
4. $y = x^2e^{-2/x}$
5. $y = \ln(\sin^2\theta)$
6. $y = \ln(\sec^2\theta)$
7. $y = \log_2(x^2/2)$
8. $y = \log_5(3x - 7)$
9. $y = 8^{-t}$
10. $y = 9^{2t}$
11. $y = 5x^{3.6}$
12. $y = \sqrt{2}x^{-\sqrt{2}}$
13. $y = (x + 2)^{x+2}$
14. $y = 2(\ln x)^{x/2}$
15. $y = \sin^{-1}\sqrt{1 - u^2}, \quad 0 < u < 1$
16. $y = \arcsin\left(\frac{1}{\sqrt{v}}\right), \quad v > 1$
17. $y = \ln\cos^{-1}x$
18. $y = z\cos^{-1}z - \sqrt{1 - z^2}$
19. $y = t\arctan t - \frac{1}{2}\ln t$
20. $y = (1 + t^2)\cot^{-1}2t$
21. $y = z\sec^{-1}z - \sqrt{z^2 - 1}, \quad z > 1$
22. $y = 2\sqrt{x - 1}\sec^{-1}\sqrt{x}$
23. $y = \csc^{-1}(\sec\theta), \quad 0 < \theta < \pi/2$
24. $y = (1 + x^2)e^{\tan^{-1}x}$

Logarithmic Differentiation

In Exercises 25–30, use logarithmic differentiation to find the derivative of y with respect to the appropriate variable.

25. $y = \dfrac{2(x^2 + 1)}{\sqrt{\cos 2x}}$
26. $y = \sqrt[10]{\dfrac{3x + 4}{2x - 4}}$
27. $y = \left(\dfrac{(t + 1)(t - 1)}{(t - 2)(t + 3)}\right)^5, \quad t > 2$
28. $y = \dfrac{2u2^u}{\sqrt{u^2 + 1}}$
29. $y = (\sin\theta)^{\sqrt{\theta}}$
30. $y = (\ln x)^{1/(\ln x)}$

Evaluating Integrals

Evaluate the integrals in Exercises 31–78.

31. $\displaystyle\int e^x \sin(e^x)\,dx$
32. $\displaystyle\int e^t \cos(3e^t - 2)\,dt$
33. $\displaystyle\int e^x \sec^2(e^x - 7)\,dx$
34. $\displaystyle\int e^y \csc(e^y + 1)\cot(e^y + 1)\,dy$
35. $\displaystyle\int \sec^2 x\, e^{\tan x}\,dx$
36. $\displaystyle\int \csc^2 x\, e^{\cot x}\,dx$
37. $\displaystyle\int_{-1}^{1} \frac{dx}{3x - 4}$
38. $\displaystyle\int_1^e \frac{\sqrt{\ln x}}{x}\,dx$
39. $\displaystyle\int_0^{\pi} \tan\frac{x}{3}\,dx$
40. $\displaystyle\int_{1/6}^{1/4} 2\cot\pi x\,dx$
41. $\displaystyle\int_0^4 \frac{2t}{t^2 - 25}\,dt$
42. $\displaystyle\int_{-\pi/2}^{\pi/6} \frac{\cos t}{1 - \sin t}\,dt$
43. $\displaystyle\int \frac{\tan(\ln v)}{v}\,dv$
44. $\displaystyle\int \frac{dv}{v\ln v}$
45. $\displaystyle\int \frac{(\ln x)^{-3}}{x}\,dx$
46. $\displaystyle\int \frac{\ln(x - 5)}{x - 5}\,dx$
47. $\displaystyle\int \frac{1}{r}\csc^2(1 + \ln r)\,dr$
48. $\displaystyle\int \frac{\cos(1 - \ln v)}{v}\,dv$
49. $\displaystyle\int x3^{x^2}\,dx$
50. $\displaystyle\int 2^{\tan x}\sec^2 x\,dx$
51. $\displaystyle\int_1^7 \frac{3}{x}\,dx$
52. $\displaystyle\int_1^{32} \frac{1}{5x}\,dx$
53. $\displaystyle\int_1^4 \left(\frac{x}{8} + \frac{1}{2x}\right)dx$
54. $\displaystyle\int_1^8 \left(\frac{2}{3x} - \frac{8}{x^2}\right)dx$
55. $\displaystyle\int_{-2}^{-1} e^{-(x+1)}\,dx$
56. $\displaystyle\int_{-\ln 2}^{0} e^{2w}\,dw$

57. $\displaystyle\int_0^{\ln 5} e^r(3e^r + 1)^{-3/2}\,dr$

58. $\displaystyle\int_0^{\ln 9} e^\theta(e^\theta - 1)^{1/2}\,d\theta$

59. $\displaystyle\int_1^e \frac{1}{x}(1 + 7\ln x)^{-1/3}\,dx$

60. $\displaystyle\int_e^{e^2} \frac{1}{x\sqrt{\ln x}}\,dx$

61. $\displaystyle\int_1^3 \frac{(\ln(v+1))^2}{v+1}\,dv$

62. $\displaystyle\int_2^4 (1 + \ln t)t\ln t\,dt$

63. $\displaystyle\int_1^8 \frac{\log_4\theta}{\theta}\,d\theta$

64. $\displaystyle\int_1^e \frac{8\ln 3\log_3\theta}{\theta}\,d\theta$

65. $\displaystyle\int_{-3/4}^{3/4} \frac{6\,dx}{\sqrt{9-4x^2}}$

66. $\displaystyle\int_{-1/5}^{1/5} \frac{6\,dx}{\sqrt{4-25x^2}}$

67. $\displaystyle\int_{-2}^{2} \frac{3\,dt}{4+3t^2}$

68. $\displaystyle\int_{\sqrt{3}}^{3} \frac{dt}{3+t^2}$

69. $\displaystyle\int \frac{dy}{y\sqrt{4y^2-1}}$

70. $\displaystyle\int \frac{24\,dy}{y\sqrt{y^2-16}}$

71. $\displaystyle\int_{\sqrt{2}/3}^{2/3} \frac{dy}{|y|\sqrt{9y^2-1}}$

72. $\displaystyle\int_{-2/\sqrt{5}}^{-\sqrt{6}/\sqrt{5}} \frac{dy}{|y|\sqrt{5y^2-3}}$

73. $\displaystyle\int \frac{dx}{\sqrt{-2x-x^2}}$

74. $\displaystyle\int \frac{dx}{\sqrt{-x^2+4x-1}}$

75. $\displaystyle\int_{-2}^{-1} \frac{2\,dv}{v^2+4v+5}$

76. $\displaystyle\int_{-1}^{1} \frac{3\,dv}{4v^2+4v+4}$

77. $\displaystyle\int \frac{dt}{(t+1)\sqrt{t^2+2t-8}}$

78. $\displaystyle\int \frac{dt}{(3t+1)\sqrt{9t^2+6t}}$

Solving Equations

In Exercises 79–84, solve for y.

79. $3^y = 2^{y+1}$

80. $4^{-y} = 3^{y+2}$

81. $9e^{2y} = x^2$

82. $3^y = 3\ln x$

83. $\ln(y-1) = x + \ln y$

84. $\ln(10\ln y) = \ln 5x$

Applying L'Hôpital's Rule

Use l'Hôpital's Rule to find the limits in Exercises 85–108.

85. $\displaystyle\lim_{x\to 1}\frac{x^2+3x-4}{x-1}$

86. $\displaystyle\lim_{x\to 1}\frac{x^a-1}{x^b-1}$

87. $\displaystyle\lim_{x\to\pi}\frac{\tan x}{x}$

88. $\displaystyle\lim_{x\to 0}\frac{\tan x}{x+\sin x}$

89. $\displaystyle\lim_{x\to 0}\frac{\sin^2 x}{\tan(x^2)}$

90. $\displaystyle\lim_{x\to 0}\frac{\sin mx}{\sin nx}$

91. $\displaystyle\lim_{x\to\pi/2^-}\sec 7x\cos 3x$

92. $\displaystyle\lim_{x\to 0^+}\sqrt{x}\sec x$

93. $\displaystyle\lim_{x\to 0}(\csc x - \cot x)$

94. $\displaystyle\lim_{x\to 0}\left(\frac{1}{x^4}-\frac{1}{x^2}\right)$

95. $\displaystyle\lim_{x\to\infty}\left(\sqrt{x^2+x+1}-\sqrt{x^2-x}\right)$

96. $\displaystyle\lim_{x\to\infty}\left(\frac{x^3}{x^2-1}-\frac{x^3}{x^2+1}\right)$

97. $\displaystyle\lim_{x\to 0}\frac{10^x-1}{x}$

98. $\displaystyle\lim_{\theta\to 0}\frac{3^\theta-1}{\theta}$

99. $\displaystyle\lim_{x\to 0}\frac{2^{\sin x}-1}{e^x-1}$

100. $\displaystyle\lim_{x\to 0}\frac{2^{-\sin x}-1}{e^x-1}$

101. $\displaystyle\lim_{x\to 0}\frac{5-5\cos x}{e^x-x-1}$

102. $\displaystyle\lim_{x\to 0}\frac{x\sin x^2}{\tan^3 x}$

103. $\displaystyle\lim_{t\to 0^+}\frac{t-\ln(1+2t)}{t^2}$

104. $\displaystyle\lim_{x\to 4}\frac{\sin^2(\pi x)}{e^{x-4}+3-x}$

105. $\displaystyle\lim_{t\to 0^+}\left(\frac{e^t}{t}-\frac{1}{t}\right)$

106. $\displaystyle\lim_{y\to 0^+} e^{-1/y}\ln y$

107. $\displaystyle\lim_{x\to\infty}\left(\frac{e^x+1}{e^x-1}\right)^{\ln x}$

108. $\displaystyle\lim_{x\to 0^+}\left(1+\frac{3}{x}\right)^x$

Comparing Growth Rates of Functions

109. Does f grow faster, slower, or at the same rate as g as $x\to\infty$? Give reasons for your answers.

a. $f(x) = \log_2 x$, $g(x) = \log_3 x$

b. $f(x) = x$, $g(x) = x + \dfrac{1}{x}$

c. $f(x) = x/100$, $g(x) = xe^{-x}$

d. $f(x) = x$, $g(x) = \tan^{-1}x$

e. $f(x) = \csc^{-1}x$, $g(x) = 1/x$

f. $f(x) = \sinh x$, $g(x) = e^x$

110. Does f grow faster, slower, or at the same rate as g as $x\to\infty$? Give reasons for your answers.

a. $f(x) = 3^{-x}$, $g(x) = 2^{-x}$

b. $f(x) = \ln 2x$, $g(x) = \ln x^2$

c. $f(x) = 10x^3 + 2x^2$, $g(x) = e^x$

d. $f(x) = \tan^{-1}(1/x)$, $g(x) = 1/x$

e. $f(x) = \sin^{-1}(1/x)$, $g(x) = 1/x^2$

f. $f(x) = \operatorname{sech} x$, $g(x) = e^{-x}$

111. True, or false? Give reasons for your answers.

a. $\dfrac{1}{x^2}+\dfrac{1}{x^4} = O\left(\dfrac{1}{x^2}\right)$

b. $\dfrac{1}{x^2}+\dfrac{1}{x^4} = O\left(\dfrac{1}{x^4}\right)$

c. $x = o(x + \ln x)$

d. $\ln(\ln x) = o(\ln x)$

e. $\tan^{-1}x = O(1)$

f. $\cosh x = O(e^x)$

112. True, or false? Give reasons for your answers.

a. $\dfrac{1}{x^4} = O\left(\dfrac{1}{x^2}+\dfrac{1}{x^4}\right)$

b. $\dfrac{1}{x^4} = o\left(\dfrac{1}{x^2}+\dfrac{1}{x^4}\right)$

c. $\ln x = o(x+1)$

d. $\ln 2x = O(\ln x)$

e. $\sec^{-1}x = O(1)$

f. $\sinh x = O(e^x)$

Theory and Applications

113. The function $f(x) = e^x + x$, being differentiable and one-to-one, has a differentiable inverse $f^{-1}(x)$. Find the value of df^{-1}/dx at the point $f(\ln 2)$.

114. Find the inverse of the function $f(x) = 1 + (1/x)$, $x \neq 0$. Then show that $f^{-1}(f(x)) = f(f^{-1}(x)) = x$ and that

$$\left.\frac{df^{-1}}{dx}\right|_{f(x)} = \frac{1}{f'(x)}.$$

In Exercises 115 and 116, find the absolute maximum and minimum values of each function on the given interval.

115. $y = x \ln 2x - x, \quad \left[\frac{1}{2e}, \frac{e}{2}\right]$

116. $y = 10x(2 - \ln x), \quad (0, e^2]$

117. Area Find the area between the curve $y = 2(\ln x)/x$ and the x-axis from $x = 1$ to $x = e$.

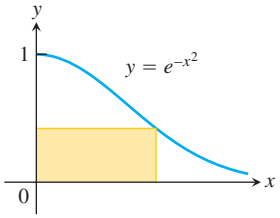
a. Show that the area between the curve $y = 1/x$ and the x-axis from $x = 10$ to $x = 20$ is the same as the area between the curve and the x-axis from $x = 1$ to $x = 2$.

b. Show that the area between the curve $y = 1/x$ and the x-axis from ka to kb is the same as the area between the curve and the x-axis from $x = a$ to $x = b$ $(0 < a < b, k > 0)$.

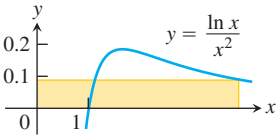
119. A particle is traveling upward and to the right along the curve $y = \ln x$. Its x-coordinate is increasing at the rate $(dx/dt) = \sqrt{x}$ m/s. At what rate is the y-coordinate changing at the point $(e^2, 2)$?

120. A girl is sliding down a slide shaped like the curve $y = 3e^{-x/3}$. Her y-coordinate is changing at the rate $dy/dt = (-1/4)\sqrt{3 - y}$ m/s. At approximately what rate is her x-coordinate changing when she reaches the bottom of the slide at $x = 3$ m?

121. The rectangle shown here has one side on the positive y-axis, one side on the positive x-axis, and its upper right-hand vertex on the curve $y = e^{-x^2}$. What dimensions give the rectangle its largest area, and what is that area?



122. The rectangle shown here has one side on the positive y-axis, one side on the positive x-axis, and its upper right-hand vertex on the curve $y = (\ln x)/x^2$. What dimensions give the rectangle its largest area, and what is that area?



T **123.** Graph the following functions and use what you see to locate and estimate the extreme values, identify the coordinates of the inflection points, and identify the intervals on which the graphs are concave up and concave down. Then confirm your estimates by working with the functions' derivatives.

a. $y = (\ln x)/\sqrt{x}$

b. $y = e^{-x^2}$

c. $y = (1 + x)e^{-x}$

T **124.** Graph $f(x) = x \ln x$. Does the function appear to have an absolute minimum value? Confirm your answer with calculus.

In Exercises 125–128 solve the differential equation.

125. $\frac{dy}{dx} = \sqrt{y}\cos^2\sqrt{y}$

126. $y' = \frac{3y(x + 1)^2}{y - 1}$

127. $yy' = \sec y^2 \sec^2 x$

128. $y\cos^2 x\, dy + \sin x\, dx = 0$

In Exercises 129–132 solve the initial value problem.

129. $\frac{dy}{dx} = e^{-x-y-2}, \quad y(0) = -2$

130. $\frac{dy}{dx} = \frac{y \ln y}{1 + x^2}, \quad y(0) = e^2$

131. $x\, dy - \left(y + \sqrt{y}\right) dx = 0, \quad y(1) = 1$

132. $y^{-2}\frac{dx}{dy} = \frac{e^x}{e^{2x} + 1}, \quad y(0) = 1$

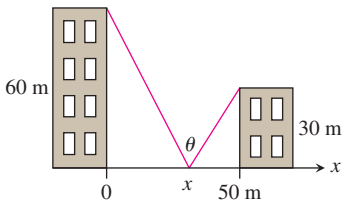
133. What is the age of a sample of charcoal in which 90% of the carbon-14 originally present has decayed?

134. Cooling a pie A deep-dish apple pie, whose internal temperature was 104°C when removed from the oven, was set out on a breezy 5°C porch to cool. Fifteen minutes later, the pie's internal temperature was 82°C. How long did it take the pie to cool from there to 21°C?

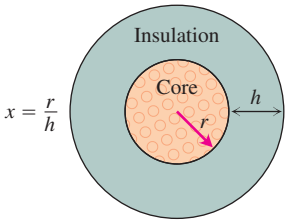
135. Locating a solar station You are under contract to build a solar station at ground level on the east–west line between the two buildings shown here. How far from the taller building should you place the station to maximize the number of hours it will be in the sun on a day when the sun passes directly overhead? Begin by observing that

$$\theta = \pi - \cot^{-1}\frac{x}{60} - \cot^{-1}\frac{50 - x}{30}.$$

Then find the value of x that maximizes θ.



136. A round underwater transmission cable consists of a core of copper wires surrounded by nonconducting insulation. If x denotes the ratio of the radius of the core to the thickness of the insulation, it is known that the speed of the transmission signal is given by the equation $v = x^2 \ln (1/x)$. If the radius of the core is 1 cm, what insulation thickness h will allow the greatest transmission speed?



CHAPTER 7 Additional and Advanced Exercises

Limits

Find the limits in Exercises 1–6.

1. $\lim_{b\to 1^-}\int_0^b \frac{dx}{\sqrt{1-x^2}}$

2. $\lim_{x\to\infty}\frac{1}{x}\int_0^x \tan^{-1} t\,dt$

3. $\lim_{x\to 0^+}(\cos\sqrt{x})^{1/x}$

4. $\lim_{x\to\infty}(x+e^x)^{2/x}$

5. $\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}\right)$

6. $\lim_{n\to\infty}\frac{1}{n}\left(e^{1/n}+e^{2/n}+\cdots+e^{(n-1)/n}+e^{n/n}\right)$

7. Let $A(t)$ be the area of the region in the first quadrant enclosed by the coordinate axes, the curve $y=e^{-x}$, and the vertical line $x=t, t>0$. Let $V(t)$ be the volume of the solid generated by revolving the region about the x-axis. Find the following limits.

a. $\lim_{t\to\infty} A(t)$ **b.** $\lim_{t\to\infty} V(t)/A(t)$ **c.** $\lim_{t\to 0^+} V(t)/A(t)$

8. Varying a logarithm's base

a. Find $\lim \log_a 2$ as $a\to 0^+, 1^-, 1^+$, and ∞.

T **b.** Graph $y=\log_a 2$ as a function of a over the interval $0<a\le 4$.

In Exercises 9 and 10, use implicit differentiation to find $\frac{dy}{dx}$.

9. $y^{e^x}=x^y+1$ **10.** $y^{\ln x}=x^{x^y}$

Theory and Examples

11. Find the areas between the curves $y=2(\log_2 x)/x$ and $y=2(\log_4 x)/x$ and the x-axis from $x=1$ to $x=e$. What is the ratio of the larger area to the smaller?

T **12.** Graph $f(x)=\tan^{-1}x+\tan^{-1}(1/x)$ for $-5\le x\le 5$. Then use calculus to explain what you see. How would you expect f to behave beyond the interval $[-5,5]$? Give reasons for your answer.

13. For what $x>0$ does $x^{(x^x)}=(x^x)^x$? Give reasons for your answer.

T **14.** Graph $f(x)=(\sin x)^{\sin x}$ over $[0,3\pi]$. Explain what you see.

15. Find $f'(2)$ if $f(x)=e^{g(x)}$ and $g(x)=\int_2^x \frac{t}{1+t^4}dt$.

16. a. Find df/dx if

$$f(x)=\int_1^{e^x}\frac{2\ln t}{t}dt.$$

b. Find $f(0)$.

c. What can you conclude about the graph of f? Give reasons for your answer.

17. Even-odd decompositions

a. Suppose that g is an even function of x and h is an odd function of x. Show that if $g(x)+h(x)=0$ for all x then $g(x)=0$ for all x and $h(x)=0$ for all x.

b. If $f(x)=f_E(x)+f_O(x)$ is the sum of an even function $f_E(x)$ and an odd function $f_O(x)$, then show that

$$f_E(x)=\frac{f(x)+f(-x)}{2} \quad \text{and} \quad f_O(x)=\frac{f(x)-f(-x)}{2}.$$

c. What is the significance of the result in part (b)?

18. Let g be a function that is differentiable throughout an open interval containing the origin. Suppose g has the following properties:

i) $g(x+y)=\frac{g(x)+g(y)}{1-g(x)g(y)}$ for all real numbers x, y, and $x+y$ in the domain of g.

ii) $\lim_{h\to 0} g(h)=0$

iii) $\lim_{h\to 0}\frac{g(h)}{h}=1$

a. Show that $g(0)=0$.

b. Show that $g'(x)=1+[g(x)]^2$.

c. Find $g(x)$ by solving the differential equation in part (b).

19. Center of mass Find the center of mass of a thin plate of constant density covering the region in the first and fourth quadrants enclosed by the curves $y=1/(1+x^2)$ and $y=-1/(1+x^2)$ and by the lines $x=0$ and $x=1$.

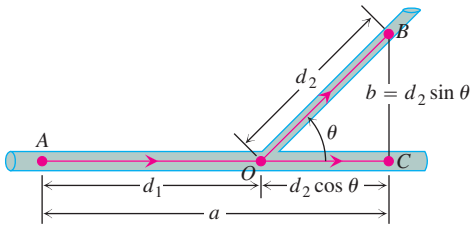
20. Solid of revolution The region between the curve $y=1/(2\sqrt{x})$ and the x-axis from $x=1/4$ to $x=4$ is revolved about the x-axis to generate a solid.

a. Find the volume of the solid.

b. Find the centroid of the region.

21. The best branching angles for blood vessels and pipes When a smaller pipe branches off from a larger one in a flow system, we may want it to run off at an angle that is best from some energy-saving point of view. We might require, for instance, that energy loss due to friction be minimized along the section AOB shown in the accompanying figure. In this diagram, B is a given point to be reached by the smaller pipe, A is a point in the larger pipe upstream from B, and O is the point where the branching occurs. A law due to Poiseuille states that the loss of energy due to friction in nonturbulent flow is proportional to the length of the path and inversely proportional to the fourth power of the radius. Thus, the loss along AO is $(kd_1)/R^4$ and along OB is $(kd_2)/r^4$, where k is a constant, d_1 is the length of AO, d_2 is the length of OB, R is the radius of the larger pipe, and r is the radius of the smaller pipe. The angle θ is to be chosen to minimize the sum of these two losses:

$$L=k\frac{d_1}{R^4}+k\frac{d_2}{r^4}.$$



In our model, we assume that $AC=a$ and $BC=b$ are fixed. Thus we have the relations

$$d_1+d_2\cos\theta=a \qquad d_2\sin\theta=b$$

so that

$$d_2 = b \csc \theta,$$
$$d_1 = a - d_2 \cos \theta = a - b \cot \theta.$$

We can express the total loss L as a function of θ:

$$L = k\left(\frac{a - b \cot \theta}{R^4} + \frac{b \csc \theta}{r^4}\right).$$

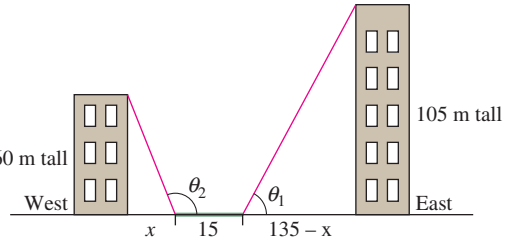
a. Show that the critical value of θ for which $dL/d\theta$ equals zero is

$$\theta_c = \cos^{-1} \frac{r^4}{R^4}.$$

b. If the ratio of the pipe radii is $r/R = 5/6$, estimate to the nearest degree the optimal branching angle given in part (a).

The mathematical analysis described here is also used to explain the angles at which arteries branch in an animal's body.

22. Urban gardening A vegetable garden 15 m wide is to be grown between two buildings, which are 150 m apart along an east-west line. If the buildings are 60 m and 105 m tall, where should the garden be placed in order to receive the maximum number of hours of sunlight exposure? (*Hint:* Determine the value of x in the accompanying figure that maximizes sunlight exposure for the garden.)



23. Consider point (a, b) on the graph of $y = \ln x$ and triangle ABC formed by the tangent line at (a, b), the y-axis, and the line $y = b$. Show that

$$(\text{area triangle } ABC) = \frac{a}{2}.$$

